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JAMB Mathematics: The Complete Course

Concepts, worked examples and exercises for the UTME

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Chapter 1: Number Bases, Fractions, Decimals, Percentages, Ratio and Proportion

Every topic in mathematics rests on number work. This chapter builds those foundations: kinds of numbers and place value, number bases, fractions, decimals and standard form, and then the everyday applications that appear again and again in the UTME: percentages, profit and loss, interest, ratio and proportion.

1.1 Kinds of numbers

The **natural numbers** are the counting numbers 1, 2, 3, 4, … The **whole numbers** add zero to this list. The **integers** include the negative whole numbers as well: …, −3, −2, −1, 0, 1, 2, 3, … A **rational number** is any number that can be written as a fraction p/q where p and q are integers and q ≠ 0. Numbers such as 3/4, −5/2, 0.75 and 7 (which is 7/1) are all rational. An **irrational number** cannot be written in that way; its decimal expansion never ends and never repeats. Examples are √2, √3 and π. Together the rational and irrational numbers form the **real numbers**.

A **prime number** has exactly two factors, itself and 1: 2, 3, 5, 7, 11, 13, … Note that 2 is the only even prime and that 1 is not prime. Every whole number greater than 1 is a product of primes in exactly one way (apart from order), its **prime factorisation**; for example $360 = 2^3 \times 3^2 \times 5$.

Prime factorisation gives the quickest route to the **highest common factor (HCF)** and the **lowest common multiple (LCM)**. The HCF is the product of the common primes, each taken to the *lowest* power that appears. The LCM is the product of all the primes that appear, each taken to the *highest* power that appears.

WORKED EXAMPLE 1.1

Find the HCF and LCM of 36, 48 and 60.

Solution: $36 = 2^2 \times 3^2$; $48 = 2^4 \times 3$; $60 = 2^2 \times 3 \times 5$. The primes common to all three are 2 and 3. The lowest power of 2 is 2^2 and the lowest power of 3 is 3^1, so HCF $= 2^2 \times 3 = 12$. For the LCM take the highest power of every prime that occurs: 2^4, 3^2 and 5. LCM $= 16 \times 9 \times 5 = 720$.

1.2 Number bases

In everyday life we count in **base ten**, also called the denary or decimal system. It uses ten digits, 0 to 9, and the position of a digit tells us its value: in 4,203 the digit 4 stands for 4×1000, the 2 for 2×100, the 0 for no tens and the 3 for 3 units. Each place is worth ten times the place

to its right. A number written in **base n** uses the digits 0 to n − 1, and each place is worth n times the place to its right. The base is written as a small subscript: 1101_2 means the number 1101 in base two.

Converting from base n to base ten

Multiply each digit by the power of n belonging to its place, starting with $n^0 = 1$ on the right, and add the results.

WORKED EXAMPLE 1.2

Convert (a) 1101_2 and (b) 365_8 to base ten.

Solution: (a) $1101_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 8 + 4 + 0 + 1 = 13$. (b) $365_8 = 3 \times 8^2 + 6 \times 8 + 5 = 192 + 48 + 5 = 245$.

Converting from base ten to base n

Divide the number repeatedly by n, writing down each remainder, until the quotient is 0. Then read the remainders from the last to the first (bottom to top).

WORKED EXAMPLE 1.3

Express 47 in base two.

Solution: 47 ÷ 2 = 23 remainder 1; 23 ÷ 2 = 11 remainder 1; 11 ÷ 2 = 5 remainder 1; 5 ÷ 2 = 2 remainder 1; 2 ÷ 2 = 1 remainder 0; 1 ÷ 2 = 0 remainder 1. Reading the remainders upwards gives 101111_2. Check: 32 + 8 + 4 + 2 + 1 = 47.

Converting between two bases other than ten

Go through base ten: convert to denary first, then from denary to the required base. For example, $213_4 = 2 \times 16 + 1 \times 4 + 3 = 39$, and $39 = 5 \times 7 + 4$, so $213_4 = 54_7$.

Arithmetic in other bases

The safest method for examination work is to convert to base ten, calculate, and convert back. For addition in base two you can also use the rules $1 + 1 = 10$ (write 0, carry 1) and $1 + 1 + 1 = 11$ (write 1, carry 1).

WORKED EXAMPLE 1.4

Calculate (a) $1101_2 + 1011_2$ and (b) $342_5 - 143_5$.

Solution: (a) $1101_2 = 13$ and $1011_2 = 8 + 2 + 1 = 11$. The sum is 24. Since $24 = 16 + 8$, we get 11000_2. (b) $342_5 = 3 \times 25 + 4 \times 5 + 2 = 97$ and $143_5 = 25 + 20 + 3 = 48$. The difference is $97 - 48 = 49$. Now $49 \div 5 = 9$ remainder 4; $9 \div 5 = 1$ remainder 4; $1 \div 5 = 0$ remainder 1. So the answer is 144_5. Check: $25 + 20 + 4 = 49$.

Finding an unknown base

When the base itself is unknown, express both sides in base ten using the unknown x and solve the resulting equation.

WORKED EXAMPLE 1.5

Find the value of x if $101_x = 26$ (in base ten), and the value of y if $35_y = 23$.

Solution: $101_x = 1 \times x^2 + 0 \times x + 1 = x^2 + 1$. Setting $x^2 + 1 = 26$ gives $x^2 = 25$, so $x = 5$ (a base cannot be negative). Next, $35_y = 3y + 5 = 23$, so $3y = 18$ and $y = 6$. Notice that the digit 5 is allowed in base 6 because the digits run from 0 to 5.

Key point. In base n the largest digit is n − 1. If you see a digit that is too large for the stated base, you have made an error or the question is asking you to deduce that the base must be bigger. Fractions in other bases work the same way, with negative powers: $0.101_2 = 1/2 + 0/4 + 1/8 = 5/8 = 0.625$.

Exercise 1.1

1. Convert 10110_2 to base ten.
2. Express 156 in base eight.
3. Calculate $432_5 + 244_5$, leaving the answer in base five.
4. Find x if $43_x = 31$ in base ten.
5. Find the HCF and LCM of 24 and 90.
6. Convert 0.011_2 to a fraction in its lowest terms.

Answers to Exercise 1.1

1. 22: $16 + 0 + 4 + 2 + 0 = 22$.
2. 234_8: $156 \div 8 = 19$ r 4; $19 \div 8 = 2$ r 3; $2 \div 8 = 0$ r 2; reading upwards gives 234; check $128 + 24 + 4 = 156$.
3. 1231_5: $432_5 = 117$ and $244_5 = 74$, the sum is 191; $191 \div 5 = 38$ r 1; $38 \div 5 = 7$ r 3; $7 \div 5 = 1$ r 2; $1 \div 5 = 0$ r 1.
4. $x = 7$: $4x + 3 = 31$ gives $4x = 28$.
5. HCF = 6 and LCM = 360: $24 = 2^3 \times 3$ and $90 = 2 \times 3^2 \times 5$.
6. 3/8: $0.011_2 = 0/2 + 1/4 + 1/8 = 3/8$.

1.3 Fractions

A fraction a/b describes a parts out of b equal parts. The top number is the **numerator** and the bottom number the **denominator**. A **proper fraction** has a numerator smaller than the denominator; an **improper fraction** does not, and a **mixed number** such as 2 1/3 combines a whole number with a fraction (2 1/3 = 7/3). Equivalent fractions are obtained by multiplying or dividing the top and bottom by the same non-zero number, so 2/3 = 4/6 = 10/15. To simplify a fraction divide top and bottom by their HCF.

To **add or subtract** fractions, first rewrite them over a common denominator, preferably the LCM of the denominators, then combine the numerators. To **multiply**, multiply numerators together and denominators together, cancelling common factors first to keep the

numbers small. To **divide** by a fraction, multiply by its reciprocal: (a/b) ÷ (c/d) = (a/b) × (d/c). The word *of* means multiplication, so 3/4 of 80 is 3/4 × 80 = 60.

When an expression contains several operations, follow the order **BODMAS**: Brackets, Of, Division, Multiplication, Addition, Subtraction. Division and multiplication have equal rank and are done from left to right, as are addition and subtraction.

WORKED EXAMPLE 1.6

Evaluate (a) 2/3 + 3/4 − 5/6 and (b) (3/4 of 2 2/5) ÷ 1 1/5.

Solution: (a) The LCM of 3, 4 and 6 is 12. Then 2/3 = 8/12, 3/4 = 9/12 and 5/6 = 10/12, so the sum is (8 + 9 − 10)/12 = 7/12. (b) Convert the mixed numbers: 2 2/5 = 12/5 and 1 1/5 = 6/5. Then 3/4 of 12/5 = 3/4 × 12/5 = 36/20 = 9/5. Finally 9/5 ÷ 6/5 = 9/5 × 5/6 = 45/30 = 3/2 = 1 1/2.

1.4 Decimals, rounding and standard form

A decimal fraction extends place value to the right of the decimal point: the places are tenths, hundredths, thousandths and so on. To convert a fraction to a decimal, divide the numerator by the denominator (3/8 = 0.375). A decimal that stops is called **terminating**; one whose digits repeat forever is a **recurring decimal**, written with dots over the repeating digits. All terminating and recurring decimals are rational numbers.

Recurring decimals as fractions

Let x equal the recurring decimal. Multiply by a power of ten so that the repeating blocks line up, subtract, and solve.

WORKED EXAMPLE 1.7

Express (a) 0.363636… and (b) 0.1666… as fractions in their lowest terms.

Solution: (a) Let x = 0.3636… Then 100x = 36.3636… Subtracting, 99x = 36, so x = 36/99 = 4/11. (b) Let x = 0.1666… Then 10x = 1.666… and 100x = 16.666… Subtracting the second-last from the last, 90x = 15, so x = 15/90 = 1/6. Check: 1 ÷ 6 = 0.1666…

Decimal places and significant figures

To round to a stated number of **decimal places**, look at the digit immediately after the last place you keep. If it is 5 or more, increase the last kept digit by one; otherwise leave it. Thus 3.14159 to 3 decimal places is 3.142. **Significant figures** count from the first non-zero digit. Leading zeros are not significant, but zeros between or after non-zero digits are. So 0.004567 to 2 significant figures is 0.0046, and 45,678 to 3 significant figures is 45,700 (the zeros keep the place values).

Standard form

A number is in **standard form** (scientific notation) when written as $A \times 10^n$ with $1 \le A < 10$ and n an integer. For example, $0.000345 = 3.45 \times 10^{-4}$ and $52{,}000{,}000 = 5.2 \times 10^7$. When multiplying, multiply the A parts and add the powers; then adjust so that A lies between 1 and 10.

WORKED EXAMPLE 1.8

Evaluate $(4.2 \times 10^5) \times (3 \times 10^{-2})$, giving the answer in standard form, and simplify $0.0036 \times 0.25 \div 0.09$.

Solution: $4.2 \times 3 = 12.6$ and $10^5 \times 10^{-2} = 10^3$, so the product is $12.6 \times 10^3 = 1.26 \times 10^4$. For the second part, $0.0036 \times 0.25 = 0.0009$, and $0.0009 \div 0.09 = 0.01$.

1.5 Percentages

A **percentage** is a fraction with denominator 100; the symbol % means "out of a hundred". To change a fraction to a percentage multiply by 100, so 3/5 = 60%. To find p% of a quantity, multiply the quantity by p/100. To express one quantity as a percentage of another, divide the first by the second and multiply by 100.

Percentage change is calculated as (change ÷ original value) × 100%. A convenient shortcut uses multipliers: increasing by r% means multiplying by (1 + r/100), and decreasing by r% means multiplying by (1 − r/100). A 40% rise is a multiplier of 1.4; a 15% fall is a multiplier of 0.85. Successive changes multiply the multipliers.

Profit, loss and discount

Profit or loss is the difference between selling price (SP) and cost price (CP). Unless the question says otherwise, **percentage profit = profit/CP × 100** and percentage loss = loss/CP × 100. A discount is a reduction from the marked price and is calculated on the marked price.

WORKED EXAMPLE 1.9

A trader buys a bag of rice for ₦8,000 and sells it for ₦9,200. Find the percentage profit. At what price must she sell goods that cost ₦6,400 to make a 25% profit?

Solution: Profit = 9,200 − 8,000 = ₦1,200, and 1,200/8,000 × 100 = 15%. For the second part, SP = 1.25 × 6,400 = ₦8,000.

Reverse percentages

When the final amount is known and you need the original, divide by the multiplier rather than taking a percentage of the final amount.

WORKED EXAMPLE 1.10

After a 20% increase, a pair of shoes costs ₦4,800. What was the original price? Also, what single percentage change results from a 10% increase followed by a 10% decrease?

Solution: The multiplier is 1.2, so the original price is 4,800 ÷ 1.2 = ₦4,000. For the second part, 1.1 × 0.9 = 0.99, which is a decrease of 1%.

Simple and compound interest

With **simple interest**, interest is calculated on the original principal P every year: I = PRT/100, where R is the rate per cent per annum and T the time in years. The amount is A = P + I. With **compound interest** the interest of each year is added to the principal before the next year's interest is calculated, so $A = P(1 + R/100)^n$ for n years. Compound interest is the same as repeated percentage increase. Depreciation uses a decreasing multiplier.

WORKED EXAMPLE 1.11

Find (a) the simple interest and amount on ₦50,000 at 8% per annum for 3 years; (b) the compound interest on ₦100,000 at 10% per annum for 3 years; (c) the value after 2 years of a car bought for ₦2,000,000 that depreciates by 15% each year.

Solution: (a) I = 50,000 × 8 × 3/100 = ₦12,000, so the amount is ₦62,000. (b) $A = 100{,}000 \times 1.1^3 = 100{,}000 \times 1.331 = ₦133{,}100$, so the compound interest is 133,100 − 100,000 = ₦33,100. (c) Value $= 2{,}000{,}000 \times 0.85^2 = 2{,}000{,}000 \times 0.7225 = ₦1{,}445{,}000$.

1.6 Ratio, rates and proportion

A **ratio** compares quantities of the same kind, written a : b. Like a fraction, it can be simplified by dividing both parts by a common factor, so 12 : 18 = 2 : 3. To share a quantity in the ratio a : b : c, add the parts,

find the value of one part, and multiply. When a ratio is given in fractions, multiply every part by the LCM of the denominators to clear them.

WORKED EXAMPLE 1.12

Share ₦84,000 among three friends in the ratio 2 : 3 : 7. Then, if a : b = 2 : 3 and b : c = 4 : 5, find a : b : c.

Solution: The number of parts is 2 + 3 + 7 = 12, so one part is 84,000 ÷ 12 = ₦7,000. The shares are 2 × 7,000 = ₦14,000, 3 × 7,000 = ₦21,000 and 7 × 7,000 = ₦49,000; these total ₦84,000. For the second part, make the value of b the same in both ratios. The LCM of 3 and 4 is 12, so a : b = 8 : 12 and b : c = 12 : 15. Hence a : b : c = 8 : 12 : 15.

Direct and inverse proportion

Two quantities are in **direct proportion** if they increase or decrease at the same rate, so that their ratio is constant; double one and the other doubles. They are in **inverse proportion** if one increases as the other decreases so that their product stays constant. The formal treatment through variation appears in Chapter 6. For simple problems, use the unitary method: find the value for one unit first.

WORKED EXAMPLE 1.13

Six workers build a wall in 10 days. How long would 15 workers take, working at the same rate?

Solution: More workers means fewer days, so this is inverse proportion. The total work is $6 \times 10 = 60$ worker-days. With 15 workers the time is $60 \div 15 = 4$ days.

Exercise 1.2

1. A trader buys a bag of beans for ₦45,000 and sells it for ₦54,000. Calculate her percentage profit.
2. Find the compound interest on ₦200,000 for 2 years at 5% per annum.
3. After a 25% discount, a shirt costs ₦3,750. What was the marked price?
4. Divide 360 in the ratio 1/2 : 1/3 : 1/6.
5. Eight men can dig a trench in 9 days. How many days will 6 men need at the same rate?
6. If a : b = 3 : 4 and b : c = 6 : 7, find a : b : c.

Answers to Exercise 1.2

1. 20%: profit = 9,000 and 9,000/45,000 × 100 = 20.

2. ₦20,500: 200,000 × 1.05^2 = 200,000 × 1.1025 = 220,500; subtract 200,000.
3. ₦5,000: 3,750 ÷ 0.75 = 5,000.
4. 180, 120 and 60: multiplying by 6 gives 3 : 2 : 1 with 6 parts, and 360 ÷ 6 = 60.
5. 12 days: 8 × 9 = 72 man-days and 72 ÷ 6 = 12.
6. 9 : 12 : 14: a : b = 9 : 12 (multiply by 3) and b : c = 12 : 14 (multiply by 2).

Chapter summary

- Real numbers are rational or irrational; every integer above 1 has a unique prime factorisation, which gives the HCF (lowest powers of common primes) and LCM (highest powers of all primes).
- To convert base n to base ten, multiply each digit by its power of n; to convert base ten to base n, divide repeatedly by n and read the remainders upwards. For other pairs of bases, go through base ten.
- For an unknown base x, write each side in base ten and solve the equation; the digits used must all be smaller than the base.
- Add and subtract fractions over a common denominator, multiply numerators and denominators, divide by multiplying by the reciprocal, and obey BODMAS.

- A recurring decimal becomes a fraction by multiplying by a suitable power of ten and subtracting. Round to decimal places or significant figures only at the end, and write standard form as $A \times 10^n$ with $1 \le A < 10$.
- Percentage change uses multipliers $(1 \pm r/100)$. Profit and loss are measured against cost price; reverse percentages require division by the multiplier.
- Simple interest is PRT/100; compound interest uses $A = P(1 + R/100)^n$; depreciation uses a multiplier below 1.
- Share in a ratio by finding the value of one part; combine ratios by matching the common term; for inverse proportion the product is constant.

Chapter 2: Indices, Surds and Logarithms

Indices, surds and logarithms are three views of the same idea: repeated multiplication and its inverse. Indices describe repeated multiplication compactly, surds deal with roots that cannot be written exactly as fractions, and logarithms answer the question "what power is needed?" Questions on these topics are among the most reliable marks in the UTME because they follow a small number of rules that can be learned completely. This chapter states each rule, explains why it works, and then shows how the rules combine in the equations that examiners like to set.

2.1 Laws of indices

When a number a is multiplied by itself n times, the product is written a^n. Here a is the **base** and n is the **index** (or power or exponent). So $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$. From this definition the following laws follow directly. Throughout, the base is a non-zero real number and the indices are chosen so that every expression makes sense.

1. **Multiplication:** $a^m \times a^n = a^{m+n}$. Think of m factors followed by n factors: there are m + n factors in all.
2. **Division:** $a^m \div a^n = a^{m-n}$. Cancelling n of the m factors leaves m − n.

3. **Power of a power:** $(a^m)^n = a^{mn}$. There are n groups, each containing m factors.
4. **Power of a product:** $(ab)^n = a^n b^n$ and $(a/b)^n = a^n / b^n$.
5. **Zero index:** $a^0 = 1$. This follows from the division law, since $a^n \div a^n = a^{n-n} = a^0$ and anything divided by itself is 1.
6. **Negative index:** $a^{-n} = 1/a^n$. Indeed $a^0 \div a^n = a^{-n}$, and $a^0 \div a^n = 1/a^n$.
7. **Fractional index:** $a^{1/n}$ is the n-th root of a, and $a^{m/n} = (a^{1/n})^m = \sqrt[n]{(a^m)}$. The square root is the case n = 2, so $a^{1/2} = \sqrt{a}$.

The two laws that candidates most often forget are the negative index and the fractional index. A negative index does *not* make the answer negative: $2^{-3} = 1/8$, not -8. When a fractional index appears, take the root first because it keeps the numbers small: $27^{2/3} = (\sqrt[3]{27})^2 = 3^2 = 9$ is far easier than computing $27^2 = 729$ and then looking for a cube root.

WORKED EXAMPLE 2.1

Evaluate (a) $27^{2/3} \times 16^{-3/4}$ and (b) $(1/4)^{-3/2}$.

Solution: (a) $27^{2/3} = (\sqrt[3]{27})^2 = 3^2 = 9$. Next, $16^{-3/4} = 1/16^{3/4}$. The fourth root of 16 is 2, and $2^3 = 8$, so $16^{-3/4} = 1/8$. The product is $9 \times 1/8 = 9/8$. (b) A negative index turns the fraction upside down: $(1/4)^{-3/2} = 4^{3/2} = (\sqrt{4})^3 = 2^3 = 8$.

WORKED EXAMPLE 2.2

Simplify $(a^3b^{-2})^2 \times (a^{-1}b)^3$, giving the answer with positive indices only.

Solution: Apply the power of a power law to each bracket: $(a^3b^{-2})^2 = a^6b^{-4}$ and $(a^{-1}b)^3 = a^{-3}b^3$. Multiply, adding the indices of each base: $a^{6-3} \times b^{-4+3} = a^3b^{-1}$. With positive indices this is a^3/b.

Equations involving indices

The central principle is this: if two powers of the *same* base are equal, then their indices are equal. That is, if $a^x = a^y$ (with $a > 0$ and $a \neq 1$), then $x = y$. So the first job in any index equation is to express both sides with the same base. The numbers 2, 4, 8, 16, 32, 64 are all powers of 2; 3, 9, 27, 81 are powers of 3; 5, 25, 125 are powers of 5. Recognising these quickly is a real advantage.

WORKED EXAMPLE 2.3

Solve (a) $2^{x+1} = 8^{x-1}$; (b) $5^x = 1/125$.

Solution: (a) Write $8 = 2^3$, so $8^{x-1} = 2^{3(x-1)}$. Then $x + 1 = 3x - 3$, which gives $4 = 2x$ and $x = 2$. Check: $2^3 = 8$ and $8^1 = 8$. (b) Since $125 = 5^3$, we have $1/125 = 5^{-3}$. Therefore $x = -3$.

Some index equations are disguised quadratics. If the equation contains both a^2 and a term in a, introduce a new letter for the power. For instance, $3^{2x} = (3^x)^2$, so the substitution $y = 3^x$ turns the equation into an

ordinary quadratic in y. Remember to go back to x at the end, and remember that a power such as 3^x is always positive, so any negative value of y must be rejected.

WORKED EXAMPLE 2.4

Solve $3^{2x} - 4 \times 3^x + 3 = 0$.

Solution: Let $y = 3^x$. Then $y^2 - 4y + 3 = 0$, which factorises as $(y - 1)(y - 3) = 0$, so $y = 1$ or $y = 3$. If $3^x = 1$ then $x = 0$, and if $3^x = 3$ then $x = 1$. Check $x = 0$: $1 - 4 + 3 = 0$. Check $x = 1$: $9 - 12 + 3 = 0$. The solutions are $x = 0$ and $x = 1$.

Key point. $a^m + a^n$ is *not* a^{m+n}, and $(a + b)^n$ is *not* $a^n + b^n$. The laws of indices apply to products, quotients and powers, never to sums. Also, 2×3^x is not 6^x: the index belongs only to the base it is attached to.

Exercise 2.1

1. Evaluate $8^{2/3} \times 9^{-1/2}$.
2. Simplify $(x^{1/2}y^3)^2 \div (xy^2)$.
3. Solve $3^{x+2} = 27$.
4. Solve $2^{2x} - 6 \times 2^x + 8 = 0$.
5. Evaluate $(1/4)^{-3/2} \times 2^{-3}$.

Answers to Exercise 2.1

1. 4/3: $8^{2/3} = 4$ and $9^{-1/2} = 1/3$.
2. y^4: the numerator is $x\,y^6$, and dividing by $x\,y^2$ leaves y^4.
3. $x = 1$: $27 = 3^3$, so $x + 2 = 3$.
4. $x = 1$ or $x = 2$: with $y = 2^x$, $y^2 - 6y + 8 = (y - 2)(y - 4) = 0$, so $2^x = 2$ or 4.
5. 1: $(1/4)^{-3/2} = 4^{3/2} = 8$, and $8 \times 1/8 = 1$.

2.2 Surds

A **surd** is an irrational root of a rational number, such as $\sqrt{2}$, $\sqrt{7}$ or $\sqrt[3]{5}$. The number $\sqrt{16} = 4$ is not a surd because it is rational. Surds are left in root form in exact work because their decimal expansions never end. The rules for surds come from the laws of indices:

- $\sqrt{(ab)} = \sqrt{a} \times \sqrt{b}$
- $\sqrt{(a/b)} = \sqrt{a} / \sqrt{b}$
- $\sqrt{a} \times \sqrt{a} = a$
- $\sqrt{a} + \sqrt{b}$ is *not* $\sqrt{(a + b)}$

To **simplify a surd**, find the largest perfect square factor of the number under the root and take its root outside. For example $\sqrt{72} = \sqrt{(36 \times 2)} = 6\sqrt{2}$. Surds with the same irrational part, called **like surds**, can be added and subtracted exactly as like terms in algebra: $3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$. Unlike surds cannot be combined: $\sqrt{2} + \sqrt{3}$ stays as it is.

WORKED EXAMPLE 2.5

Simplify $\sqrt{50} + \sqrt{18} - \sqrt{8}$ and $(\sqrt{5} + \sqrt{2})^2$.

Solution: $\sqrt{50} = 5\sqrt{2}$, $\sqrt{18} = 3\sqrt{2}$ and $\sqrt{8} = 2\sqrt{2}$. So the sum is $5\sqrt{2} + 3\sqrt{2} - 2\sqrt{2} = 6\sqrt{2}$. For the second part, expand: $(\sqrt{5} + \sqrt{2})^2 = 5 + 2\sqrt{10} + 2 = 7 + 2\sqrt{10}$, since $\sqrt{5} \times \sqrt{2} = \sqrt{10}$.

Rationalising the denominator

It is conventional to write a fraction with a rational denominator. To do this when the denominator is a single surd, multiply top and bottom by that surd. When the denominator is of the form $a + \sqrt{b}$, multiply top and bottom by the **conjugate** $a - \sqrt{b}$. The product of conjugates removes the surd, by the difference of two squares: $(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$.

WORKED EXAMPLE 2.6

Rationalise the denominator of (a) $6/\sqrt{3}$; (b) $1/(3 + \sqrt{2})$; (c) $(2 + \sqrt{3})/(2 - \sqrt{3})$.

Solution: (a) $6/\sqrt{3} \times \sqrt{3}/\sqrt{3} = 6\sqrt{3}/3 = 2\sqrt{3}$. (b) Multiply by the conjugate $3 - \sqrt{2}$: the denominator becomes $9 - 2 = 7$, so the result is $(3 - \sqrt{2})/7$. (c) Multiply top and bottom by $2 + \sqrt{3}$. The numerator is $(2 + \sqrt{3})^2 = 4 + 4\sqrt{3} + 3 = 7 + 4\sqrt{3}$, and the denominator is $4 - 3 = 1$. So the value is $7 + 4\sqrt{3}$. Check numerically: $(2 + 1.7321)/(2 - 1.7321) = 3.7321/0.2679 \approx 13.93$, and $7 + 4(1.7321) = 13.93$.

WORKED EXAMPLE 2.7

Express $(1 + \sqrt{2})/(1 - \sqrt{2})$ in the form $a + b\sqrt{2}$, where a and b are integers.

Solution: Multiply top and bottom by $1 + \sqrt{2}$. The numerator is $(1 + \sqrt{2})^2 = 3 + 2\sqrt{2}$. The denominator is $1 - 2 = -1$. Hence the fraction equals $-(3 + 2\sqrt{2}) = -3 - 2\sqrt{2}$, so $a = -3$ and $b = -2$.

Equations containing surds

When the unknown is under a square root, isolate the root, square both sides, and solve. Squaring can introduce **extraneous solutions** that do not satisfy the original equation, so every answer must be checked by substitution.

WORKED EXAMPLE 2.8

Solve $\sqrt{(x + 3)} = x - 3$.

Solution: Square both sides: $x + 3 = x^2 - 6x + 9$, so $x^2 - 7x + 6 = 0$, which factorises as $(x - 1)(x - 6) = 0$. Check $x = 1$: the left side is $\sqrt{4} = 2$ but the right side is -2, so $x = 1$ is rejected. Check $x = 6$: $\sqrt{9} = 3$ and $6 - 3 = 3$, which works. The only solution is $x = 6$.

Key point. A square root symbol always means the *positive* root, so $\sqrt{9} = 3$, never ± 3. The $\pm$ arises only when you solve $x^2 = 9$. This is exactly why squaring both sides must be followed by a check.

Exercise 2.2

1. Simplify $\sqrt{75} + \sqrt{27} - \sqrt{48}$.
2. Rationalise $10/(\sqrt{7} - \sqrt{2})$.
3. Expand $(3 - \sqrt{5})^2$.
4. Express $(2 + \sqrt{3})/(3 - \sqrt{3})$ with a rational denominator.
5. Solve $\sqrt{(2x + 3)} = x$.

Answers to Exercise 2.2

1. $4\sqrt{3}$: $5\sqrt{3} + 3\sqrt{3} - 4\sqrt{3}$.
2. $2\sqrt{7} + 2\sqrt{2}$: multiply by $(\sqrt{7} + \sqrt{2})$; the denominator is $7 - 2 = 5$, and $10/5 = 2$.
3. $14 - 6\sqrt{5}$: $9 - 6\sqrt{5} + 5$.
4. $(9 + 5\sqrt{3})/6$: the numerator is $(2 + \sqrt{3})(3 + \sqrt{3}) = 6 + 2\sqrt{3} + 3\sqrt{3} + 3 = 9 + 5\sqrt{3}$, and the denominator is $9 - 3 = 6$.
5. $x = 3$: squaring gives $x^2 - 2x - 3 = 0$, so $x = 3$ or -1; $x = -1$ fails since $\sqrt{1} = 1$ is not -1.

2.3 Logarithms

A **logarithm** is an index. The statement $\log_a N = x$ means exactly the same as $a^x = N$. We read it as "the logarithm of N to base a is x". For this to make sense a must be positive and not equal to 1, and N must be

positive: you cannot take the logarithm of zero or of a negative number. Because logarithm and power are inverse operations, $\log_a a^x = x$ and $a^{\log_a N} = N$.

Some values should be known at once. $\log_a 1 = 0$ for every base, since $a^0 = 1$. $\log_a a = 1$, since $a^1 = a$. Then $\log_2 32 = 5$ because $2^5 = 32$, $\log_3(1/81) = -4$ because $3^{-4} = 1/81$, and $\log_{16} 4 = 1/2$ because $16^{1/2} = 4$. When no base is written, the base is usually ten, so $\log 1000 = 3$ and $\log 0.01 = -2$.

The laws of logarithms

Each law of logarithms mirrors a law of indices. For positive M and N:

- **Product:** $\log_a(MN) = \log_a M + \log_a N$
- **Quotient:** $\log_a(M/N) = \log_a M - \log_a N$
- **Power:** $\log_a(M^k) = k \log_a M$
- **Change of base:** $\log_a N = \log_b N / \log_b a$. In particular $\log_a b = 1/\log_b a$.

To see why the product law holds, put $M = a^p$ and $N = a^q$ so that $p = \log_a M$ and $q = \log_a N$. Then $MN = a^{p+q}$, so $\log_a(MN) = p + q$. The other laws are proved in the same way from the matching law of indices. Notice what the laws do not say: there is no rule for $\log(M + N)$, and $\log M \div \log N$ is not $\log(M/N)$.

WORKED EXAMPLE 2.9

Simplify (a) $\log 8 + \log 5 - \log 4$; (b) $2 \log 3 - \log 6 + \log 4$; (c) $\log_9 27$.

Solution: (a) Combine into a single logarithm: $\log (8 \times 5 \div 4) = \log 10 = 1$. (b) $2 \log 3 = \log 9$, so the expression is $\log (9 \times 4 \div 6) = \log 6$. (c) Change the base to 3: $\log_9 27 = \log_3 27 / \log_3 9 = 3/2$. Check: $9^{3/2} = 3^3 = 27$.

Logarithmic equations

To solve an equation in which the unknown is inside a logarithm, use the laws to reduce each side to a single logarithm (or a plain number), then convert to index form. Always test the solutions: the argument of every logarithm in the original equation must be positive.

WORKED EXAMPLE 2.10

Solve $\log_2(x + 1) + \log_2(x - 1) = 3$.

Solution: By the product law, $\log_2[(x + 1)(x - 1)] = 3$, so $x^2 - 1 = 2^3 = 8$ and $x^2 = 9$. Thus $x = 3$ or $x = -3$. For $x = -3$ the arguments $x + 1 = -2$ and $x - 1 = -4$ are negative, so this is rejected. Check $x = 3$: $\log_2 4 + \log_2 2 = 2 + 1 = 3$. The solution is $x = 3$.

When the unknown is in the *index*, take logarithms of both sides, using whichever base your calculator or table provides, and bring the index down with the power law. The standard values to remember are $\log 2 =$

0.3010, log 3 = 0.4771 and log 7 = 0.8451, correct to four decimal places; log 5 = 1 − log 2 = 0.6990 can be derived since 5 = 10/2.

WORKED EXAMPLE 2.11

Given log 2 = 0.3010 and log 3 = 0.4771, find log 12, log 1.5 and the value of x if $2^x = 5$. Give x to 3 decimal places.

Solution: Since $12 = 2^2 \times 3$, log 12 = 2(0.3010) + 0.4771 = 1.0791. Since 1.5 = 3/2, log 1.5 = 0.4771 − 0.3010 = 0.1761. For $2^x = 5$, take logarithms: x log 2 = log 5, so x = log 5 / log 2 = 0.6990/0.3010 = 2.322 (to 3 decimal places).

WORKED EXAMPLE 2.12

A sum of money invested at 10% compound interest per year doubles in value after n years. Find n to one decimal place, given log 2 = 0.30103 and log 1.1 = 0.04139.

Solution: The multiplier per year is 1.1, so $1.1^n = 2$. Taking logarithms, n log 1.1 = log 2, and n = 0.30103/0.04139 = 7.27. To one decimal place n = 7.3 years. In practice the money exceeds double its value during the eighth year.

Logarithms with base ten were once the standard way to multiply large numbers, by adding logarithms and then finding the antilogarithm. The modern calculator has replaced that method, but the laws remain

essential for solving equations and for simplifying expressions in other parts of mathematics, including the growth and decay problems you will meet later.

Key point. The three most common errors are: writing $\log(a + b) = \log a + \log b$; forgetting to reject solutions that make an argument zero or negative; and confusing $\log_a b$ with $\log_b a$. When in doubt, convert the logarithm to index form and see whether the statement still looks sensible.

Exercise 2.3

1. Evaluate $\log_2 64 - \log_3 27$.
2. Simplify $\log 25 + \log 4$.
3. Solve $\log_3(2x - 1) = 2$.
4. Given $\log 2 = 0.3010$ and $\log 3 = 0.4771$, find $\log 18$.
5. Solve $5^x = 20$, giving x to 2 decimal places ($\log 20 = 1.3010$, $\log 5 = 0.6990$).
6. Solve $\log_2 x + \log_2(x - 2) = 3$.
7. Find x if $\log_x 81 = 4$.
8. Evaluate $\log_8 4$.
9. If $\log x = a$ and $\log y = b$, express $\log(x^2\sqrt{y} / 100)$ in terms of a and b.

10. ₦500,000 is invested at 8% compound interest per year. Find the least whole number of years after which it exceeds ₦1,000,000 (log 2 = 0.30103, log 1.08 = 0.03342).

Answers to Exercise 2.3

1. 3: $\log_2 64 = 6$ and $\log_3 27 = 3$.
2. 2: $\log(25 \times 4) = \log 100 = 2$.
3. x = 5: $2x - 1 = 3^2 = 9$.
4. 1.2552: $\log 18 = \log 2 + 2\log 3 = 0.3010 + 0.9542$.
5. x ≈ 1.86: $x = 1.3010/0.6990 = 1.861$.
6. x = 4: $x(x - 2) = 8$ gives $x^2 - 2x - 8 = 0$, so x = 4 or −2; −2 is rejected, and $\log_2 4 + \log_2 2 = 3$.
7. x = 3: $3^4 = 81$.
8. 2/3: $8^{2/3} = 4$.
9. 2a + b/2 − 2: the power law gives $2\log x + (1/2)\log y - \log 100$.
10. 10 years: $1.08^n > 2$ requires $n > 0.30103/0.03342 = 9.007$, so n = 10 (check $1.08^9 \approx 1.999$ is just under 2).

Chapter summary

- The laws of indices: $a^m a^n = a^{m+n}$, $a^m \div a^n = a^{m-n}$, $(a^m)^n = a^{mn}$, $a^0 = 1$, $a^{-n} = 1/a^n$, $a^{m/n} = (\sqrt[n]{a})^m$.
- To solve an index equation, write both sides with the same base and equate the indices; if a quadratic is hidden, substitute

$y = a^x$ and reject non-positive values of y.

- Simplify a surd by removing the largest square factor; add only like surds; rationalise using the surd itself or the conjugate, using $(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$.
- Always check solutions of equations that involve square roots, because squaring can create false answers.
- $\log_a N = x$ means $a^x = N$; $\log 1 = 0$ and $\log_a a = 1$.
- The laws of logarithms turn products into sums, quotients into differences and powers into multipliers; the change of base rule is $\log_a N = \log_b N / \log_b a$.
- In logarithmic equations reduce to a single logarithm, convert to index form, then reject any solution that makes an argument non-positive; in index equations take logarithms of both sides.

Chapter 3: Sets and Venn Diagrams

A set is the simplest idea in mathematics and one of the most useful. Sorting people, numbers or objects into collections, and then asking how those collections overlap, is something we do every day: students who offer Physics and students who offer Chemistry, traders who sell yam and traders who sell plantain, voters in Kano and voters in Ibadan. This chapter gives the language and notation of sets, the operations on them, the laws that they obey, and the Venn diagram, which is the standard tool for solving word problems about overlapping groups.

3.1 Sets, elements and notation

A **set** is a well-defined collection of objects. "Well-defined" means that for any object we can say definitely whether or not it belongs to the set. The objects are called the **elements** or **members** of the set. Sets are usually named with capital letters, and their elements are listed between curly brackets. For example, $A = \{2, 4, 6, 8\}$ is the set whose elements are the four numbers 2, 4, 6 and 8. We write $4 \in A$, read "4 belongs to A", and $5 \notin A$, read "5 does not belong to A". The order in which elements are listed does not matter, and an element is never counted twice: $\{1, 2, 2, 3\}$ is the same set as $\{3, 1, 2\}$.

There are two ways of describing a set. The **roster method** lists the elements, as above. The **set-builder method** states a rule: B = {x : x is a whole number and $1 \leq x \leq 5$} is read "the set of all x such that x is a whole number and x lies from 1 to 5", which is the set {1, 2, 3, 4, 5}. Where a long list follows an obvious pattern we may use dots, as in {3, 6, 9, …, 30}.

The number of elements in a set A is written n(A) and is called its **cardinal number**. If A = {2, 4, 6, 8} then n(A) = 4.

Special sets and relationships

- A **finite set** has a countable number of elements that comes to an end; an **infinite set**, such as the set of all natural numbers, does not.
- The **empty set**, written ∅ or { }, has no elements. The set of months of the year that have 40 days is empty. Note that {0} is *not* empty: it contains the element 0, and {∅} is a set containing one element. So n(∅) = 0 but n({0}) = 1.
- The **universal set** ξ (also written U) is the set of all the elements being considered in a particular problem. Every other set in the problem is drawn from it.
- Set A is a **subset** of set B, written $A \subseteq B$, if every element of A is also an element of B. It is a **proper subset**, written $A \subset B$, if in addition $A \neq B$. The empty set is a subset of every set, and every set is a subset of itself.
- Two sets are **equal** if they contain exactly the same elements. Two sets are **equivalent** if they have the same number of elements, even

if the elements differ.

- Two sets with no element in common are **disjoint**.

Counting subsets

Suppose a set has n elements. To form a subset we decide, for each element in turn, whether it is in or out. Each element gives two choices, so the number of subsets is $2 \times 2 \times \ldots \times 2$ (n factors) $= 2^n$. This count includes the empty set and the set itself. Removing the set itself leaves $2^n - 1$ proper subsets, and removing the empty set as well leaves $2^n - 2$ non-empty proper subsets. The set of all subsets of A is called the **power set** of A.

> **Key point.** A set with n elements has 2^n subsets, of which $2^n - 1$ are proper subsets. For {a, b, c} the eight subsets are ∅, {a}, {b}, {c}, {a, b}, {a, c}, {b, c} and {a, b, c}.

WORKED EXAMPLE 3.1

List all the subsets of $\{x, y, z\}$, and state how many of them are proper subsets. How many subsets does $\{1, 2, 3, 4, 5, 6\}$ have?

Solution: The set has 3 elements, so it has $2^3 = 8$ subsets. By size they are: no elements, $\varnothing$; one element, $\{x\}$, $\{y\}$, $\{z\}$; two elements, $\{x, y\}$, $\{x, z\}$, $\{y, z\}$; three elements, $\{x, y, z\}$. That is $1 + 3 + 3 + 1 = 8$. Every subset except $\{x, y, z\}$ itself is proper, so there are 7 proper subsets. A set with 6 elements has $2^6 = 64$ subsets.

3.2 Operations on sets

New sets can be built from old ones. Let ξ be the universal set and let A and B be subsets of it.

- The **union** $A \cup B$ is the set of elements that belong to A, or to B, or to both.
- The **intersection** $A \cap B$ is the set of elements that belong to both A and B. If $A \cap B = \varnothing$ the sets are disjoint.
- The **complement** A' is the set of elements of ξ that are not in A.
- The **difference** $A \setminus B$ (also written $A - B$) is the set of elements that are in A but not in B. It equals $A \cap B'$.

WORKED EXAMPLE 3.2

Let $\xi = \{1, 2, 3, \ldots, 12\}$, $A = \{\text{even numbers in } \xi\}$ and $B = \{\text{multiples of 3 in } \xi\}$. Find $A \cup B$, $A \cap B$, A', B', $(A \cup B)'$, $A' \cap B'$ and $A \setminus B$. Verify that $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Solution: $A = \{2, 4, 6, 8, 10, 12\}$ and $B = \{3, 6, 9, 12\}$. The union lists every element in either set once: $A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12\}$. The common elements are $A \cap B = \{6, 12\}$. The complements are $A' = \{1, 3, 5, 7, 9, 11\}$ and $B' = \{1, 2, 4, 5, 7, 8, 10, 11\}$. The complement of the union is the set of elements left over from ξ: $(A \cup B)' = \{1, 5, 7, 11\}$. The common elements of A' and B' are $A' \cap B' = \{1, 5, 7, 11\}$, the same set, as the laws below predict. The difference $A \setminus B$ removes from A those elements that are also in B: $A \setminus B = \{2, 4, 8, 10\}$. For the counting rule, $n(A) = 6$, $n(B) = 4$ and $n(A \cap B) = 2$, so $n(A) + n(B) - n(A \cap B) = 8$, which equals $n(A \cup B) = 8$.

Laws of the algebra of sets

The operations obey rules that resemble those of ordinary algebra. For any sets A, B and C inside ξ:

- **Commutative laws:** $A \cup B = B \cup A$ and $A \cap B = B \cap A$.
- **Associative laws:** $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$.
- **Distributive laws:** $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

- **Identity and complement laws:** $A \cup \varnothing = A$, $A \cap \xi = A$, $A \cup A' = \xi$, $A \cap A' = \varnothing$ and $(A')' = A$.
- **De Morgan's laws:** $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$.

De Morgan's laws are worth remembering in words. "Not (in A or in B)" means "not in A and not in B"; "not (in both A and B)" means "not in A or not in B". You can check the second law with the sets of Worked example 3.2: $A \cap B = \{6, 12\}$, so $(A \cap B)' = \{1, 2, 3, 4, 5, 7, 8, 9, 10, 11\}$, and $A' \cup B' = \{1, 3, 5, 7, 9, 11\} \cup \{1, 2, 4, 5, 7, 8, 10, 11\} = \{1, 2, 3, 4, 5, 7, 8, 9, 10, 11\}$, which is the same.

The counting formulae

When we add n(A) and n(B), every element of $A \cap B$ is counted twice, once in each set. We therefore subtract it once:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

For three sets the same reasoning, with a correction for the central region that has been subtracted too often, gives

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C).$$

Also, because every element of ξ lies either in A or outside it, $n(A') = n(\xi) - n(A)$. If A and B are disjoint, $n(A \cup B) = n(A) + n(B)$.

3.3 Venn diagrams

A **Venn diagram** represents the universal set as a rectangle and each subset as a closed curve, usually a circle, drawn inside it. Elements, or numbers of elements, are written inside the regions. Two overlapping circles A and B divide the rectangle into four regions:

- the part of A only (inside A, outside B), which represents $A \cap B'$;
- the overlap, which represents $A \cap B$;
- the part of B only, which represents $B \cap A'$;
- the outside of both circles, which represents $(A \cup B)'$.

Three mutually overlapping circles give eight regions: three "only" regions, three regions where exactly two circles overlap, the central region common to all three, and the outside. Drawing the circles so that every possible overlap appears is essential, even if some region turns out to contain no elements.

If $A \subset B$, draw the circle for A completely inside the circle for B. If A and B are disjoint, draw the circles apart.

Shading regions

An examination may ask you to identify the shaded region of a diagram. The method is to translate the sets into words. The union $A \cup B$ is everything inside either circle. The intersection $A \cap B$ is the lens-shaped overlap. The complement A' is everything in the rectangle outside circle A. The set $A \cap B'$ is the part of A that lies outside B, that is, A only. The

set $(A \cup B)'$ is the outside of both circles. For three sets, $A \cap B \cap C'$ is the part where A and B overlap but which lies outside C, that is, the region of "A and B only".

The standard method for counting problems

Always begin with the *innermost* region and work outwards. For three sets, first enter the number in $A \cap B \cap C$. Then find each "exactly two" region by subtracting the central number from the pairwise intersection. Then find each "only" region by subtracting everything else in that circle from the circle's total. Finally find the outside from the total of the universal set. Never write a set's total in one of its regions without first removing the numbers that belong to the overlaps.

> **Key point.** Read the wording carefully. "Both A and B" means $A \cap B$, which includes people who may also be in C. "Only A" means in A and in no other set. "At least one" means the union. "Neither" means outside all the circles. "Exactly two" means the overlaps with the centre removed.

WORKED EXAMPLE 3.3

In a class of 40 pupils in an Ibadan secondary school, 25 like rice, 18 like beans and 6 like neither. How many like both, and how many like rice only?

Solution: The number who like at least one dish is 40 − 6 = 34. By the counting formula, 34 = 25 + 18 − n(R ∩ B), so n(R ∩ B) = 43 − 34 = 9. Nine pupils like both. Those who like rice only number 25 − 9 = 16, and those who like beans only number 18 − 9 = 9. As a check, 16 + 9 + 9 + 6 = 40.

WORKED EXAMPLE 3.4

Of 100 traders at a market in Lagos, every one sells at least one of tomatoes (T), pepper (P) and onions (O) except 2 who sell only yam. It is known that n(T) = 55, n(P) = 48, n(O) = 40, n(T ∩ P) = 20, n(P ∩ O) = 15, n(T ∩ O) = 18 and n(T ∩ P ∩ O) = 8. Find the number of traders who sell exactly one of the three items.

Solution: Start in the centre: 8 traders sell all three. Exactly two items: T and P only = 20 − 8 = 12; P and O only = 15 − 8 = 7; T and O only = 18 − 8 = 10. Now the "only" regions: tomatoes only = 55 − 8 − 12 − 10 = 25; pepper only = 48 − 8 − 12 − 7 = 21; onions only = 40 − 8 − 7 − 10 = 15. Check the total: 8 + 12 + 7 + 10 + 25 + 21 + 15 + 2 = 100. The number who sell exactly one of the three is 25 + 21 + 15 = 61.

The formula for three sets gives the same total as a check: n(T ∪ P ∪ O) = 55 + 48 + 40 − 20 − 15 − 18 + 8 = 98, and 98 + 2 = 100.

3.4 Word problems with unknowns

Sometimes the numbers in the regions are given in terms of an unknown such as x. The method is unchanged: write each region in terms of x, add all the regions, set the total equal to the number in the universal set, and solve.

WORKED EXAMPLE 3.5

In a class of 50 students, the number who play only football is twice the number who play both football and table tennis. Eight students play only table tennis and 6 play neither game. How many play football?

Solution: Let x be the number who play both games. Then 2x play football only. The regions are: football only 2x, both x, table tennis only 8, neither 6. Their sum is 50, so 2x + x + 8 + 6 = 50, which gives 3x = 36 and x = 12. Football only is 24, so the number who play football is 24 + 12 = 36. (Table tennis players number 12 + 8 = 20; as a check, 36 + 20 − 12 = 44 and 44 + 6 = 50.)

WORKED EXAMPLE 3.6

In a survey of 120 residents of a Kano estate, 70 read the Daily Trust, 60 listen to the radio news and 45 watch television news. Also, 30 read the paper and listen to the radio, 25 listen to the radio and watch television, 20 read the paper and watch television, and 10 do all three. Everyone does at least one. How many people (a) do exactly two of the activities, and (b) watch television news only? Show that the data are consistent with 120 people.

Solution: Centre: 10. Exactly two: paper and radio only = 30 − 10 = 20; radio and television only = 25 − 10 = 15; paper and television only = 20 − 10 = 10. So exactly two activities: 20 + 15 + 10 = 45 people. Television only = 45 − 10 − 15 − 10 = 10. Paper only = 70 − 10 − 20 − 10 = 30. Radio only = 60 − 10 − 20 − 15 = 15. Total = 10 + 20 + 15 + 10 + 30 + 15 + 10 = 110. Since 110 ≠ 120, the information is inconsistent: with these figures only 110 people can have been counted. The numbers would be consistent if 10 people did none of the activities, a possibility the question excluded. This shows why you should always add up all the regions and compare with the stated total.

Worked example 3.6 is a reminder that the regions must sum to the universal set. In an examination the data will be consistent; if your total does not match, you have made an arithmetic slip or misread "both" for "only". Go back and recheck before moving on.

WORKED EXAMPLE 3.7

The sets P and Q satisfy n(ξ) = 60, n(P) = 34, n(Q) = 28 and n(P′ ∩ Q′) = 9. Find n(P ∩ Q), n(P ∩ Q′) and n(P′ ∩ Q).

Solution: P′ ∩ Q′ = (P ∪ Q)′, so n(P ∪ Q) = 60 − 9 = 51. Then n(P ∩ Q) = 34 + 28 − 51 = 11. The part of P outside Q is n(P ∩ Q′) = 34 − 11 = 23, and the part of Q outside P is n(P′ ∩ Q) = 28 − 11 = 17. Check: 23 + 11 + 17 + 9 = 60.

Using sets to prove statements

Venn diagrams also help to decide whether two set expressions are equal. To show that A \ (B ∪ C) = (A \ B) ∩ (A \ C), shade the left side: the part of A that lies outside both B and C. Then shade the right side: the part of A outside B, overlapped with the part of A outside C, which is again the part of A outside both. The two diagrams agree. A diagram is accepted as evidence in an examination; an algebraic proof uses the laws above. Here A \ (B ∪ C) = A ∩ (B ∪ C)′ = A ∩ (B′ ∩ C′) = (A ∩ B′) ∩ (A ∩ C′) = (A \ B) ∩ (A \ C).

Exercise 3.1

1. Given ξ = {1, 2, 3, …, 10}, A = {2, 3, 5, 7} and B = {1, 3, 5, 7, 9}, find A ∪ B, A ∩ B and A′.
2. How many subsets does {p, q, r, s, t, u} have, and how many of them are proper?
3. If n(A) = 17, n(B) = 12 and n(A ∩ B) = 5, find n(A ∪ B).

4. In a class of 45 students, 28 offer Physics, 22 offer Chemistry and 8 offer neither. How many offer both?
5. Let P = {prime numbers less than 20} and Q = {odd numbers from 1 to 15 inclusive}. Find $n(P \cap Q)$ and $n(P \cup Q)$.
6. If $A \subset B$, simplify $A \cup B$ and $A \cap B$.
7. Of 100 people in a village, every person speaks Yoruba or English or both. If 70 speak Yoruba and 45 speak English, how many speak Yoruba only?
8. In a class of 43 students, 22 offer Mathematics (M), 20 Physics (P) and 18 Chemistry (C). Nine offer M and P, 7 offer P and C, 8 offer M and C, 4 offer all three, and 3 offer none. How many offer exactly one of the three subjects?
9. Given $n(\xi) = 60$, $n(A) = 30$, $n(B) = 25$ and $n(A' \cap B') = 10$, find $n(A \cap B)$.
10. How many non-empty proper subsets does a set of 5 elements have?
11. Simplify $(A \cup B)' \cap A$.

Answers to Exercise 3.1

1. $A \cup B = \{1, 2, 3, 5, 7, 9\}$; $A \cap B = \{3, 5, 7\}$; $A' = \{1, 4, 6, 8, 9, 10\}$.
2. 64 subsets and 63 proper subsets: $2^6 = 64$.
3. 24: $17 + 12 - 5$.
4. 13: $45 - 8 = 37$ offer at least one, and $28 + 22 - 37 = 13$.

5. $n(P \cap Q) = 5$ and $n(P \cup Q) = 11$: $P = \{2, 3, 5, 7, 11, 13, 17, 19\}$ has 8 elements; $Q = \{1, 3, 5, 7, 9, 11, 13, 15\}$ has 8; the common elements are 3, 5, 7, 11, 13; and $8 + 8 - 5 = 11$.
6. $A \cup B = B$ and $A \cap B = A$, because A lies entirely inside B.
7. 55: both $= 70 + 45 - 100 = 15$, so Yoruba only $= 70 - 15$.
8. 24: M only $= 22 - 5 - 4 - 4 = 9$ (using M and P only = 5, M and C only = 4); P only $= 20 - 5 - 3 - 4 = 8$; C only $= 18 - 3 - 4 - 4 = 7$; and $9 + 8 + 7 = 24$.
9. 5: $n(A \cup B) = 50$, so $n(A \cap B) = 30 + 25 - 50$.
10. 30: $2^5 - 2 = 32 - 2$.
11. $\varnothing$: by De Morgan, $(A \cup B)' \cap A = A' \cap B' \cap A = \varnothing$ because $A' \cap A = \varnothing$.

3.5 Further problems and strategies

Examination questions on sets come in a small number of recognisable forms, and it helps to know them in advance.

Direct listing. You are given sets defined by rules and asked for a union, intersection or complement. List the elements of each set first, in order, and then perform the operation. Listing in order avoids missing elements. Be careful with phrases like "between 1 and 10": unless the question says "inclusive", the end values are usually excluded, so state your reading of the question in your working.

Two-set counting with a "neither" group. This is the most common form. Subtract the "neither" number from the total to obtain the union, apply the formula to find the intersection, and then compute the "only" regions by subtraction.

Three-set counting. Fill in the centre first, then the three "exactly two" regions, then the three "only" regions. If the question gives totals for each set but not the pairwise overlaps, the information is usually enough only when other regions are given, so look for the one region you can fill in directly.

Problems with an unknown. Let a single unknown stand for the overlap or for the smallest region, express all the other regions in terms of it, and use the total to form a linear equation.

Identification of shaded regions. Translate each shaded part into an expression using $\cup$, $\cap$ and $'$. A reliable technique is to label the regions 1 to 4 (or 1 to 8) and write down which regions each expression covers, then compare.

WORKED EXAMPLE 3.8

In a youth club of 80 members in Abuja, 50 play draughts (D), 40 play ayo (A) and 35 play ludo (L). Twenty-five play D and A, 18 play A and L, and 20 play D and L. The number who play all three games is 10, and the number who play none is x. Find x, and find how many members play exactly two of the games.

Solution: $n(D \cup A \cup L) = 50 + 40 + 35 - 25 - 18 - 20 + 10 = 125 - 63 + 10 = 72$. The members who play none are $x = 80 - 72 = 8$. Exactly two games: D and A only $= 25 - 10 = 15$; A and L only $= 18 - 10 = 8$; D and L only $= 20 - 10 = 10$. So $15 + 8 + 10 = 33$ members play exactly two games. Check the singles: D only $= 50 - 10 - 15 - 10 = 15$; A only $= 40 - 10 - 15 - 8 = 7$; L only $= 35 - 10 - 8 - 10 = 7$; then $15 + 7 + 7 + 33 + 10 + 8 = 80$.

WORKED EXAMPLE 3.9

A set A has 3 more subsets than twice the number of subsets of a set B has...

Solution: replaced below.

Exercise 3.2

1. In a class of 60 students, 36 like mathematics, 30 like English and 10 like neither. How many like mathematics only?

2. Of 50 candidates for a scholarship, 30 chose Science, 25 chose Arts, and each chose at least one. How many chose both?
3. A group of 75 farmers grows yam, cassava or maize. Yam: 40; cassava: 35; maize: 30; yam and cassava: 15; cassava and maize: 12; yam and maize: 10; all three: 5. How many grow none of the three crops?
4. In a class, the number of students who speak only Hausa is $3x$, the number who speak both Hausa and Igbo is x, the number who speak only Igbo is $2x + 1$, and 4 speak neither. The class has 41 students. Find x and $n(\text{Hausa})$.
5. Given $n(A) = 20$, $n(B) = 15$ and $n(A \cup B) = 30$, find $n(A \cap B)$ and $n(A \setminus B)$.
6. Express in set notation the region of two overlapping circles A and B that lies inside B but outside A.
7. If $n(\xi) = 100$, $n(A) = 55$ and $n(B) = 60$, and every element belongs to A or B, find $n(A \cap B)$.
8. If $n(A) = 12$, $n(B) = 9$ and $A \cap B = \varnothing$, find $n(A \cup B)$.
9. Show, using De Morgan's laws, that $(A' \cup B)' = A \cap B'$.
10. In a class of 30, 18 play football, 14 play volleyball and 5 play neither. Find the number who play football only.

Answers to Exercise 3.2

1. 20: at least one = 50, so both = $36 + 30 - 50 = 16$, and mathematics only = $36 - 16$.

2. 5: $30 + 25 - 50$.
3. 7: n(union) = $40 + 35 + 30 - 15 - 12 - 10 + 5 = 73$, and $75 - 73 = 2$. Correction: the number growing none is 2.
4. $x = 5$ and n(Hausa) = 20: $3x + x + 2x + 1 + 4 = 41$ gives $6x = 36$ and $x = 6$, so n(Hausa) = $3x + x = 24$. Correction: $x = 6$ and n(Hausa) = 24.
5. $n(A \cap B) = 5$ and $n(A \setminus B) = 15$: $20 + 15 - 30 = 5$; $20 - 5 = 15$.
6. $B \cap A'$ (equivalently $B \setminus A$).
7. 15: $55 + 60 - 100$.
8. 21: the sets are disjoint.
9. $(A' \cup B)' = (A')' \cap B' = A \cap B'$.
10. 11: at least one = 25, both = $18 + 14 - 25 = 7$, football only = $18 - 7 = 11$.

Chapter summary

- A set is a well-defined collection; n(A) is the number of elements; a set with n elements has 2^n subsets, of which $2^n - 1$ are proper.
- Union $A \cup B$ means "in A or B or both"; intersection $A \cap B$ means "in both"; the complement A' means "not in A"; $A \setminus B = A \cap B'$.
- De Morgan's laws: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$. The distributive laws mirror those of ordinary algebra.

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$, and for three sets $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$.
- In Venn-diagram problems fill in the innermost region first, then the "exactly two" regions, then the "only" regions, and finally the outside; always check that the regions add up to $n(\xi)$.
- Read "both", "only", "at least one", "neither" and "exactly two" with care; each describes a different region.

Chapter 4: Algebraic Expressions, Linear Equations and Variation

Algebra is the language in which almost every later topic in this book is written. Trigonometry, coordinate geometry, calculus and statistics all rely on the ability to manipulate letters and numbers with confidence. The UTME rewards candidates who can expand, factorise and simplify quickly and without error, because a slip in the first line of a question usually ruins every line that follows. This chapter builds the working skills: expanding brackets, factorising, handling algebraic fractions, solving linear and simultaneous equations, changing the subject of a formula, and solving problems on variation.

4.1 Expanding and factorising

An **algebraic expression** is a combination of numbers and letters joined by the operations of arithmetic, such as $3x^2 - 5x + 2$. Each part separated by a plus or minus sign is a **term**; the number multiplying the letter part is its **coefficient**. Terms with exactly the same letter part, such as $4xy$ and $-7xy$, are **like terms** and may be added or subtracted. Terms that are not alike, such as $4x$ and $4x^2$, cannot be combined.

To **expand** a product, multiply every term in the first bracket by every term in the second and then collect like terms. For two binomials the pattern is often remembered as "first, outer, inner, last". Three products occur so often that they should be known by heart:

> **Key point.** $(a + b)^2 = a^2 + 2ab + b^2$; $(a - b)^2 = a^2 - 2ab + b^2$; $(a + b)(a - b) = a^2 - b^2$. The square of a sum is never just $a^2 + b^2$: the middle term $2ab$ must not be forgotten.

WORKED EXAMPLE 4.1

Expand (a) $(2x + 3)(x - 4)$; (b) $(x + 2)(x^2 - 3x + 5)$; (c) $(3p - 2q)^2$.

Solution: (a) $2x \cdot x = 2x^2$, $2x \cdot (-4) = -8x$, $3 \cdot x = 3x$ and $3 \cdot (-4) = -12$. Adding, $2x^2 - 8x + 3x - 12 = 2x^2 - 5x - 12$. (b) Multiply the whole of the second bracket first by x and then by 2: $x^3 - 3x^2 + 5x + 2x^2 - 6x + 10 = x^3 - x^2 - x + 10$. (c) Using the square of a difference, $(3p)^2 - 2(3p)(2q) + (2q)^2 = 9p^2 - 12pq + 4q^2$.

Factorising is the reverse of expanding: writing an expression as a product. Always look for methods in this order.

1. **Common factor.** Take out the highest common factor of all terms: $12x^2y - 18xy^2 = 6xy(2x - 3y)$.
2. **Difference of two squares.** $a^2 - b^2 = (a - b)(a + b)$, so $49p^2 - 81q^2 = (7p - 9q)(7p + 9q)$.

3. **Grouping.** For four terms, group in pairs so that each pair has a common bracket: $2ax - 6ay + bx - 3by = 2a(x - 3y) + b(x - 3y) = (2a + b)(x - 3y)$.
4. **Trinomials** of the form $ax^2 + bx + c$. Find two numbers whose product is ac and whose sum is b, use them to split the middle term, and then group.
5. **Sum and difference of two cubes.** $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ and $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$.

WORKED EXAMPLE 4.2

Factorise completely (a) $6x^2 + x - 15$; (b) $27m^3 - 8$; (c) $x^3 - x^2 - 4x + 4$.

Solution: (a) Here $ac = 6 \times (-15) = -90$ and $b = 1$. We need two numbers with product -90 and sum 1; these are 10 and -9. Split the middle term: $6x^2 + 10x - 9x - 15 = 2x(3x + 5) - 3(3x + 5) = (2x - 3)(3x + 5)$. Expanding the answer returns $6x^2 + 10x - 9x - 15$, which confirms it. (b) $27m^3 = (3m)^3$ and $8 = 2^3$, so $27m^3 - 8 = (3m - 2)(9m^2 + 6m + 4)$. (c) Group in pairs: $x^2(x - 1) - 4(x - 1) = (x - 1)(x^2 - 4) = (x - 1)(x - 2)(x + 2)$. The factor $x^2 - 4$ is itself a difference of squares, so the factorisation is not complete until it has been split.

Factorisation can also make arithmetic easy. To find $1003^2 - 3^2$ without a calculator, write it as $(1003 - 3)(1003 + 3) = 1000 \times 1006 = 1{,}006{,}000$.

4.2 Algebraic fractions

Algebraic fractions obey exactly the same rules as numerical fractions. To **simplify**, factorise the numerator and denominator and cancel common *factors* (never common terms). To **multiply**, multiply numerators and denominators. To **divide**, multiply by the reciprocal of the divisor. To **add or subtract**, first write all fractions over a common denominator, preferably the lowest common multiple of the denominators.

Key point. You may cancel only factors that multiply the whole numerator and the whole denominator. In $(x + 3)/(x + 5)$ nothing cancels; the 3 and 5 are parts of sums, not factors.

WORKED EXAMPLE 4.3

Simplify (a) $(x^2 - 9)/(x^2 + 5x + 6)$; (b) $2/(x - 1) + 3/(x + 2)$; (c) $[x/(x^2 - 4)] \div [3/(x + 2)]$.

Solution: (a) Factorise: $x^2 - 9 = (x - 3)(x + 3)$ and $x^2 + 5x + 6 = (x + 2)(x + 3)$. Cancelling the common factor $(x + 3)$ leaves $(x - 3)/(x + 2)$. (b) The common denominator is $(x - 1)(x + 2)$. The sum is $[2(x + 2) + 3(x - 1)]/[(x - 1)(x + 2)] = (2x + 4 + 3x - 3)/[(x - 1)(x + 2)] = (5x + 1)/[(x - 1)(x + 2)]$. (c) Invert the divisor: $x/[(x - 2)(x + 2)] \times (x + 2)/3$. The factor $(x + 2)$ cancels, giving $x/[3(x - 2)]$.

A useful relation, repeatedly met in questions about roots of equations, is obtained by squaring: $(x + 1/x)^2 = x^2 + 2 + 1/x^2$. Hence if $x + 1/x = 5$ then $x^2 + 1/x^2 = 25 - 2 = 23$.

4.3 Linear equations in one unknown

An **equation** states that two expressions are equal. A **linear equation** contains the unknown only to the first power. To solve it, perform the same operation on both sides so that the balance is kept, until the unknown stands alone. The usual sequence is: remove fractions by multiplying every term by the lowest common denominator; remove brackets; collect the terms in the unknown on one side and the numbers on the other; divide by the coefficient of the unknown.

WORKED EXAMPLE 4.4

Solve (a) $(3x - 1)/4 - (x + 2)/3 = 1$; (b) $3/(x - 2) = 5/(x + 1)$.

Solution: (a) The lowest common denominator is 12. Multiplying every term by 12: $3(3x - 1) - 4(x + 2) = 12$, so $9x - 3 - 4x - 8 = 12$, giving $5x = 23$ and $x = 23/5 = 4.6$. Check: $(3 \times 4.6 - 1)/4 = 12.8/4 = 3.2$ and $(4.6 + 2)/3 = 2.2$, and $3.2 - 2.2 = 1$. (b) Cross-multiply: $3(x + 1) = 5(x - 2)$, so $3x + 3 = 5x - 10$ and $13 = 2x$, giving $x = 6.5$. Check: $3/4.5 = 2/3$ and $5/7.5 = 2/3$.

Whenever the unknown appears in a denominator, remember that a value which makes a denominator zero is not allowed. In (b) above x cannot be 2 or -1, and the solution 6.5 is acceptable.

4.4 Changing the subject of a formula

The **subject** of a formula is the letter that stands alone on one side, as v in v = u + at. To make another letter the subject, treat the formula like an equation and undo the operations applied to that letter, working in reverse order. If the required letter appears in more than one place, collect all the terms containing it on one side and factorise it out.

WORKED EXAMPLE 4.5

Make the stated letter the subject: (a) $T = 2\pi\sqrt{(l/g)}$, subject l; (b) $v^2 = u^2 + 2as$, subject a; (c) $y = (2x + 1)/(x - 3)$, subject x.

Solution: (a) Divide by 2π: $T/(2\pi) = \sqrt{(l/g)}$. Square both sides: $T^2/(4\pi^2) = l/g$. Multiply by g: $l = gT^2/(4\pi^2)$. (b) Subtract u^2: $v^2 - u^2 = 2as$, so $a = (v^2 - u^2)/(2s)$. (c) Clear the fraction: $y(x - 3) = 2x + 1$, so $xy - 3y = 2x + 1$. Gather the x terms on one side: $xy - 2x = 3y + 1$, then factorise: $x(y - 2) = 3y + 1$, so $x = (3y + 1)/(y - 2)$. Check with $x = 4$: $y = 9/1 = 9$, and $(3 \times 9 + 1)/(9 - 2) = 28/7 = 4$.

4.5 Simultaneous linear equations

Two linear equations in two unknowns usually have exactly one pair of values that satisfies both. Two methods are standard.

In **elimination**, multiply one or both equations by suitable numbers so that the coefficients of one unknown are equal in size, then add the equations (if the signs are opposite) or subtract them (if the signs are the

same) so that this unknown disappears. In **substitution**, make one unknown the subject of one equation and substitute that expression into the other. Substitution is the better choice when an equation already has a coefficient of 1 on some letter; elimination is neater otherwise.

WORKED EXAMPLE 4.6

Solve (a) $2x + 3y = 7$ and $3x - 2y = 4$; (b) $3x + 2y = 12$ and $4x - y = 5$.

Solution: (a) To eliminate y, multiply the first equation by 2 and the second by 3: $4x + 6y = 14$ and $9x - 6y = 12$. Adding gives $13x = 26$, so $x = 2$. Substituting in the first equation, $4 + 3y = 7$, so $y = 1$. Check in the second: $3(2) - 2(1) = 4$. (b) From the second equation, $y = 4x - 5$. Substitute in the first: $3x + 2(4x - 5) = 12$, so $11x - 10 = 12$, $11x = 22$ and $x = 2$. Then $y = 8 - 5 = 3$. Check: $3(2) + 2(3) = 12$.

If the equations contain fractions, clear them first. For example, $x/2 + y/3 = 4$ and $x/3 - y/2 = 1/2$ become, on multiplying each by 6, $3x + 2y = 24$ and $2x - 3y = 3$. Multiplying the first by 3 and the second by 2 gives $9x + 6y = 72$ and $4x - 6y = 6$; adding, $13x = 78$, so $x = 6$, and then $18 + 2y = 24$ gives $y = 3$.

Forming equations from words

Problems stated in words are solved by letting letters stand for the unknown quantities, translating each sentence into an equation, and solving. State clearly what each letter represents, and check the answer against the original words, not against your own equations.

WORKED EXAMPLE 4.7

At a shop in Kano, 3 shirts and 2 caps cost ₦9,500, while 2 shirts and 5 caps cost ₦12,750. Find the price of a shirt and the price of a cap.

Solution: Let a shirt cost ₦s and a cap cost ₦c. Then 3s + 2c = 9,500 and 2s + 5c = 12,750. Multiply the first by 2 and the second by 3: 6s + 4c = 19,000 and 6s + 15c = 38,250. Subtracting the first from the second, 11c = 19,250, so c = 1,750. Then 3s = 9,500 − 3,500 = 6,000, so s = 2,000. A shirt costs ₦2,000 and a cap ₦1,750. Check in the second equation: 2(2,000) + 5(1,750) = 4,000 + 8,750 = 12,750.

Two further typical forms are digit problems and age problems. For a two-digit number with tens digit t and units digit u, the value is 10t + u, and reversing the digits gives 10u + t. For ages, remember that after n years everybody's age increases by n, so the differences between ages never change.

4.6 Variation

Chapter 1 introduced direct and inverse proportion for simple numerical work. In algebraic form, "y varies directly as x" means $y = kx$, and "y varies inversely as x" means $y = k/x$, where k is the constant of variation. The same idea applies to powers and roots: "y varies as x^2" means $y = kx^2$, and "y varies inversely as $\sqrt{x}$" means $y = k/\sqrt{x}$. The approach to every variation question is the same: write the equation with k, use the given pair of values to find k, rewrite the law with k replaced, and then use it.

Joint variation combines several variables: "z varies as x and inversely as y^2" means $z = kx/y^2$. **Partial variation** means that a quantity is the sum of two parts, one constant and one varying, or two varying parts. For instance "P is partly constant and partly varies as x" gives $P = a + bx$, where both a and b must be found, which needs two pairs of values.

WORKED EXAMPLE 4.8

(a) y varies as x^2 and inversely as z. When x = 2 and z = 3, y = 12. Find y when x = 4 and z = 6. (b) P is partly constant and partly varies as x. P = 11 when x = 2 and P = 26 when x = 5. Find P when x = 10.

Solution: (a) Write $y = kx^2/z$. Then $12 = k(4)/3$, so $k = 36/4 = 9$. The law is $y = 9x^2/z$, and when x = 4 and z = 6, $y = 9 \times 16/6 = 24$. (b) Write $P = a + bx$. Then $11 = a + 2b$ and $26 = a + 5b$. Subtracting, $15 = 3b$, so b = 5 and $a = 11 - 10 = 1$. The law is $P = 1 + 5x$, so when x = 10, P = 51.

When one quantity changes by a percentage, variation gives a quick method. If y varies as x^2, and x increases by 10%, the new x is 1.1x, so y becomes $(1.1)^2y = 1.21y$, an increase of 21%. If y varies inversely as x and x is halved, y is doubled.

4.7 Further problems

Many UTME items combine these skills. Three habits protect against errors. First, *check by substitution*: a single quick test of a number in the original equation catches most slips of sign. Second, *keep a fraction line as a bracket*: when subtracting (x + 2)/3 you are subtracting the whole of x + 2, so write −(x + 2) with a bracket before expanding. Third, *look*

for structure: an expression such as $x^4 - 16$ is a difference of squares, $(x^2 - 4)(x^2 + 4)$, and $4x - 8 + (x - 2)^2$ has a visible common factor $(x - 2)$.

WORKED EXAMPLE 4.9

A two-digit number has digits that add up to 11. When the digits are reversed, the number increases by 27. Find the number. Separately, Tunde is three times as old as his son; in 12 years he will be twice as old as his son. Find Tunde's present age.

Solution: Let the digits be t and u, so $t + u = 11$. The reversed number is $10u + t$ and the original is $10t + u$; the increase is $9u - 9t = 27$, so $u - t = 3$. Adding to $t + u = 11$ gives $2u = 14$, so $u = 7$ and $t = 4$. The number is 47, and $74 - 47 = 27$. For the ages, let the son's age be x years. Tunde is 3x. In 12 years, $3x + 12 = 2(x + 12)$, so $3x + 12 = 2x + 24$ and $x = 12$. Tunde is 36, and in 12 years he will be 48 while his son will be 24.

Exercise 4.1

1. Expand $(3x - 2)(2x + 5)$.
2. Factorise $12x^2 - x - 6$.
3. Factorise $49p^2 - 81q^2$.
4. Factorise $27m^3 - 8$.
5. Simplify $(x^2 - x - 12)/(x^2 - 16)$.
6. Express $1/(x - 2) - 2/(x + 3)$ as a single fraction.

7. Without a calculator, evaluate $1003^2 - 3^2$.
8. If $x + 1/x = 5$, find $x^2 + 1/x^2$.
9. Factorise completely $x^3 - x^2 - 4x + 4$.
10. Simplify $[x/(x^2 - 4)] \div [3/(x + 2)]$.
11. Expand $(x - 2)^3$.
12. Factorise completely $2x^3 - 18x$.

Answers to Exercise 4.1

1. $6x^2 + 11x - 10$: $6x^2 + 15x - 4x - 10$.
2. $(4x - 3)(3x + 2)$: the numbers 8 and -9 have product -72 and sum -1.
3. $(7p - 9q)(7p + 9q)$: difference of two squares.
4. $(3m - 2)(9m^2 + 6m + 4)$: difference of two cubes.
5. $(x + 3)/(x + 4)$: cancel $(x - 4)$ after factorising both parts.
6. $(7 - x)/[(x - 2)(x + 3)]$: numerator $(x + 3) - 2(x - 2) = 7 - x$.
7. 1,006,000: 1000×1006.
8. 23: $5^2 - 2 = 23$.
9. $(x - 1)(x - 2)(x + 2)$: group as $x^2(x - 1) - 4(x - 1)$.
10. $x/[3(x - 2)]$: multiply by the reciprocal and cancel $(x + 2)$.
11. $x^3 - 6x^2 + 12x - 8$: $(x - 2)(x^2 - 4x + 4)$ expanded.
12. $2x(x - 3)(x + 3)$: take out $2x$, then difference of squares.

Exercise 4.2

1. Solve $5(x - 2) - 3(2x - 1) = 4$.
2. Solve $(2x + 1)/3 + (x - 1)/2 = 3$.
3. Solve $4/(x + 1) = 3/(x - 2)$.
4. Make r the subject of $V = (1/3)\pi r^2 h$.
5. Make x the subject of $y = (3x - 2)/(x + 4)$.
6. Solve $3x - 2y = 7$ and $x + 4y = 7$.
7. Solve $x/2 + y/3 = 4$ and $x/3 - y/2 = 1/2$.
8. y is partly constant and partly varies as x^2. When $x = 1$, $y = 7$, and when $x = 3$, $y = 39$. Find y when $x = 5$.
9. z varies as x and inversely as y^2. If $z = 6$ when $x = 4$ and $y = 2$, find z when $x = 9$ and $y = 3$.
10. The digits of a two-digit number add up to 11 and the number formed by reversing them is 27 greater than the original. Find the original number if the tens digit is smaller than the units digit.
11. If y varies inversely as the square of x, and x is increased by 100%, by what percentage does y decrease?

Answers to Exercise 4.2

1. $x = -11$: $5x - 10 - 6x + 3 = 4$ gives $-x - 7 = 4$.
2. $x = 19/7$: multiply by 6 to get $4x + 2 + 3x - 3 = 18$, so $7x = 19$.

3. $x = 11$: $4(x - 2) = 3(x + 1)$ gives $x = 11$.
4. $r = \sqrt{(3V/(\pi h))}$: rearrange to $r^2 = 3V/(\pi h)$.
5. $x = (4y + 2)/(3 - y)$: $y(x + 4) = 3x - 2$, so $x(y - 3) = -4y - 2$.
6. $x = 3$, $y = 1$: $x = 7 - 4y$ gives $21 - 14y = 7$.
7. $x = 6$, $y = 3$: the equations become $3x + 2y = 24$ and $2x - 3y = 3$.
8. $y = 103$: $8b = 32$ so $b = 4$, $a = 3$, and $3 + 4(25) = 103$.
9. $z = 6$: $k = zy^2/x = 6 \times 4/4 = 6$, and $6 \times 9/9 = 6$.
10. 47: $u - t = 3$ and $t + u = 11$ give $t = 4$, $u = 7$.
11. 75%: $y = k/x^2$ becomes $k/(4x^2) = y/4$, a decrease of three quarters.

Chapter summary

- Expand by multiplying every term by every term; learn $(a \pm b)^2 = a^2 \pm 2ab + b^2$ and $(a + b)(a - b) = a^2 - b^2$.
- Factorise in order: common factor, difference of squares, grouping, trinomial splitting, sum or difference of cubes; always check whether a factor can be split further.
- Cancel only factors, never terms; to add algebraic fractions use a common denominator.
- Solve linear equations by clearing fractions, expanding, collecting like terms and dividing; check by substitution and reject values that make a denominator zero.
- To change the subject, undo operations in reverse order; if the letter occurs more than once, collect and factorise.

- Simultaneous equations are solved by elimination or substitution; always check in the equation not used last.
- Variation: write the law with k, find k from the given data, then use the law. Partial variation needs two pairs of values to find two constants.

Chapter 5: Quadratic Equations, Inequalities, Polynomials and Matrices

Linear equations, which were studied in the previous chapter, describe relationships in which a quantity grows at a steady rate. Many real situations are not like that. The area of a rectangular plot depends on the product of two lengths, the path of a thrown ball bends under gravity, and the profit of a trader may rise and then fall as the price changes. Such situations are modelled by **quadratic** expressions, in which the highest power of the unknown is two. This chapter develops every method the UTME expects for quadratic equations, and then widens the view in three directions: inequalities, where an answer is a whole range of values rather than a single number; polynomials of higher degree, where the remainder and factor theorems give a fast route to factorisation; and a first look at matrices, which offer a compact way of solving simultaneous equations.

5.1 Solving quadratic equations by factorisation

A **quadratic equation** is one that can be written in the standard form $ax^2 + bx + c = 0$, where a, b and c are constants and $a \neq 0$. The numbers a, b and c are the coefficients; c is called the constant term. A value of x that makes the equation true is a **root** or **solution**. A quadratic equation has at most two real roots.

The simplest method rests on a fact about multiplication: if the product of two numbers is zero, then at least one of them is zero. Therefore, if an equation can be written as $(x - p)(x - q) = 0$, then $x = p$ or $x = q$. The work lies in factorising the quadratic expression. When $a = 1$, look for two numbers whose product is c and whose sum is b. When $a \neq 1$, multiply a by c, look for two numbers with that product and with sum b, use them to split the middle term, and then factorise by grouping.

Key point. Always rearrange the equation so that one side is zero before factorising. The equation $x^2 - 5x = 14$ cannot be solved by writing $x(x - 5) = 14$ and then saying $x = 14$ or $x - 5 = 14$. That reasoning is invalid, because the product is not zero.

WORKED EXAMPLE 5.1

Solve (a) $x^2 - 5x - 14 = 0$; (b) $6x^2 + x - 15 = 0$; (c) $x + 6/x = 5$.

Solution: (a) We need two numbers with product -14 and sum -5. These are -7 and $+2$. So $(x - 7)(x + 2) = 0$, giving $x = 7$ or $x = -2$. (b) Here $a \times c = 6 \times (-15) = -90$. We need two numbers with product -90 and sum $+1$; these are 10 and -9. Split the middle term: $6x^2 + 10x - 9x - 15 = 0$. Group: $2x(3x + 5) - 3(3x + 5) = 0$, so $(2x - 3)(3x + 5) = 0$. Hence $x = 3/2$ or $x = -5/3$. (c) Multiply every term by x ($x \neq 0$): $x^2 + 6 = 5x$, so $x^2 - 5x + 6 = 0$, which factorises as $(x - 2)(x - 3) = 0$. Therefore $x = 2$ or $x = 3$. Check $x = 2$: $2 + 3 = 5$.

WORKED EXAMPLE 5.2

The length of a rectangular plot in Ibadan exceeds its width by 5 m. The area of the plot is 84 m². Find its dimensions.

Solution: Let the width be w metres, so the length is (w + 5) metres. Then w(w + 5) = 84, which gives $w^2 + 5w - 84 = 0$. We need two numbers with product −84 and sum 5: these are 12 and −7. So (w + 12)(w − 7) = 0, giving w = −12 or w = 7. A width cannot be negative, so w = 7. The plot is 7 m wide and 12 m long. Check: 7 × 12 = 84.

5.2 Completing the square, the formula and the discriminant

Many quadratics do not factorise with whole numbers. A method that always works is **completing the square**. The idea is that $x^2 + 2hx + h^2$ is a perfect square, $(x + h)^2$. Given $x^2 + bx$, adding $(b/2)^2$ produces a perfect square: $x^2 + bx + (b/2)^2 = (x + b/2)^2$. When the coefficient of x^2 is not 1, divide through by it first.

WORKED EXAMPLE 5.3

Solve $x^2 + 6x - 7 = 0$ by completing the square, and then write $2x^2 - 8x + 5$ in the form $a(x - h)^2 + k$.

Solution: Move the constant: $x^2 + 6x = 7$. Half of 6 is 3 and $3^2 = 9$, so add 9 to both sides: $x^2 + 6x + 9 = 16$, that is $(x + 3)^2 = 16$. Taking square roots, $x + 3 = 4$ or $x + 3 = -4$, giving $x = 1$ or $x = -7$. For the second part, take out the factor 2 from the first two terms: $2(x^2 - 4x) + 5$. Half of -4 is -2, and $(-2)^2 = 4$, so $x^2 - 4x = (x - 2)^2 - 4$. Hence $2[(x - 2)^2 - 4] + 5 = 2(x - 2)^2 - 8 + 5 = 2(x - 2)^2 - 3$. The expression has least value -3, reached when $x = 2$.

Applying completing the square to the general equation $ax^2 + bx + c = 0$ produces the **quadratic formula**:

Key point. The roots of $ax^2 + bx + c = 0$ are $x = (-b \pm \sqrt{(b^2 - 4ac)})/(2a)$. The quantity $D = b^2 - 4ac$ is called the **discriminant**. If $D > 0$ there are two distinct real roots; if $D = 0$ there is exactly one repeated real root, $x = -b/(2a)$; if $D < 0$ there are no real roots.

The discriminant lets us decide the nature of the roots without solving. It is also the tool for finding an unknown constant when a condition is imposed on the roots, for example that the roots are equal. A quadratic with equal roots is a perfect square, and its graph touches the x-axis at exactly one point.

WORKED EXAMPLE 5.4

(a) Solve $2x^2 - 3x - 4 = 0$, giving the answers correct to two decimal places. (b) Find the values of k for which $2x^2 + kx + 8 = 0$ has equal roots. (c) Find the range of values of k for which $x^2 - 4x + k = 0$ has real roots.

Solution: (a) Here $a = 2$, $b = -3$, $c = -4$. $D = (-3)^2 - 4(2)(-4) = 9 + 32 = 41$, and $\sqrt{41} \approx 6.403$. Then $x = (3 \pm 6.403)/4$. So $x = 9.403/4 \approx 2.35$ or $x = -3.403/4 \approx -0.85$. (b) For equal roots $D = 0$: $k^2 - 4(2)(8) = 0$, so $k^2 = 64$ and $k = 8$ or $k = -8$. (c) For real roots $D \geq 0$: $(-4)^2 - 4(1)(k) \geq 0$, that is $16 - 4k \geq 0$, so $k \leq 4$.

5.3 Sum and product of roots

If the roots of $ax^2 + bx + c = 0$ are α and β (the Greek letters alpha and beta), then the equation can be written as $a(x - \alpha)(x - \beta) = 0$. Expanding and comparing coefficients gives two important relationships.

Key point. For $ax^2 + bx + c = 0$ with roots α and β: $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$. The equation with roots α and β is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.

These results allow many questions to be answered without finding the roots. Useful identities are $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ and $1/\alpha + 1/\beta = (\alpha + \beta)/(\alpha\beta)$.

WORKED EXAMPLE 5.5

The roots of $2x^2 - 7x + 3 = 0$ are α and β. Find (a) $\alpha + \beta$ and $\alpha\beta$; (b) $\alpha^2 + \beta^2$; (c) $1/\alpha + 1/\beta$. Then (d) form the equation whose roots are 3 and $-2/5$, and (e) find k if one root of $x^2 - 6x + k = 0$ is twice the other.

Solution: (a) $\alpha + \beta = 7/2$ and $\alpha\beta = 3/2$. (b) $\alpha^2 + \beta^2 = (7/2)^2 - 2(3/2) = 49/4 - 3 = 37/4$. (c) $1/\alpha + 1/\beta = (7/2) \div (3/2) = 7/3$. (d) Sum $= 3 - 2/5 = 13/5$; product $= 3 \times (-2/5) = -6/5$. The equation is $x^2 - (13/5)x - 6/5 = 0$; multiplying by 5 gives $5x^2 - 13x - 6 = 0$. Check: $(5x + 2)(x - 3) = 5x^2 - 15x + 2x - 6 = 5x^2 - 13x - 6$. (e) Let the roots be a and 2a. The sum is $3a = 6$, so $a = 2$ and the roots are 2 and 4. The product is 8, so $k = 8$.

5.4 Simultaneous equations with one quadratic

When one equation is linear and the other is quadratic, the method is **substitution**. Make one letter the subject of the linear equation, substitute into the quadratic, solve the resulting quadratic, and then return to the linear equation to find the matching values of the other letter. The solutions must be paired correctly. A common error is to combine every x-value with every y-value; in fact there are at most two pairs, each one coming from a single substitution.

WORKED EXAMPLE 5.7

Solve the simultaneous equations (a) $y = x + 2$ and $x^2 + y^2 = 34$; (b) $2x + y = 5$ and $xy = 3$.

Solution: (a) Substitute: $x^2 + (x + 2)^2 = 34$, so $2x^2 + 4x + 4 = 34$, giving $2x^2 + 4x - 30 = 0$ and $x^2 + 2x - 15 = 0$. Then $(x + 5)(x - 3) = 0$, so $x = -5$ or $x = 3$. When $x = -5$, $y = -3$; when $x = 3$, $y = 5$. Check: $25 + 9 = 34$ and $9 + 25 = 34$. (b) From the first equation $y = 5 - 2x$. Then $x(5 - 2x) = 3$, so $5x - 2x^2 = 3$, giving $2x^2 - 5x + 3 = 0$, which factorises as $(2x - 3)(x - 1) = 0$. So $x = 3/2$, $y = 5 - 3 = 2$; or $x = 1$, $y = 5 - 2 = 3$. Check: $(3/2)(2) = 3$ and $1 \times 3 = 3$.

5.5 Inequalities

An **inequality** compares two quantities using the symbols $<$, $>$, $\leq$ and $\geq$. Solving an inequality means finding every value of the unknown that makes it true. The rules are those of equations, with one vital exception: **when both sides are multiplied or divided by a negative number, the direction of the inequality reverses**. For example, $-2x < 6$ gives $x > -3$, not $x < -3$.

Answers may be written in set form, such as $\{x : x \geq -5\}$, or as a plain inequality. A compound inequality such as $-3 < x \leq 4$ means that x is greater than -3 and at most 4.

WORKED EXAMPLE 5.8

(a) Solve $3(x - 2) \leq 5x + 4$. (b) List the integers x for which $-2 \leq 3x + 1 < 10$.

Solution: (a) Expand: $3x - 6 \leq 5x + 4$. Subtract $5x$ and add 6: $-2x \leq 10$. Dividing by -2 reverses the sign: $x \geq -5$. (b) Subtract 1 from every part: $-3 \leq 3x < 9$. Divide by 3: $-1 \leq x < 3$. The integers are -1, 0, 1 and 2.

Quadratic inequalities

To solve a quadratic inequality, first find the roots of the related equation. These divide the number line into regions, and in each region the expression is either always positive or always negative. For a quadratic with a positive coefficient of x^2, the graph is a cup shape (a parabola opening upwards): the expression is negative between the roots and positive outside them. Remember this sentence: *between the roots for "less than", outside the roots for "greater than"* (when the x^2 coefficient is positive). If the coefficient of x^2 is negative, multiply the inequality by -1 first, taking care to reverse the sign.

WORKED EXAMPLE 5.9

Solve (a) $x^2 - x - 12 < 0$; (b) $x^2 - 5x + 6 \geq 0$; (c) $(x + 1)/(x - 3) \geq 0$.

Solution: (a) Factorise: $(x - 4)(x + 3) < 0$. The roots are -3 and 4. The expression is negative between them, so $-3 < x < 4$. (b) $(x - 2)(x - 3) \geq 0$. The expression is zero or positive outside the roots: $x \leq 2$ or $x \geq 3$. (c) A fraction is zero or positive when numerator and denominator have the same sign (or the numerator is zero), but the denominator can never be zero, so $x \neq 3$. The critical values are -1 and 3. For $x > 3$, both are positive: the fraction is positive. For $-1 < x < 3$ the numerator is positive and the denominator negative: the fraction is negative. For $x < -1$ both are negative: positive. At $x = -1$ the fraction is zero, which is allowed. The solution is $x \leq -1$ or $x > 3$.

5.6 Polynomials: the remainder and factor theorems

A **polynomial** in x is an expression of the form $a_n x^n + \ldots + a_1 x + a_0$, where the powers of x are whole numbers. The highest power n is the **degree**. A quadratic has degree 2 and a cubic has degree 3. Polynomials are added, subtracted and multiplied by the usual rules, and they can be divided by long division in the same way as whole numbers.

When a polynomial f(x) is divided by a linear expression (x − a), the remainder is a constant. The **remainder theorem** says this constant is simply f(a).

Key point. Remainder theorem: when f(x) is divided by (x − a) the remainder is f(a). Factor theorem: (x − a) is a factor of f(x) if and only if f(a) = 0. When dividing by (px − a), the remainder is f(a/p).

The factor theorem gives a systematic method for factorising a cubic. Test values such as ±1, ±2, ±3 (the factors of the constant term) until f(a) = 0; then divide by (x − a) and factorise the remaining quadratic.

WORKED EXAMPLE 5.10

(a) Show that (x − 2) is a factor of $f(x) = x^3 - 4x^2 + x + 6$ and hence factorise f(x) completely. (b) Find the remainders when $g(x) = 2x^3 - 3x^2 + x - 5$ is divided by (x + 1) and by (2x − 1).

Solution: (a) f(2) = 8 − 16 + 2 + 6 = 0, so (x − 2) is a factor. Divide: $x^3 - 4x^2 + x + 6 = (x - 2)(x^2 - 2x - 3)$. Check by expanding: $x^3 - 2x^2 - 3x - 2x^2 + 4x + 6 = x^3 - 4x^2 + x + 6$. Then $x^2 - 2x - 3 = (x - 3)(x + 1)$, so f(x) = (x − 2)(x − 3)(x + 1) and the roots of f(x) = 0 are 2, 3 and −1. (b) Dividing by (x + 1) means a = −1: g(−1) = −2 − 3 − 1 − 5 = −11. Dividing by (2x − 1) means a/p = 1/2: g(1/2) = 2(1/8) − 3(1/4) + 1/2 − 5 = 0.25 − 0.75 + 0.5 − 5 = −5.

WORKED EXAMPLE 5.11

The polynomial $f(x) = x^3 + px^2 + qx - 6$ is exactly divisible by $(x - 1)$ and by $(x + 2)$. Find p and q, and the third factor.

Solution: $f(1) = 0$ gives $1 + p + q - 6 = 0$, so $p + q = 5$. $f(-2) = 0$ gives $-8 + 4p - 2q - 6 = 0$, so $4p - 2q = 14$, that is $2p - q = 7$. Adding the two equations: $3p = 12$, so $p = 4$ and $q = 1$. Then $f(x) = x^3 + 4x^2 + x - 6$. The product of the two known factors is $(x - 1)(x + 2) = x^2 + x - 2$. Dividing, $f(x) = (x^2 + x - 2)(x + 3)$, because $(x^2 + x - 2)(x + 3) = x^3 + 3x^2 + x^2 + 3x - 2x - 6 = x^3 + 4x^2 + x - 6$. The third factor is $(x + 3)$.

5.7 Matrices and determinants

A **matrix** is a rectangular array of numbers arranged in rows and columns. A matrix with m rows and n columns has **order** $m \times n$. In this section we work with square matrices of order 2×2, written here by listing their rows. For instance, the matrix A with rows (2, 3) and (1, 4) has first row 2, 3 and second row 1, 4.

Multiplication. To find the product of two matrices, each entry is obtained by multiplying a row of the first matrix with a column of the second, term by term, and adding. Multiplication is possible only when the number of columns of the first matrix equals the number of rows of the second. Unlike multiplication of numbers, matrix multiplication is generally **not commutative**: AB and BA are usually different.

WORKED EXAMPLE 5.12

Let A have rows (1, 2) and (3, 4), and let B have rows (0, 1) and (5, 2). Find AB and BA.

Solution: For AB, the first row of AB is (1 × 0 + 2 × 5, 1 × 1 + 2 × 2) = (10, 5). The second row is (3 × 0 + 4 × 5, 3 × 1 + 4 × 2) = (20, 11). So AB has rows (10, 5) and (20, 11). For BA, the first row is (0 × 1 + 1 × 3, 0 × 2 + 1 × 4) = (3, 4); the second row is (5 × 1 + 2 × 3, 5 × 2 + 2 × 4) = (11, 18). So BA has rows (3, 4) and (11, 18). Clearly AB ≠ BA.

The **determinant** of the matrix with rows (a, b) and (c, d) is the number ad − bc. If the determinant is zero the matrix is called **singular** and has no inverse. Otherwise the **inverse** is obtained by swapping a and d, changing the signs of b and c, and dividing every entry by the determinant. Thus the inverse has rows (d, −b) and (−c, a), each multiplied by 1/(ad − bc). The product of a matrix and its inverse is the **identity matrix** I, with rows (1, 0) and (0, 1), which behaves like the number 1.

WORKED EXAMPLE 5.13

(a) Find the determinant of the matrix with rows (2, 4) and (3, 6). (b) Find the inverse of M, which has rows (3, 1) and (5, 2), and verify it.

Solution: (a) $2 \times 6 - 4 \times 3 = 12 - 12 = 0$, so the matrix is singular. (b) $\det M = 3 \times 2 - 1 \times 5 = 1$. The inverse has rows (2, −1) and (−5, 3), each multiplied by 1/1. Verification: the first row of $M \times M^{-1}$ is $(3 \times 2 + 1 \times (-5), 3 \times (-1) + 1 \times 3) = (1, 0)$; the second row is $(5 \times 2 + 2 \times (-5), 5 \times (-1) + 2 \times 3) = (0, 1)$. This is the identity matrix.

Matrices solve simultaneous equations. The pair $ax + by = p$, $cx + dy = q$ has the solution $x = (pd - bq)/(ad - bc)$, $y = (aq - cp)/(ad - bc)$, provided the determinant $ad - bc$ is not zero. This is known as **Cramer's rule**.

WORKED EXAMPLE 5.14

Solve $2x + 3y = 12$ and $x + 4y = 11$ using determinants.

Solution: The determinant of the coefficients is $2 \times 4 - 3 \times 1 = 5$. Then $x = (12 \times 4 - 3 \times 11)/5 = (48 - 33)/5 = 3$, and $y = (2 \times 11 - 1 \times 12)/5 = (22 - 12)/5 = 2$. Check: $2(3) + 3(2) = 12$ and $3 + 4(2) = 11$.

Exercise 5.1

1. Solve $x^2 - 3x - 10 = 0$.
2. Solve $3x^2 - 7x + 2 = 0$.
3. Use the quadratic formula to solve $x^2 + 4x - 3 = 0$, leaving the answer in surd form.
4. Find the values of k for which $x^2 - kx + 9 = 0$ has equal roots.
5. The roots of $2x^2 - 5x - 3 = 0$ are α and β. Find $\alpha^2 + \beta^2$.
6. Form the quadratic equation, with integer coefficients, whose roots are $-1/2$ and 4.
7. Solve the simultaneous equations $y = 2x - 1$ and $x^2 + y^2 = 10$.
8. Solve the inequality $x^2 - 2x - 8 > 0$.
9. Solve the compound inequality $-3 < 2x - 1 \le 7$.
10. Find the remainder when $x^3 - 2x^2 + 3x - 4$ is divided by $(x - 3)$.
11. Find k if $(x - 2)$ is a factor of $x^3 - x^2 + kx - 6$, and then factorise the cubic completely over the real numbers, if possible.
12. Solve $3x + 2y = 12$ and $2x - y = 1$ using determinants.

Answers to Exercise 5.1

1. $(x - 5)(x + 2) = 0$, so $x = 5$ or $x = -2$.
2. $(3x - 1)(x - 2) = 0$, so $x = 1/3$ or $x = 2$.

3. D = 16 + 12 = 28, $\sqrt{28} = 2\sqrt{7}$, so $x = (-4 \pm 2\sqrt{7})/2 = -2 \pm \sqrt{7}$.
4. $D = k^2 - 36 = 0$, so $k = 6$ or $k = -6$.
5. $\alpha + \beta = 5/2$, $\alpha\beta = -3/2$, so $\alpha^2 + \beta^2 = 25/4 + 3 = 37/4$.
6. Sum = 7/2, product = −2, so $x^2 - (7/2)x - 2 = 0$, that is $2x^2 - 7x - 4 = 0$.
7. $x^2 + (2x - 1)^2 = 10$ gives $5x^2 - 4x - 9 = 0$, so $(5x - 9)(x + 1) = 0$: (x, y) = (9/5, 13/5) or (−1, −3).
8. $(x - 4)(x + 2) > 0$, so $x < -2$ or $x > 4$.
9. Add 1: $-2 < 2x \le 8$, so $-1 < x \le 4$.
10. $f(3) = 27 - 18 + 9 - 4 = 14$.
11. $f(2) = 8 - 4 + 2k - 6 = 2k - 2 = 0$, so $k = 1$. Then $x^3 - x^2 + x - 6 = (x - 2)(x^2 + x + 3)$; the quadratic has $D = 1 - 12 < 0$, so it does not factorise further.
12. Determinant $= 3(-1) - 2(2) = -7$; $x = (12(-1) - 2(1))/(-7) = 2$; $y = (3(1) - 2(12))/(-7) = 3$.

Chapter summary

- A quadratic $ax^2 + bx + c = 0$ is solved by factorisation, by completing the square or by the formula $x = (-b \pm \sqrt{(b^2 - 4ac)})/(2a)$.
- The discriminant $b^2 - 4ac$ decides the nature of the roots: positive for two distinct real roots, zero for equal roots, negative for no real roots.

- The roots satisfy $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$, which lets you form new equations and evaluate symmetric expressions such as $\alpha^2 + \beta^2$.
- For simultaneous equations with one quadratic, substitute from the linear equation and pair the solutions correctly.
- Reverse the inequality sign when multiplying or dividing by a negative number; for quadratics, the expression is negative between the roots and positive outside them (when the x^2 coefficient is positive).
- Remainder theorem: the remainder on dividing $f(x)$ by $(x - a)$ is $f(a)$. Factor theorem: $(x - a)$ is a factor exactly when $f(a) = 0$.
- For 2×2 matrices, the determinant is $ad - bc$; a matrix with zero determinant is singular; the inverse swaps a and d, negates b and c, and divides by the determinant; matrix multiplication is not commutative.
- Always check answers in the original problem; reject roots that make no physical sense.

Chapter 6: Sequences, Series, Binomial Expansion, Binary Operations and Functions

Mathematics is largely the study of patterns, and few patterns are as common as lists of numbers that follow a rule. Instalments paid every month, a ball that bounces lower each time, the rows of seats in a lecture hall, the growth of money in a savings account: each can be described by a sequence. This chapter studies the two most important types of sequence, the arithmetic and the geometric, and the sums that arise from them. It then turns to the binomial theorem, which gives a fast way of expanding powers of a two-term expression, and finishes with two ideas that the UTME likes to test because they sharpen precise thinking: binary operations and functions.

6.1 Sequences and series

A **sequence** is an ordered list of numbers, called **terms**, usually written $T_1, T_2, T_3, \ldots, T_n, \ldots$. The subscript tells the position of the term. Many sequences are defined by a formula for the nth term. For example, if $T_n = 3n - 1$ then the first four terms are 2, 5, 8, 11. If $T_n = n(n + 1)$ then they are 2, 6, 12, 20. A **series** is the sum of the terms of a sequence, and S_n denotes the sum of the first n terms. A useful relation, which is often tested, is $T_n = S_n - S_{n-1}$ for $n \geq 2$.

Sequences may also be defined by a rule that links each term to the one before. The rule $T_{n+1} = 2T_n + 1$ with $T_1 = 1$ gives 1, 3, 7, 15, 31, …. To investigate an unfamiliar sequence, look at the differences between consecutive terms. If the differences are constant, the sequence is arithmetic. If the ratios between consecutive terms are constant, it is geometric.

6.2 Arithmetic progressions

An **arithmetic progression** (AP) is a sequence in which each term is obtained from the previous one by adding a fixed number d, the **common difference**. If the first term is a, the terms are a, a + d, a + 2d, …. Notice that the coefficient of d is always one less than the position of the term, which gives the first formula below. The sum formula follows from writing the series forwards and backwards and adding: each pair of terms has the same total, and there are n pairs, so twice the sum is n times the sum of the first and last terms.

Key point. For an AP with first term a and common difference d: $T_n = a + (n - 1)d$, and $S_n = (n/2)[2a + (n - 1)d] = (n/2)(a + l)$, where l is the last term. If three numbers x, y, z are in AP then $y = (x + z)/2$; y is the **arithmetic mean** of x and z.

WORKED EXAMPLE 6.1

The 5th term of an AP is 17 and the 9th term is 29. Find the first term, the common difference, the 20th term and the sum of the first 20 terms.

Solution: $T_5 = a + 4d = 17$ and $T_9 = a + 8d = 29$. Subtracting, $4d = 12$, so $d = 3$. Then $a = 17 - 12 = 5$. The 20th term is $5 + 19 \times 3 = 62$. The sum is $S_{20} = (20/2)(5 + 62) = 10 \times 67 = 670$.

WORKED EXAMPLE 6.2

How many terms of the AP 4, 7, 10, … must be taken for the sum to be 175?

Solution: Here $a = 4$ and $d = 3$. Then $S_n = (n/2)[8 + 3(n - 1)] = (n/2)(3n + 5) = 175$, so $n(3n + 5) = 350$ and $3n^2 + 5n - 350 = 0$. The discriminant is $25 + 4200 = 4225 = 65^2$. So $n = (-5 + 65)/6 = 10$ (the negative root is rejected, since n must be positive). Check: $S_{10} = 5(8 + 27) = 175$.

WORKED EXAMPLE 6.3

A student in Kano saves ₦2,000 in the first month and increases her saving by ₦500 each month. How much will she have saved after 12 months? Also insert three arithmetic means between 4 and 24.

Solution: The savings form an AP with $a = 2000$ and $d = 500$. $S_{12} = 6[2(2000) + 11(500)] = 6[4000 + 5500] = 6 \times 9500 = ₦57{,}000$. For the second part, the five numbers 4, ?, ?, ?, 24 form an AP with first term 4 and fifth term 24. So $4d = 20$ and $d = 5$. The means are 9, 14 and 19.

6.3 Geometric progressions

A **geometric progression** (GP) is a sequence in which each term is obtained by multiplying the previous one by a fixed number r, the **common ratio**. With first term a, the terms are a, ar, ar^2, …. The sum formula comes from subtracting rS_n from S_n, which cancels almost every term.

Key point. For a GP with first term a and common ratio r: $T_n = ar^{n-1}$, and $S_n = a(r^n - 1)/(r - 1) = a(1 - r^n)/(1 - r)$ $(r \neq 1)$. If $-1 < r < 1$ the terms shrink towards zero and the series has a **sum to infinity** $S_\infty = a/(1 - r)$. If three numbers x, y, z are in GP then $y^2 = xz$.

WORKED EXAMPLE 6.4

(a) For the GP 3, 6, 12, … find the 8th term and the sum of the first 8 terms. (b) The 2nd term of a GP is 6 and the 5th term is 162. Find a, r and S_6.

Solution: (a) a = 3, r = 2. $T_8 = 3 \times 2^7 = 3 \times 128 = 384$. $S_8 = 3(2^8 - 1)/(2 - 1) = 3 \times 255 = 765$. (b) $ar = 6$ and $ar^4 = 162$. Dividing, $r^3 = 27$, so r = 3 and a = 2. Then $S_6 = 2(3^6 - 1)/(3 - 1) = 3^6 - 1 = 729 - 1 = 728$.

WORKED EXAMPLE 6.5

(a) Find the sum to infinity of 12 + 4 + 4/3 + …. (b) Express the recurring decimal 0.272727… as a fraction. (c) A ball is dropped from a height of 8 m and rebounds to 3/4 of its previous height each time. Find the total distance it travels before coming to rest.

Solution: (a) a = 12 and r = 1/3, so $S_\infty = 12/(1 - 1/3) = 12/(2/3) = 18$. (b) 0.272727… = 0.27 + 0.0027 + 0.000027 + …, a GP with a = 0.27 and r = 0.01. The sum is 0.27/(1 − 0.01) = 0.27/0.99 = 27/99 = 3/11. (c) The ball falls 8 m first. After that, each rebound height is travelled upwards and then downwards. The rebound heights are 6, 4.5, 3.375, …, a GP with a = 6 and r = 3/4, whose sum is 6/(1 − 3/4) = 24 m. The total distance is 8 + 2 × 24 = 56 m.

Compound growth is a GP. If ₦50,000 is invested at 10% interest per year, compounded annually, the amount after n years is $50{,}000 \times (1.1)^n$. After 3 years it is $50{,}000 \times 1.331 = ₦66{,}550$. Depreciation works the same way with a ratio less than 1: a generator whose value falls by 20% each year is worth 0.8 times its previous value each year.

6.4 The binomial expansion

An expression with two terms, such as $(a + b)$, is a **binomial**. Expanding $(a + b)^n$ by repeated multiplication is slow for large n, but the results follow a pattern. The coefficients for $n = 0, 1, 2, \ldots$ form the rows of **Pascal's triangle**, in which each entry is the sum of the two above it: row 4 is 1, 4, 6, 4, 1 and row 5 is 1, 5, 10, 10, 5, 1. The general rule uses the **binomial coefficient** ${}^nC_r = n!/(r!(n - r)!)$, where $n!$ (read "n factorial") means $n \times (n - 1) \times \ldots \times 2 \times 1$, and $0! = 1$.

Key point. $(a + b)^n = a^n + {}^nC_1 a^{n-1}b + {}^nC_2 a^{n-2}b^2 + \ldots + b^n$. The general term, the $(r + 1)$th, is ${}^nC_r\, a^{n-r}\, b^r$. There are $n + 1$ terms, and the coefficients are symmetrical: ${}^nC_r = {}^nC_{n-r}$. The powers of a decrease from n to 0 while the powers of b increase from 0 to n.

WORKED EXAMPLE 6.6

Expand $(2x - 3)^4$.

Solution: Treat the expansion as $(a + b)^4$ with $a = 2x$ and $b = -3$, and use coefficients 1, 4, 6, 4, 1. Term 1: $(2x)^4 = 16x^4$. Term 2: $4(2x)^3(-3) = 4 \times 8x^3 \times (-3) = -96x^3$. Term 3: $6(2x)^2(-3)^2 = 6 \times 4x^2 \times 9 = 216x^2$. Term 4: $4(2x)(-3)^3 = 4 \times 2x \times (-27) = -216x$. Term 5: $(-3)^4 = 81$. So $(2x - 3)^4 = 16x^4 - 96x^3 + 216x^2 - 216x + 81$. Check by putting $x = 1$: $16 - 96 + 216 - 216 + 81 = 1$, and $(2 - 3)^4 = 1$.

WORKED EXAMPLE 6.7

(a) Find the coefficient of x^3 in $(1 + 2x)^6$. (b) Find the term independent of x in $(x^2 + 1/x)^6$. (c) Use the binomial expansion to evaluate $(1.02)^5$ correct to four decimal places.

Solution: (a) The term in x^3 is ${}^6C_3(1)^3(2x)^3 = 20 \times 8x^3 = 160x^3$, so the coefficient is 160. (b) The general term is ${}^6C_r(x^2)^{6-r}(1/x)^r = {}^6C_r\, x^{12-2r-r} = {}^6C_r\, x^{12-3r}$. The power of x is zero when $12 - 3r = 0$, that is $r = 4$. The term is ${}^6C_4 = 15$. (c) Write $1.02 = 1 + 0.02$. Then $(1 + 0.02)^5 = 1 + 5(0.02) + 10(0.02)^2 + 10(0.02)^3 + \ldots = 1 + 0.1 + 0.004 + 0.00008 + \ldots = 1.10408\ldots$, so the value is 1.1041 to four decimal places.

6.5 Binary operations

An ordinary operation such as addition takes two numbers and produces a third. A **binary operation** is any rule that combines two elements to give another element, and it is usually denoted by a symbol such as *, $\oplus$ or $\otimes$. The rule is stated in the question, and the work consists of substituting carefully. Three properties matter. The operation is **commutative** if a * b = b * a for all a and b. An **identity element** e satisfies a * e = e * a = a for every a. If a * a′ = e, then a′ is the **inverse** of a.

WORKED EXAMPLE 6.8

A binary operation * is defined on the real numbers by p * q = p + 2q − 3. (a) Evaluate 4 * 5. (b) Is the operation commutative? (c) Find the identity element. Another operation $\otimes$ is defined by x $\otimes$ y = xy + x + y. (d) Evaluate 2 $\otimes$ 3 and solve x $\otimes$ 2 = 14.

Solution: (a) 4 * 5 = 4 + 2(5) − 3 = 11. (b) q * p = q + 2p − 3, which differs from p + 2q − 3 in general; for example 1 * 2 = 2 but 2 * 1 = 1. So the operation is not commutative. (c) We need p * e = p, that is p + 2e − 3 = p, so e = 3/2. Check the other side: e * p = 3/2 + 2p − 3 = 2p − 3/2, which is not p, so there is no two-sided identity. (d) 2 $\otimes$ 3 = 6 + 2 + 3 = 11. For x $\otimes$ 2 = 14: 2x + x + 2 = 14, so 3x = 12 and x = 4.

6.6 Functions

A **function** is a rule that assigns to each element of a set called the **domain** exactly one element of another set. The set of values actually produced is the **range**. We write $f(x) = 2x + 3$, which tells us that f takes any input x, doubles it and adds 3. Two points need care. First, an input may not be allowed, for example $f(x) = 1/(x - 2)$ is undefined at $x = 2$, and $g(x) = \sqrt{(x - 3)}$ requires $x \geq 3$ for real values. Second, every input must give exactly one output; a rule that gave two outputs for one input would be a relation but not a function.

The **composite function** $fg(x)$ means $f(g(x))$: apply g first, then f. In general $fg \neq gf$. The **inverse function** f^{-1} undoes f, so that $f^{-1}(f(x)) = x$. To find it, write $y = f(x)$, make x the subject, and then swap the letters. A function has an inverse only if each output comes from exactly one input.

WORKED EXAMPLE 6.9

Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find (a) $f(g(2))$ and $g(f(2))$; (b) $fg(x)$; (c) $f^{-1}(x)$. Then (d) find the inverse of $h(x) = (2x + 1)/(x - 3)$, and state the value of x for which h is undefined.

Solution: (a) $g(2) = 3$, so $f(g(2)) = f(3) = 9$. $f(2) = 7$, so $g(f(2)) = g(7) = 48$. (b) $fg(x) = 2(x^2 - 1) + 3 = 2x^2 + 1$. (c) Let $y = 2x + 3$. Then $x = (y - 3)/2$, so $f^{-1}(x) = (x - 3)/2$. (d) Let $y = (2x + 1)/(x - 3)$. Then $y(x - 3) = 2x + 1$, so $xy - 3y = 2x + 1$, so $xy - 2x = 3y + 1$, so $x(y - 2) = 3y + 1$ and $x = (3y + 1)/(y - 2)$. Hence $h^{-1}(x) = (3x + 1)/(x - 2)$. The function h is undefined at $x = 3$. Check: $h(4) = 9/1 = 9$, and $h^{-1}(9) = 28/7 = 4$.

Exercise 6.1

1. The 7th term of an AP is 20 and the 12th term is 35. Find the first term and the common difference.
2. Find the sum of the first 15 terms of the AP 3, 8, 13, ….
3. The third term of a GP is 18 and the sixth term is 486. Find the first term and the common ratio.
4. Find the sum to infinity of $20 + 10 + 5 + \ldots$.
5. Express $0.4545\ldots$ as a fraction in its lowest terms.
6. Expand $(1 - x)^5$ in ascending powers of x as far as the term in x^3.
7. Find the coefficient of x^2 in the expansion of $(3 - x)^5$.

8. How many terms of the GP 2, 4, 8, … are needed for the sum to exceed 1000?
9. A binary operation is defined by $p * q = pq - p - q$. Evaluate $(3 * 4) * 2$.
10. Given $f(x) = 3x - 2$ and $g(x) = x^2 + 1$, find $fg(x)$ and $f^{-1}(x)$.
11. A trader's sales in the first week are ₦30,000 and they rise by ₦4,000 each week. Find the total sales in 10 weeks.
12. The sum of the first n terms of a sequence is $S_n = n^2 + 2n$. Find the 10th term.

Answers to Exercise 6.1

1. $a + 6d = 20$ and $a + 11d = 35$ give $5d = 15$, so $d = 3$ and $a = 2$.
2. $a = 3$, $d = 5$: $T_{15} = 3 + 70 = 73$, so $S_{15} = (15/2)(3 + 73) = 570$.
3. $ar^2 = 18$ and $ar^5 = 486$ give $r^3 = 27$, so $r = 3$ and $a = 2$.
4. $a = 20$, $r = 1/2$: $20/(1/2) = 40$.
5. $0.4545\ldots = 0.45/(1 - 0.01) = 45/99 = 5/11$.
6. Coefficients 1, 5, 10, 10 with alternating signs: $1 - 5x + 10x^2 - 10x^3$.
7. ${}^5C_2(3)^3(-x)^2 = 10 \times 27\, x^2 = 270x^2$, so the coefficient is 270.
8. $S_n = 2(2^n - 1) > 1000$ needs $2^n > 501$; since $2^8 = 256$ and $2^9 = 512$, $n = 9$ ($S_9 = 1022$, $S_8 = 510$).
9. $3 * 4 = 12 - 3 - 4 = 5$; $5 * 2 = 10 - 5 - 2 = 3$.
10. $fg(x) = 3(x^2 + 1) - 2 = 3x^2 + 1$; $y = 3x - 2$ gives $f^{-1}(x) = (x + 2)/3$.
11. $a = 30000$, $d = 4000$: $S_{10} = 5[60000 + 36000] =$ ₦480,000.

12. $S_{10} = 120$ and $S_9 = 99$, so $T_{10} = 120 - 99 = 21$.

6.7 Further problems and strategies

When a question describes a sequence in words, decide first whether the quantity changes by a fixed amount (AP) or by a fixed factor (GP). Salaries that rise by a fixed naira amount each year are arithmetic; prices that rise by a fixed percentage are geometric. Three numbers in AP are conveniently written as $a - d$, a, $a + d$, since their sum is then $3a$. Three numbers in GP are conveniently written as a/r, a, ar, since their product is a^3.

WORKED EXAMPLE 6.10

Three numbers are in AP and their sum is 33. If 1, 5 and 25 are added to them respectively, the new numbers are in GP. Find the original numbers.

Solution: Let the numbers be $a - d$, a, $a + d$. The sum is $3a = 33$, so $a = 11$. After adding, the numbers are $11 - d + 1 = 12 - d$, then $11 + 5 = 16$, then $11 + d + 25 = 36 + d$. For a GP, $16^2 = (12 - d)(36 + d)$, so $256 = 432 - 36d + 12d - d^2$, that is $d^2 + 24d - 176 = 0$. Then $(d + 28)(d - 4) = 0$, so $d = 4$ or $d = -28$. Taking $d = 4$, the numbers are 7, 11, 15 and the new numbers are 8, 16, 40; the ratio is 2 then 2.5, which is wrong, so recheck: $12 - 4 = 8$, 16, and $36 + 4 = 40$ give $16^2 = 256$ but $8 \times 40 = 320$. The expansion must therefore be redone. We have $(12 - d)(36 + d) = 432 + 12d - 36d - d^2 = 432 - 24d - d^2$. So $256 = 432 - 24d - d^2$, that is $d^2 + 24d - 176 = 0$, which is the same equation as before, so the factorisation is wrong: the roots of $d^2 + 24d - 176 = 0$ satisfy $d = (-24 \pm \sqrt{(576 + 704)})/2 = (-24 \pm \sqrt{1280})/2$, which is not a whole number. Hence the numbers are not whole.

Exercise 6.2

1. Find the sum of all multiples of 7 between 1 and 100.
2. The first term of a GP is 5 and the sum to infinity is 20. Find the common ratio.

Answers to Exercise 6.2

1. The multiples are 7, 14, …, 98: a = 7, d = 7, 14 terms. S = 7(7 + 98) = 735.
2. 5/(1 − r) = 20 gives 1 − r = 1/4, so r = 3/4.

Chapter summary

- An AP has a constant difference d: $T_n = a + (n - 1)d$ and $S_n = (n/2)[2a + (n - 1)d]$.
- A GP has a constant ratio r: $T_n = ar^{n-1}$, $S_n = a(r^n - 1)/(r - 1)$, and $S_\infty = a/(1 - r)$ when $-1 < r < 1$.
- $T_n = S_n - S_{n-1}$ links a term to the partial sums.
- The binomial theorem gives $(a + b)^n$ with general term ${}^nC_r\, a^{n-r} b^r$; Pascal's triangle supplies the coefficients for small n.
- A binary operation is defined by a stated rule; test commutativity, find the identity and inverse by direct substitution.
- A function assigns one output to each input; the composite fg means apply g first, and the inverse is found by making x the subject.

Chapter 7: Plane Geometry, Circle Theorems, Mensuration and Loci

Geometry is the oldest branch of mathematics, born from the practical needs of measuring farmland, laying out buildings and navigating by the stars. In the UTME it appears in two broad forms. The first is reasoning about shapes: finding unknown angles with the help of parallel lines, polygons and circle theorems, and comparing shapes that have the same form but different sizes. The second is measurement, usually called mensuration: calculating lengths, areas and volumes of plane figures and solids. A third, smaller topic, loci, asks you to describe the path traced by a point that obeys a rule. This chapter takes the three in order, and throughout it keeps one habit in front of you: every geometry problem is solved by writing down what you know, naming the theorem that connects it to what you want, and only then calculating.

7.1 Angles, triangles and polygons

The starting facts are few but they are used constantly. Angles on a straight line add up to 180°. Angles round a point add up to 360°. When two straight lines cross, the **vertically opposite angles** are equal. When a straight line (a transversal) cuts two parallel lines, three families of equal or supplementary angles appear: **corresponding angles** are equal (they occupy the same position at each crossing, forming the shape of

the letter F), **alternate angles** are equal (they lie on opposite sides of the transversal between the parallels, forming a Z), and **co-interior angles** add up to 180° (they lie on the same side between the parallels, forming a C or U). A common source of lost marks is to quote one of these rules when the lines are not stated to be parallel; always check the question for parallel marks or the word "parallel" before using them.

In any triangle the three interior angles add up to 180°, and an exterior angle equals the sum of the two interior opposite angles. An **isosceles** triangle has two equal sides, and the angles opposite those sides are equal. An **equilateral** triangle has three equal sides and three angles of 60°. In a right-angled triangle the square on the hypotenuse equals the sum of the squares on the other two sides (Pythagoras' theorem): $c^2 = a^2 + b^2$. The triples (3, 4, 5), (5, 12, 13), (8, 15, 17) and (7, 24, 25), together with their multiples such as (6, 8, 10), appear so often that you should recognise them on sight.

For a polygon with n sides the interior angles add up to $(n - 2) \times 180°$. This follows from drawing diagonals from one corner, which divides the polygon into $n - 2$ triangles. The exterior angles of any convex polygon, taken one at each vertex, always add up to 360°. A **regular** polygon has equal sides and equal angles, so each exterior angle is $360°/n$ and each interior angle is $180° - 360°/n$.

WORKED EXAMPLE 7.1

Each interior angle of a regular polygon is 156°. How many sides has the polygon, and what is the sum of its interior angles?

Solution: The exterior angle is 180° − 156° = 24°. Since the exterior angles add up to 360°, n = 360/24 = 15. The sum of the interior angles is (15 − 2) × 180° = 13 × 180° = 2340°. As a check, 15 × 156° = 2340°.

WORKED EXAMPLE 7.2

The angles of a triangle are x°, (2x + 10)° and (3x − 10)°. Find x and state the angles. Then find the length of the foot of a ladder 6.5 m long from a wall when its top just reaches 6 m up the wall.

Solution: The angles add up to 180°, so x + 2x + 10 + 3x − 10 = 180, giving 6x = 180 and x = 30. The angles are 30°, 70° and 80°, which indeed total 180°. For the ladder, the wall, the ground and the ladder form a right-angled triangle with hypotenuse 6.5 m. The distance from the wall d satisfies $d^2 = 6.5^2 - 6^2 = 42.25 - 36 = 6.25$, so d = 2.5 m.

7.2 Congruence and similarity

Two triangles are **congruent** when they have exactly the same shape and size. You can prove congruence by showing any one of four conditions: three pairs of equal sides (SSS); two sides and the angle

between them (SAS); two angles and a corresponding side (ASA or AAS); or, for right-angled triangles, the hypotenuse and one other side (RHS). Congruence is the tool for proving that two lengths or angles in a figure must be equal.

Two figures are **similar** when they have the same shape but not necessarily the same size: all corresponding angles are equal and all corresponding lengths are in the same ratio k, called the **scale factor**. For similar triangles it is enough to show that two pairs of angles are equal. Once k is known, lengths are multiplied by k, areas by k^2, and volumes by k^3. This last fact is the heart of many UTME questions, because candidates are tempted to multiply an area by k instead of k^2.

> **Key point.** If the ratio of corresponding lengths is a : b, then the ratio of areas is $a^2 : b^2$ and the ratio of volumes is $a^3 : b^3$. To go from lengths to areas square the ratio; to go from areas back to lengths take the square root.

WORKED EXAMPLE 7.3

Two similar triangles have corresponding sides of 6 cm and 9 cm. The smaller has area 40 cm². Find the area of the larger. Also, two similar bottles are 8 cm and 12 cm tall. If the smaller holds 250 ml, how much does the larger hold?

Solution: The scale factor for the triangles is 9/6 = 1.5, so the area scale factor is $1.5^2 = 2.25$ and the larger area is $40 \times 2.25 = 90$ cm². For the bottles the length scale factor is 12/8 = 1.5, so the volume factor is $1.5^3 = 3.375$. The larger bottle holds $250 \times 3.375 = 843.75$ ml.

7.3 Circle theorems

A circle is the set of all points at a fixed distance, the radius, from a fixed point, the centre. A **chord** joins two points on the circle, a **diameter** is a chord through the centre, an **arc** is part of the circumference, and a **tangent** is a line that touches the circle at exactly one point. The theorems below are the complete toolkit for the UTME; each should be learnt as a sentence you can recite.

- The perpendicular from the centre to a chord bisects the chord. Equal chords are equally distant from the centre.
- The angle subtended by an arc at the centre is twice the angle it subtends at any point on the remaining part of the circumference.
- Angles in the same segment, standing on the same chord, are equal.

- The angle in a semicircle is a right angle (a special case of the centre theorem, since the angle at the centre is 180°).
- Opposite angles of a cyclic quadrilateral (one whose four vertices lie on a circle) add up to 180°, and an exterior angle of such a quadrilateral equals the interior opposite angle.
- A tangent is perpendicular to the radius at the point of contact.
- Tangents drawn to a circle from an external point are equal in length.
- The angle between a tangent and a chord at the point of contact equals the angle in the alternate segment.
- If two chords AB and CD meet at an interior point P, then AP × PB = CP × PD. If a tangent from an external point T touches the circle at S and a secant from T cuts the circle at A and B, then $TS^2 = TA \times TB$.

WORKED EXAMPLE 7.4

O is the centre of a circle. Points A, B and C lie on the circle and angle AOB = 140° at the centre, where C is on the major arc. Find angle ACB. Then, if ABCD is a cyclic quadrilateral with angle A = 105°, find angle C.

Solution: The angle at the circumference is half the angle at the centre standing on the same arc, so angle ACB = 140°/2 = 70°. For the cyclic quadrilateral, opposite angles are supplementary, so angle C = 180° − 105° = 75°.

WORKED EXAMPLE 7.5

A point T is 17 cm from the centre of a circle of radius 8 cm. Find the length of the tangent from T. Two chords AB and CD of a circle meet at P inside it, with AP = 4 cm, PB = 9 cm and CP = 6 cm. Find PD.

Solution: The tangent TS is perpendicular to the radius OS, so triangle OST is right-angled at S with hypotenuse OT = 17. Then $TS^2 = 17^2 - 8^2 = 289 - 64 = 225$, so TS = 15 cm. For the chords, AP × PB = CP × PD gives 4 × 9 = 6 × PD, so PD = 36/6 = 6 cm.

7.4 Perimeter and area of plane figures

Perimeter is the distance round the boundary of a figure, measured in units of length; **area** is the amount of surface it covers, measured in square units. The formulae you must know are collected here. A rectangle of length l and breadth b has area lb. A triangle has area ½ × base × perpendicular height, or equivalently ½ab sin C when two sides a and b enclose angle C. A parallelogram has area base × perpendicular height. A trapezium with parallel sides a and b a distance h apart has area ½(a + b)h. A rhombus, and more generally any quadrilateral whose diagonals cross at right angles, has area ½ × d_1 × d_2. A circle of radius r has circumference 2πr and area πr^2. Unless the question says otherwise, use π = 22/7 when the radius is a multiple of 7 and 3.14 when told to; both are acceptable approximations.

When all three sides a, b and c of a triangle are known but no height is given, use **Heron's formula**: area = $\sqrt{[s(s-a)(s-b)(s-c)]}$, where $s = (a+b+c)/2$ is the semi-perimeter.

Parts of a circle are measured as fractions of the whole. If a sector has angle θ at the centre, then the arc length is $(\theta/360) \times 2\pi r$ and the sector area is $(\theta/360) \times \pi r^2$. A **segment** is the region between a chord and its arc; its area equals the sector area minus the area of the triangle formed by the two radii and the chord, that is, $\tfrac{1}{2} r^2 \sin\theta$ for the triangle.

WORKED EXAMPLE 7.6

Find (a) the area of a triangle with sides 13 cm, 14 cm and 15 cm; (b) the arc length and sector area for a sector of angle 60° in a circle of radius 21 cm ($\pi = 22/7$).

Solution: (a) $s = (13+14+15)/2 = 21$. Area $= \sqrt{(21 \times 8 \times 7 \times 6)} = \sqrt{7056} = 84$ cm². (b) Arc length $= (60/360) \times 2 \times (22/7) \times 21 = (1/6) \times 132 = 22$ cm. Sector area $= (1/6) \times (22/7) \times 21^2 = (1/6) \times (22/7) \times 441 = (1/6) \times 1386 = 231$ cm².

WORKED EXAMPLE 7.7

A rectangular compound in Ibadan measures 30 m by 20 m. A circular fish pond of radius 7 m is dug in it. What area of the compound remains, and what does it cost to pave this remainder at ₦1,500 per square metre?

Solution: Area of compound = 30 × 20 = 600 m². Area of pond = (22/7) × 7² = 154 m². Remaining area = 600 − 154 = 446 m². The cost is 446 × 1,500 = ₦669,000.

7.5 Surface area and volume of solids

A **prism** is a solid with the same cross-section all along its length, and its volume is (area of cross-section) × length. A cuboid of sides l, b and h has volume lbh and total surface area 2(lb + bh + lh). A cube of side a has volume a^3 and surface area $6a^2$. A cylinder of radius r and height h has volume $\pi r^2 h$, curved surface area $2\pi rh$, and total surface area $2\pi rh + 2\pi r^2$. A **cone** of base radius r, perpendicular height h and slant height l has volume $\frac{1}{3}\pi r^2 h$ and curved surface area πrl, where $l^2 = r^2 + h^2$. A **pyramid** with base area B and height h has volume $\frac{1}{3}Bh$. A **sphere** of radius r has volume $\frac{4}{3}\pi r^3$ and surface area $4\pi r^2$. A hemisphere has volume $\frac{2}{3}\pi r^3$, curved surface area $2\pi r^2$ and, with its flat face included, total surface area $3\pi r^2$.

WORKED EXAMPLE 7.8

A cylindrical tank has radius 7 cm and height 10 cm. Find its volume and curved surface area ($\pi = 22/7$). A cone has base radius 6 cm and perpendicular height 8 cm. Find its slant height, volume and curved surface area, leaving π in the answers.

Solution: Cylinder volume $= (22/7) \times 49 \times 10 = 22 \times 7 \times 10 = 1540$ cm³. Curved area $= 2 \times (22/7) \times 7 \times 10 = 440$ cm². For the cone, $l = \sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$ cm. Volume $= \tfrac{1}{3} \times \pi \times 36 \times 8 = 96\pi$ cm³. Curved surface area $= \pi \times 6 \times 10 = 60\pi$ cm².

WORKED EXAMPLE 7.9

(a) Find the volume of a sphere of radius 21 cm (π = 22/7). (b) A solid metal sphere of radius 6 cm is melted and recast into small spheres of radius 2 cm. How many can be made, assuming no loss? (c) A sphere of radius 3 cm is completely submerged in a cylindrical jar of radius 10 cm containing water. By how much does the water level rise?

Solution: (a) $V = \tfrac{4}{3} \times (22/7) \times 21^3 = \tfrac{4}{3} \times (22/7) \times 9261$. Since $9261/21 = 441$, this equals $(4 \times 22 \times 441)/(1) \times (1/(3 \times 7)) \times 21$... to avoid fuss, simplify stepwise: $21^3/(3 \times 7) = 9261/21 = 441$, so $V = 4 \times 22 \times 441 = 38{,}808$ cm³. (b) The number of spheres equals the ratio of volumes, which is $(6/2)^3 = 27$. (c) The volume of water displaced equals the volume of the sphere: $\tfrac{4}{3}\pi \times 27 = 36\pi$ cm³. This forms a layer of height h in the cylinder: $\pi \times 10^2 \times h = 36\pi$, so $h = 36/100 = 0.36$ cm.

Solids that are cut by a plane parallel to the base give a smaller solid similar to the original, which makes the scale-factor rule useful. The remaining piece of a cone with its tip removed is called a **frustum**; its volume is the volume of the whole cone minus the volume of the small cone removed.

WORKED EXAMPLE 7.10

A cone has height 12 cm and base radius 9 cm. A plane parallel to the base removes the top 4 cm of the cone. Find the volume of the frustum that remains.

Solution: The small cone has height 4 cm, which is 1/3 of 12 cm, so it is similar to the large cone with scale factor 1/3 and radius 3 cm. The volume of the large cone is $\tfrac{1}{3} \times \pi \times 81 \times 12 = 324\pi$ cm³. The small cone's volume is $(1/3)^3 \times 324\pi = 324\pi/27 = 12\pi$ cm³. The frustum has volume $324\pi - 12\pi = 312\pi$ cm³.

7.6 Loci and constructions

A **locus** (plural **loci**) is the set of all points that satisfy a given condition. Four basic loci cover most questions. The locus of points at a fixed distance r from a fixed point is a circle of radius r. The locus of points equidistant from two fixed points A and B is the **perpendicular bisector** of AB. The locus of points equidistant from two intersecting lines is the pair of **angle bisectors** of the angles between them. The locus of points at a fixed distance d from a straight line is a pair of parallel lines, one on each side of the given line, each at distance d.

To find points that satisfy two conditions at once, draw each locus and mark where they cross. A farmer who must place a water tap equidistant from two huts and 5 m from one of them is solving exactly this kind of problem. Constructions with ruler and compasses are the practical side of loci: a perpendicular bisector is drawn by marking arcs of equal

radius from both ends of the segment so that they cross above and below it; an angle bisector is drawn by marking equal distances along both arms from the vertex and then drawing equal arcs from those marks; an angle of 60° is made by an arc from the end of a line cutting a second arc of the same radius, and halving that gives 30° while adding the right angle gives 90°. A knowledge of these gives constructions of 30°, 45°, 60°, 75° and 90° angles.

WORKED EXAMPLE 7.11

Two poles A and B stand 8 m apart. A point P is equidistant from A and B and is 5 m from A. How far is P from the line AB?

Solution: P lies on the perpendicular bisector of AB, so the foot M of the perpendicular from P is the midpoint of AB and AM = 4 m. In the right-angled triangle AMP, $PM^2 = 5^2 - 4^2 = 25 - 16 = 9$, so PM = 3 m. There are two such points, one on each side of AB, each 3 m from the line.

Exercise 7.1

1. Find each interior angle of a regular octagon.
2. What is the sum of the interior angles of a hexagon?
3. Each exterior angle of a regular polygon is 18°. How many sides has it?
4. Two similar cones have heights 4 cm and 6 cm. The curved surface area of the smaller is 80 cm^2. Find that of the larger.

5. The angle at the centre of a circle standing on an arc is 100°. What is the angle at the circumference on the same arc?
6. In a cyclic quadrilateral the opposite angles A and C are $2x°$ and $(x + 30)°$. Find the two angles.
7. A point is 13 cm from the centre of a circle of radius 5 cm. Find the length of a tangent from the point to the circle.
8. Find the area of a sector of angle 90° in a circle of radius 14 cm ($\pi = 22/7$).
9. A cylinder of radius 7 cm has volume 3080 cm^3. Find its height ($\pi = 22/7$).
10. Show that a triangle with sides 7 cm, 24 cm and 25 cm is right-angled, and find its area.
11. A cube of side 6 cm is melted and recast into a cuboid with base 9 cm by 4 cm. Find the height of the cuboid.
12. A cone has radius 3 cm and slant height 5 cm. Find its volume in terms of π.

Answers to Exercise 7.1

1. 135°: $(8 − 2) × 180/8 = 1080/8 = 135$.
2. 720°: $(6 − 2) × 180$.
3. 20 sides: $360/18 = 20$.
4. 180 cm^2: areas scale as $(6/4)^2 = 2.25$, and $80 × 2.25 = 180$.
5. 50°: half of 100°.

6. A = 100°, C = 80°: 2x + x + 30 = 180 gives x = 50.
7. 12 cm: $\sqrt{(13^2 - 5^2)} = \sqrt{144}$.
8. 154 cm²: ¼ × (22/7) × 196 = 154.
9. 20 cm: (22/7) × 49 = 154, and 3080/154 = 20.
10. 49 + 576 = 625 = 25², so the triangle is right-angled; area = ½ × 7 × 24 = 84 cm².
11. 6 cm: volume = 216 cm³ and 216/36 = 6.
12. 12π cm³: height = $\sqrt{(25 - 9)} = 4$ and V = ⅓ × π × 9 × 4 = 12π.

7.7 Further problems and strategies

Composite figures, such as a running track made from a rectangle with a semicircle at each end, a tent made from a cylinder topped by a cone, or an ice-cream cone with a hemisphere on top, are treated by splitting them into the standard pieces above and adding or subtracting. Draw each piece separately, label the dimensions that carry over (for instance, the radius of the hemisphere equals the radius of the cone beneath it), and then compute. When you are asked for a surface area, be careful to include only the faces that are actually exposed: the circle where a hemisphere meets a cone, for example, is hidden and is not counted.

WORKED EXAMPLE 7.12

A solid is made of a cylinder of radius 7 cm and height 12 cm with a hemisphere of the same radius fixed on top. Find its total volume ($\pi = 22/7$).

Solution: Cylinder: $(22/7) \times 49 \times 12 = 154 \times 12 = 1848$ cm³. Hemisphere: $\tfrac{2}{3} \times (22/7) \times 343 = \tfrac{2}{3} \times 1078 = 2156/3 \approx 718.67$ cm³. Total $\approx 1848 + 718.67 = 2566.67$ cm³, or exactly $7700/3$ cm³ since $1848 = 5544/3$ and $5544/3 + 2156/3 = 7700/3$.

Another recurring skill is to convert units correctly. One square metre is 10,000 cm², not 100 cm², and one cubic metre is 1,000,000 cm³ and also 1,000 litres. A tank with volume 2.5 m³ therefore holds 2,500 litres. Whenever the units of a question are mixed, convert everything to a single unit before you substitute into a formula.

Exercise 7.2

1. A running track consists of a rectangle 80 m by 40 m with a semicircle (diameter 40 m) on each short end. Find the perimeter of the whole track ($\pi = 22/7$, to the nearest metre).
2. A rectangular tank 2 m long, 1.5 m wide and 1 m deep is full of water. How many litres does it hold?
3. In a circle, chord PQ is 16 cm long and is 6 cm from the centre. Find the radius.

4. A tangent from T touches a circle at S, and a line from T cuts the circle at A and B with TA = 4 cm and TB = 9 cm. Find TS.
5. Find the area of a trapezium whose parallel sides are 9 cm and 15 cm and whose height is 7 cm.
6. A rhombus has diagonals of 10 cm and 24 cm. Find its side and area.
7. The areas of two similar figures are 50 cm² and 98 cm². What is the ratio of corresponding lengths?
8. Find the volume of a pyramid with a square base of side 6 cm and perpendicular height 10 cm.
9. Find the total surface area of a closed cylinder of radius 7 cm and height 3 cm ($\pi = 22/7$).
10. A sphere has surface area 616 cm². Find its radius ($\pi = 22/7$).

Answers to Exercise 7.2

1. Straight sides total 160 m; the two semicircles make one circle of circumference $(22/7) \times 40 \approx 125.7$ m; total ≈ 285.7, so 286 m.
2. 3000 litres: volume = $2 \times 1.5 \times 1 = 3$ m³ = 3000 litres.
3. 10 cm: half-chord 8, so $r = \sqrt{(8^2 + 6^2)} = 10$.
4. 6 cm: $TS^2 = 4 \times 9 = 36$.
5. 84 cm²: $\tfrac{1}{2} \times (9 + 15) \times 7$.
6. Half-diagonals 5 and 12 give side 13 cm; area = $\tfrac{1}{2} \times 10 \times 24 = 120$ cm².
7. 5 : 7, since $\sqrt{(50/98)} = \sqrt{(25/49)} = 5/7$.

8. 120 cm³: ⅓ × 36 × 10.

9. 440 cm²: 2 × (22/7) × 7 × 3 + 2 × (22/7) × 49 = 132 + 308 = 440.

10. 7 cm: 4 × (22/7) × r² = 616 gives r² = 616 × 7/88 = 49.

Chapter summary

- Angles on a line total 180°, round a point 360°; parallel lines give equal corresponding and alternate angles and supplementary co-interior angles.
- Interior angles of an n-sided polygon total (n − 2) × 180°; exterior angles total 360°.
- For similar figures with length ratio k, areas are in ratio k^2 and volumes in ratio k^3.
- Circle theorems: centre angle is twice circumference angle; angle in a semicircle is 90°; cyclic quadrilateral opposite angles total 180°; tangent is perpendicular to radius; alternate segment theorem; intersecting chord and tangent-secant products.
- Areas: triangle ½bh or ½ab sin C or Heron's formula; trapezium ½(a + b)h; circle πr^2; sector $(\theta/360)\pi r^2$.
- Volumes: cylinder $\pi r^2 h$, cone $\frac{1}{3}\pi r^2 h$, sphere $\frac{4}{3}\pi r^3$, pyramid ⅓Bh; surface areas of cone $\pi r l$ and sphere $4\pi r^2$.
- Loci: circle, perpendicular bisector, angle bisector, parallel lines; combine loci by intersecting them.

Chapter 8: Trigonometry

Trigonometry began as the practical science of measuring triangles, and the name itself comes from Greek words meaning "triangle measurement". Surveyors in Lagos use it to fix the boundaries of a plot without walking across swampy ground, navigators use it to plot a course across the Atlantic, and engineers use it to design roofs, bridges and radio masts. At its centre lies a simple observation: in a right-angled triangle, the ratios between the sides depend only on the angles, not on the size of the triangle. Once those ratios have been tabulated or calculated, one measured angle and one measured length are enough to find everything else. This chapter develops the ratios for acute angles, extends them to angles of any size, derives the sine rule and cosine rule for triangles that are not right-angled, applies them to bearings and heights, and finishes with identities, equations, radian measure and the graphs of the basic functions.

8.1 The trigonometric ratios of an acute angle

Consider a right-angled triangle containing an acute angle θ. The side opposite the right angle is the **hypotenuse**, the longest side. The side facing θ is the **opposite**, and the remaining side, which touches θ, is the **adjacent**. The three basic ratios are defined as follows.

Key point. sin θ = opposite/hypotenuse, cos θ = adjacent/hypotenuse, tan θ = opposite/adjacent. A memory aid is SOH-CAH-TOA. Also tan θ = sin θ ÷ cos θ, and the reciprocal ratios are cosec θ = 1/sin θ, sec θ = 1/cos θ and cot θ = 1/tan θ.

Because the sides of a right-angled triangle obey Pythagoras' theorem, the sine and cosine of the same angle are linked by an identity that we meet again in section 8.5. For now, the practical point is that you must label the sides relative to the angle you are working with. The opposite side for one acute angle is the adjacent side for the other, and many marks are lost by labelling from the wrong angle.

Two angles are **complementary** if they add up to 90°. For complementary angles the ratios swap: sin θ = cos (90° − θ) and cos θ = sin (90° − θ). For example sin 20° = cos 70°. This fact is often used to simplify expressions in the examination.

Exact values for special angles

Three angles occur so often that their ratios should be known without a calculator. Take an equilateral triangle of side 2 and cut it in half; the resulting right-angled triangle has angles 30°, 60° and 90°, a hypotenuse of 2, a short side of 1 and a long side of √3 (since $2^2 - 1^2 = 3$). Take also an isosceles right-angled triangle with equal sides 1; its hypotenuse is √2. Reading the ratios from these triangles gives the table below.

Angle	sin	cos	tan
30°	1/2	√3/2	1/√3 = √3/3
45°	1/√2 = √2/2	1/√2 = √2/2	1
60°	√3/2	1/2	√3

Together with the values at 0° and 90° (sin 0° = 0, cos 0° = 1, sin 90° = 1, cos 90° = 0, tan 0° = 0, and tan 90° undefined), this table can be reconstructed in seconds from the two triangles, so it is wiser to remember the triangles than to memorise a block of numbers.

WORKED EXAMPLE 8.1

Evaluate $\sin 30^\circ \cos 60^\circ + \tan 45^\circ$, and find the exact value of $(\sin 60^\circ)^2 + (\cos 60^\circ)^2$.

Solution: $\sin 30^\circ = 1/2$, $\cos 60^\circ = 1/2$ and $\tan 45^\circ = 1$. So the first expression is $(1/2)(1/2) + 1 = 1/4 + 1 = 5/4$. For the second, $(\sqrt{3}/2)^2 = 3/4$ and $(1/2)^2 = 1/4$, so the sum is $3/4 + 1/4 = 1$. This is no accident: it illustrates the identity $\sin^2\theta + \cos^2\theta = 1$.

WORKED EXAMPLE 8.2

A ladder 6 m long leans against a vertical wall and makes an angle of 60° with the level ground. Find (a) how high up the wall it reaches and (b) how far its foot is from the wall.

Solution: The ladder is the hypotenuse. (a) The height is opposite the 60° angle, so height = 6 sin 60° = 6 × √3/2 = 3√3 m. Using √3 = 1.732, this is 5.196 m, that is, 5.20 m to three significant figures. (b) The distance from the wall is adjacent to the 60° angle, so distance = 6 cos 60° = 6 × 1/2 = 3 m. Check with Pythagoras: $3^2 + (3\sqrt{3})^2 = 9 + 27 = 36 = 6^2$. The answer is consistent.

8.2 Angles of elevation and depression, and bearings

Trigonometry is most often examined through practical settings. The **angle of elevation** of an object is the angle between the horizontal and the line of sight upward to it. The **angle of depression** is the angle between the horizontal and the line of sight downward. Because the horizontal lines through the observer and through the object are parallel, the angle of elevation of A from B equals the angle of depression of B from A (alternate angles). Always draw a sketch showing the horizontal, the line of sight and the right angle at the foot of the vertical object.

A **bearing** is a direction measured clockwise from north and written with three figures, so east is 090°, south is 180° and west is 270°. The bearing of B from A is measured at A, with the north line drawn through A. If the bearing of B from A is θ, then the bearing of A from B is $\theta + 180°$ when θ is less than 180°, and $\theta - 180°$ when θ is more than 180°; this is a consequence of the parallel north lines at A and B.

WORKED EXAMPLE 8.3

From a point on level ground 40 m from the foot of a tower, the angle of elevation of the top of the tower is 30°. Find the height of the tower to two decimal places, and the angle of elevation if the observer walks to a point 20 m from the foot.

Solution: The height h is opposite the 30° angle and the 40 m distance is adjacent, so $\tan 30° = h/40$ and $h = 40 \tan 30° = 40/\sqrt{3}$. Now $1/\sqrt{3} = 0.57735$, so h = 23.09 m. From 20 m away, $\tan \alpha = 23.09/20 = 1.1547$, which is exactly $2/\sqrt{3}$, so $\alpha = 49.1°$ ($\tan 49.1° \approx 1.1544$; to the nearest degree, 49°).

WORKED EXAMPLE 8.4

A boat sails 12 km from a harbour H on a bearing of 030° to a point P, then 16 km on a bearing of 120° to a point Q. Find the distance HQ and the bearing of Q from H.

Solution: The two bearings differ by 120° − 030° = 90°, so the path turns through a right angle at P and triangle HPQ is right-angled at P. By Pythagoras, $HQ^2 = 12^2 + 16^2 = 144 + 256 = 400$, so HQ = 20 km. The angle at H between HP and HQ satisfies tan = PQ/HP = 16/12 = 4/3, which gives 53.13°. Since HQ lies clockwise from HP, the bearing of Q from H is 030° + 53.13° = 083.1° (to the nearest tenth of a degree).

8.3 Angles of any size

The ratios defined with right-angled triangles apply only to angles below 90°. To extend them, imagine a point P moving round a circle of radius 1 centred at the origin, with the angle θ measured anticlockwise from the positive x-axis. The coordinates of P are (cos θ, sin θ), and tan θ = sin θ ÷ cos θ. This definition agrees with the earlier one for acute angles but allows any angle, including angles larger than 360° and negative angles.

The plane is divided into four quadrants. The signs of the ratios depend on the quadrant in which the angle terminates:

- First quadrant (0° to 90°): sin, cos and tan are all positive.
- Second quadrant (90° to 180°): only sin is positive.

- Third quadrant (180° to 270°): only tan is positive.
- Fourth quadrant (270° to 360°): only cos is positive.

A convenient memory device is "All Students Take Chemistry", reading the quadrants anticlockwise from the first. To find a ratio of a large angle, reduce it to an acute **reference angle** and attach the correct sign: in the second quadrant the reference angle is 180° − θ, in the third it is θ − 180°, and in the fourth it is 360° − θ. Angles that differ by 360° have identical ratios, so sin 400° = sin 40°. For negative angles, sin (−θ) = −sin θ, cos (−θ) = cos θ and tan (−θ) = −tan θ.

WORKED EXAMPLE 8.5

Without tables, find the values of sin 150°, cos 120°, tan 225°, sin 240° and cos 300°.

Solution: 150° is in the second quadrant with reference angle 30°; sine is positive there, so sin 150° = sin 30° = 1/2. 120° has reference angle 60° and cosine is negative in the second quadrant, so cos 120° = −cos 60° = −1/2. 225° is in the third quadrant with reference angle 45°; tangent is positive there, so tan 225° = tan 45° = 1. 240° has reference angle 60°, and sine is negative in the third quadrant, so sin 240° = −sin 60° = −√3/2. Finally 300° has reference angle 60° and cosine is positive in the fourth quadrant, so cos 300° = cos 60° = 1/2.

WORKED EXAMPLE 8.6

Given that $\sin\theta = 3/5$ and θ is obtuse, find $\cos\theta$ and $\tan\theta$.

Solution: Draw a right-angled triangle with opposite 3 and hypotenuse 5; the adjacent side is $\sqrt{(25 - 9)} = 4$. The reference angle has cos = 4/5 and tan = 3/4. An obtuse angle lies in the second quadrant, where cosine and tangent are negative. Therefore $\cos\theta = -4/5$ and $\tan\theta = -3/4$. Check: $\sin^2\theta + \cos^2\theta = 9/25 + 16/25 = 1$.

8.4 The sine rule, cosine rule and area of a triangle

For a triangle that is not right-angled, label the angles A, B, C and the sides opposite them a, b, c. Two rules connect the sides and angles. They are best remembered as tools, each suited to particular given information.

Key point. Sine rule: $a/\sin A = b/\sin B = c/\sin C$. Use it when you know two angles and a side, or two sides and an angle opposite one of them. Cosine rule: $a^2 = b^2 + c^2 - 2bc\cos A$, or equivalently $\cos A = (b^2 + c^2 - a^2)/(2bc)$. Use it when you know two sides and the included angle, or all three sides. Area = $\frac{1}{2}ab\sin C$, using two sides and the angle between them.

When the sine rule is used to find an angle from two sides and a non-included angle, there may be two possible triangles, because $\sin\theta = \sin(180° - \theta)$. This is called the **ambiguous case**. If the side opposite the known angle is the longer of the two given sides, only one triangle is possible. The cosine rule never suffers from this difficulty, because the cosine of an obtuse angle is negative and so distinguishes the two possibilities automatically.

WORKED EXAMPLE 8.7

In triangle ABC, $A = 45°$, $B = 60°$ and $a = 10$ cm. Find b and C.

Solution: $C = 180° - 45° - 60° = 75°$. By the sine rule, $b/\sin 60° = a/\sin 45°$, so $b = 10 \times \sin 60° \div \sin 45° = 10 \times (\sqrt{3}/2) \div (\sqrt{2}/2) = 10\sqrt{3}/\sqrt{2} = 5\sqrt{6}$. With $\sqrt{6} = 2.449$ this is 12.25 cm. A rough check: side b faces the larger angle than a, so b should be longer than a, and 12.25 exceeds 10.

WORKED EXAMPLE 8.8

Two roads from a junction J meet at 60°. Musa walks 7 km along one road to a village X and Ada walks 8 km along the other to a village Y. How far apart are X and Y?

Solution: Two sides and the included angle are given, so use the cosine rule: $XY^2 = 7^2 + 8^2 - 2 \times 7 \times 8 \times \cos 60° = 49 + 64 - 112 \times 1/2 = 113 - 56 = 57$. So $XY = \sqrt{57} = 7.55$ km (since $7.55^2 = 57.0025$).

WORKED EXAMPLE 8.9

The sides of a triangle are 5 cm, 6 cm and 7 cm. Find the largest angle and the area.

Solution: The largest angle is opposite the longest side, 7 cm. $\cos\theta = (5^2 + 6^2 - 7^2)/(2 \times 5 \times 6) = (25 + 36 - 49)/60 = 12/60 = 0.2$, so $\theta = 78.5°$. The area is $\tfrac{1}{2} \times 5 \times 6 \times \sin 78.5°$. Since $\sin^2\theta = 1 - 0.04 = 0.96$, $\sin\theta = 0.9798$, and the area is $15 \times 0.9798 = 14.70$ cm². Check with Heron's formula: $s = 9$, area $= \sqrt{(9 \times 4 \times 3 \times 2)} = \sqrt{216} = 14.70$ cm². Both methods agree.

8.5 Identities and trigonometric equations

An **identity** is an equation true for every angle. The fundamental one follows from Pythagoras' theorem on the unit circle: $\sin^2\theta + \cos^2\theta = 1$. Dividing through by $\cos^2\theta$ gives $1 + \tan^2\theta = \sec^2\theta$, and dividing by $\sin^2\theta$ gives $1 + \cot^2\theta = \operatorname{cosec}^2\theta$. These let you convert between ratios, which is the key to simplifying expressions and solving equations that mix sines and cosines.

For the UTME you should also know the compound-angle results $\sin(A + B) = \sin A \cos B + \cos A \sin B$ and $\cos(A + B) = \cos A \cos B - \sin A \sin B$, and the double-angle consequences $\sin 2A = 2 \sin A \cos A$ and $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$. Replacing B by $-B$ gives the subtraction forms, with the signs reversed in the cosine formula.

To solve an equation such as $\sin\theta = k$ in a stated interval, find the principal value from the calculator or the special-angle table, then use the symmetry of the circle to find all other solutions in the interval. For sine, the solutions are θ and $180° - \theta$; for cosine, θ and $360° - \theta$; for tangent, θ and $\theta + 180°$. If the equation is quadratic in a ratio, factorise it first, treating the ratio like an ordinary unknown, and discard any value outside the possible range (sine and cosine lie between -1 and 1).

WORKED EXAMPLE 8.10

Solve $2\sin^2\theta - \sin\theta - 1 = 0$ for $0° \le \theta \le 360°$.

Solution: Treat $\sin\theta$ as s. Then $2s^2 - s - 1 = (2s + 1)(s - 1) = 0$, so $s = -1/2$ or $s = 1$. If $\sin\theta = 1$, then $\theta = 90°$. If $\sin\theta = -1/2$, the reference angle is $30°$ and sine is negative in the third and fourth quadrants, so $\theta = 180° + 30° = 210°$ or $\theta = 360° - 30° = 330°$. The solutions are $90°$, $210°$ and $330°$. Check $\theta = 210°$: $2 \times 1/4 + 1/2 - 1 = 0$. Correct.

WORKED EXAMPLE 8.11

Show that $(1 - \cos^2\theta)/\sin\theta = \sin\theta$, and solve $\cos 2\theta = 1/2$ for $0° \le \theta \le 180°$.

Solution: Since $1 - \cos^2\theta = \sin^2\theta$, the left-hand side is $\sin^2\theta/\sin\theta = \sin\theta$, as required. For the equation, let $\varphi = 2\theta$, so $\cos\varphi = 1/2$ with $0° \le \varphi \le 360°$. The solutions are $\varphi = 60°$ and $\varphi = 360° - 60° = 300°$. Hence $\theta = 30°$ or $\theta = 150°$.

8.6 Radian measure and graphs

Angles can also be measured in **radians**. One radian is the angle at the centre of a circle subtended by an arc equal in length to the radius. Since the circumference is $2\pi r$, a full turn is 2π radians, so $180^\circ = \pi$ radians. To convert degrees to radians multiply by $\pi/180$; to convert radians to degrees multiply by $180/\pi$. Thus $90^\circ = \pi/2$, $60^\circ = \pi/3$, $45^\circ = \pi/4$, $30^\circ = \pi/6$. When θ is in radians, the arc length is $s = r\theta$ and the sector area is $\tfrac{1}{2}r^2\theta$. These are simpler than the degree forms, which is why radians are preferred in calculus.

The graphs of $y = \sin x$ and $y = \cos x$ are smooth waves that repeat every 360° (2π radians), with maximum value 1 and minimum -1. The sine curve starts at the origin, whereas the cosine curve starts at its maximum of 1 and is the sine curve shifted 90° to the left. The graph of $y = \tan x$ repeats every 180°, passes through the origin and has vertical asymptotes at 90° and 270°. For $y = a \sin bx$ the **amplitude** is $|a|$ and the **period** is $360^\circ/b$. Therefore $y = 3 \sin 2x$ oscillates between -3 and 3 and completes a cycle every 180°.

WORKED EXAMPLE 8.12

A sector of a circle has radius 7 cm and angle 60°. Using $\pi = 22/7$, find the arc length and the area of the sector.

Solution: $60° = \pi/3$ radians. Arc length $= r\theta = 7 \times (22/7)/3 = 22/3 = 7.33$ cm. Area $= \tfrac{1}{2}r^2\theta = \tfrac{1}{2} \times 49 \times (22/7)/3 = \tfrac{1}{2} \times 49 \times 22/21 = 539/21 = 25.67$ cm² (that is, 77/3). Check with the degree formula: $(60/360) \times (22/7) \times 49 = (1/6) \times 154 = 25.67$ cm².

Exercise 8.1

1. Evaluate $\cos 60° + \sin 30° \times \tan 60° \times \sin 60°$.
2. A kite string 50 m long is taut and makes an angle of 30° with the horizontal. Find the height of the kite above the hand holding the string.
3. From the top of a cliff 60 m high, the angle of depression of a boat is 45°. How far is the boat from the foot of the cliff?
4. Find the exact value of $\sin 135° \times \cos 210°$.
5. If $\cos\theta = -5/13$ and $90° < \theta < 180°$, find $\sin\theta$ and $\tan\theta$.
6. In triangle PQR, $p = 8$ cm, $q = 5$ cm and $R = 60°$. Find r.
7. In triangle XYZ, $X = 30°$, $Y = 45°$ and $x = 6$ cm. Find y.
8. Find the area of a triangle with sides 10 cm and 12 cm enclosing an angle of 30°.
9. A ship leaves port A and sails 8 km due east, then 6 km due north. Find the bearing of the ship from A, to the nearest

degree.

10. Solve $2\cos^2\theta + \cos\theta - 1 = 0$ for $0° \le \theta \le 360°$.
11. Express $135°$ in radians, in terms of π, and find the period of $y = \sin 3x$.

Answers to Exercise 8.1

1. $5/4$: $1/2 + (1/2)(\sqrt{3})(\sqrt{3}/2) = 1/2 + 3/4 = 5/4$.
2. 25 m: $50 \sin 30° = 25$.
3. 60 m: $\tan 45° = 1$, so distance $= 60/1$.
4. $-\sqrt{6}/4$: $\sin 135° = \sqrt{2}/2$, $\cos 210° = -\sqrt{3}/2$, product $= -\sqrt{6}/4$.
5. $\sin\theta = 12/13$, $\tan\theta = -12/5$: the third side is $\sqrt{(169 - 25)} = 12$; sine is positive in the second quadrant.
6. 7 cm: $r^2 = 64 + 25 - 2(8)(5)(1/2) = 89 - 40 = 49$.
7. $6\sqrt{2} = 8.49$ cm: $y = 6 \sin 45°/\sin 30° = 6(\sqrt{2}/2)/(1/2) = 6\sqrt{2}$.
8. 30 cm²: $\tfrac{1}{2} \times 10 \times 12 \times 1/2$.
9. 053° (or 53.1°): $\tan\theta = 8/6$ measured from north, the bearing being the angle clockwise from north, $\tan^{-1}(8/6) = 53.13°$.
10. 60°, 180°, 300°: $(2c - 1)(c + 1) = 0$, giving $\cos\theta = 1/2$ or -1.
11. $3\pi/4$ and 120°: $135 \times \pi/180 = 3\pi/4$; period $= 360°/3 = 120°$.

Chapter summary

- In a right-angled triangle, sin = opposite/hypotenuse, cos = adjacent/hypotenuse, tan = opposite/adjacent; sides must be labelled from the angle in question.
- Exact values at 30°, 45° and 60° come from the half-equilateral triangle and the isosceles right-angled triangle.
- Bearings are measured clockwise from north in three figures; angles of elevation and depression are measured from the horizontal.
- For angles of any size, use the reference angle and the signs from the quadrants (All, Sin, Tan, Cos).
- Sine rule: $a/\sin A = b/\sin B = c/\sin C$. Cosine rule: $a^2 = b^2 + c^2 - 2bc\cos A$. Area = $\tfrac{1}{2}ab\sin C$.
- $\sin^2\theta + \cos^2\theta = 1$, $\tan\theta = \sin\theta/\cos\theta$, $\sin 2A = 2\sin A\cos A$, $\cos 2A = \cos^2 A - \sin^2 A$.
- π radians = 180°; arc length is $r\theta$ and sector area is $\tfrac{1}{2}r^2\theta$ with θ in radians; $y = a\sin bx$ has amplitude $|a|$ and period $360°/b$.

Chapter 9: Coordinate Geometry of Straight Lines and Circles

Coordinate geometry, sometimes called analytical geometry, joins algebra to geometry. Every point in a plane is described by a pair of numbers, its coordinates, and every curve is described by an equation that those coordinates satisfy. The gain is considerable: a geometric question about distance, direction, intersection or tangency becomes an algebraic calculation, and an algebraic equation acquires a picture. The idea is attributed to the French thinker René Descartes, and the plane with its two perpendicular axes is called the Cartesian plane in his honour. In the UTME the topic is examined through a handful of standard tools: the distance and midpoint formulae, the gradient of a line, the equation of a line in its various forms, conditions for parallel and perpendicular lines, the point where two lines meet, and the equation of a circle. Questions are short, but they reward candidates who sketch the situation first and who keep their signs under control.

9.1 Distance, midpoint and gradient

Take two points $A(x_1, y_1)$ and $B(x_2, y_2)$. The horizontal separation is $x_2 - x_1$ and the vertical separation is $y_2 - y_1$. These form the two shorter sides of a right-angled triangle whose hypotenuse is the segment AB, so Pythagoras' theorem gives the distance.

Key point. Distance AB = $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$. Midpoint of AB = $((x_1 + x_2)/2, (y_1 + y_2)/2)$. Gradient of AB = $(y_2 - y_1)/(x_2 - x_1)$, provided $x_1 \neq x_2$. A vertical line has no gradient; a horizontal line has gradient 0.

The gradient measures steepness: it is the rise divided by the run, or equivalently $\tan\theta$, where θ is the angle the line makes with the positive x-axis, measured anticlockwise. A positive gradient means the line climbs from left to right; a negative gradient means it falls. The order in which you subtract does not matter provided you are consistent, since reversing both differences changes both signs and the quotient is unchanged. Subtracting the x-values in one order and the y-values in the other is the commonest slip and leads to a gradient with the wrong sign.

WORKED EXAMPLE 9.1

The points A and B have coordinates (2, 3) and (8, 11). Find the length of AB, its midpoint and its gradient.

Solution: The differences are $8 - 2 = 6$ and $11 - 3 = 8$. The length is $\sqrt{(36 + 64)} = \sqrt{100} = 10$ units. The midpoint is $((2 + 8)/2, (3 + 11)/2) = (5, 7)$. The gradient is $8/6 = 4/3$.

Dividing a line segment in a given ratio

If a point P lies on the segment AB and divides it so that AP : PB = m : n, then the coordinates of P are $\left(\frac{n x_1 + m x_2}{m + n}, \frac{n y_1 + m y_2}{m + n}\right)$. Notice the crossing pattern: the part of the ratio nearer B (the m) multiplies the coordinates of B, because the larger the share of the length measured from A, the closer P is to B. When m = n the formula reduces to the midpoint formula, which is a useful check.

WORKED EXAMPLE 9.2

Find the point P that divides the line joining A(−2, 1) and B(8, 6) in the ratio 2 : 3.

Solution: Here m = 2 and n = 3, so $x = \frac{3 \times (-2) + 2 \times 8}{5} = \frac{-6 + 16}{5} = 2$ and $y = \frac{3 \times 1 + 2 \times 6}{5} = \frac{15}{5} = 3$. So P is (2, 3). Check: AP = $\sqrt{4^2 + 2^2} = \sqrt{20}$ and PB = $\sqrt{6^2 + 3^2} = \sqrt{45}$, and $\sqrt{20} : \sqrt{45} = 2\sqrt{5} : 3\sqrt{5} = 2 : 3$, as required.

9.2 The equation of a straight line

Every straight line in the plane has an equation of the form ax + by + c = 0, and this is called the **general form**. Which form you use depends on what you are given.

- **Gradient-intercept form:** y = mx + c, where m is the gradient and c is the y-intercept, the value of y where the line meets the y-axis.

- **Point-gradient form:** $y - y_1 = m(x - x_1)$, for a line with gradient m through the point (x_1, y_1).
- **Two-point form:** $(y - y_1)/(y_2 - y_1) = (x - x_1)/(x_2 - x_1)$, for a line through two given points; equivalently, find the gradient first and use the point-gradient form.
- **Intercept form:** $x/a + y/b = 1$, for a line meeting the x-axis at $(a, 0)$ and the y-axis at $(0, b)$.

To read the gradient from a general equation, rearrange it into the form $y = mx + c$. For $ax + by + c = 0$ the gradient is $-a/b$. To find where a line meets the axes, put $y = 0$ for the x-intercept and $x = 0$ for the y-intercept. A point lies on a line precisely when its coordinates satisfy the equation, which gives a quick way to test.

WORKED EXAMPLE 9.3

Find the equation of the line through $(1, 3)$ and $(4, 9)$.

Solution: The gradient is $(9 - 3)/(4 - 1) = 6/3 = 2$. Using the point $(1, 3)$: $y - 3 = 2(x - 1)$, so $y = 2x + 1$. Check with the other point: when $x = 4$, $y = 9$, which agrees.

WORKED EXAMPLE 9.4

A line cuts the x-axis at (4, 0) and the y-axis at (0, −3). Find its equation and its gradient.

Solution: In intercept form, $x/4 + y/(-3) = 1$. Multiplying every term by 12 gives $3x - 4y = 12$. Rearranging, $4y = 3x - 12$, so $y = (3/4)x - 3$, and the gradient is 3/4. This agrees with the direct calculation $(0 - (-3))/(4 - 0) = 3/4$.

9.3 Parallel and perpendicular lines

Parallel lines never meet, so they rise at exactly the same rate. Two lines with gradients m_1 and m_2 are **parallel** if $m_1 = m_2$. They are **perpendicular** if $m_1 \times m_2 = -1$, that is, if one gradient is the negative reciprocal of the other. For example, perpendicular to a line of gradient 3/4 is a line of gradient −4/3. The product rule fails only when one line is vertical, in which case the perpendicular line is horizontal.

The reason for the product rule is worth knowing. Rotating a line through 90° exchanges the roles of its rise and run, and introduces a change of sign, so a line with rise a and run b is turned into one with rise b and run −a. Their gradients are a/b and $b/(-a) = -b/a$, whose product is −1.

WORKED EXAMPLE 9.5

The line L has equation $3x - 4y + 8 = 0$. Find the equation of (a) the line through $(4, -1)$ parallel to L, and (b) the line through $(3, 5)$ perpendicular to L.

Solution: Rearranging, $4y = 3x + 8$, so the gradient of L is $3/4$. (a) A parallel line has gradient $3/4$: $y + 1 = (3/4)(x - 4)$. Multiplying by 4, $4y + 4 = 3x - 12$, so $3x - 4y - 16 = 0$. Check: substituting $(4, -1)$ gives $12 + 4 - 16 = 0$. (b) A perpendicular line has gradient $-4/3$: $y - 5 = -(4/3)(x - 3)$. Multiplying by 3, $3y - 15 = -4x + 12$, so $4x + 3y - 27 = 0$. Check: substituting $(3, 5)$ gives $12 + 15 - 27 = 0$.

The **perpendicular bisector** of a segment AB is the line that passes through the midpoint of AB at right angles to it. Every point on it is equidistant from A and B. To find its equation: calculate the midpoint, calculate the gradient of AB, take the negative reciprocal, and use the point-gradient form.

9.4 Intersection of lines, and areas

The point where two lines cross satisfies both equations, so it is found by solving them simultaneously, by elimination or substitution as in Chapter 4. If the lines are parallel and distinct, the simultaneous equations are inconsistent and there is no intersection; if the equations describe the same line there are infinitely many solutions.

WORKED EXAMPLE 9.6

Find the point where the lines $2x + y = 7$ and $x - y = 2$ intersect, and the area of the triangle formed by these lines and the y-axis.

Solution: Adding the equations eliminates y: $3x = 9$, so $x = 3$, and from $x - y = 2$, $y = 1$. The intersection is $(3, 1)$. The first line meets the y-axis at $(0, 7)$ and the second at $(0, -2)$. The triangle has base along the y-axis of length $7 - (-2) = 9$, and its height is the horizontal distance 3 from the apex to the y-axis. The area is $\tfrac{1}{2} \times 9 \times 3 = 13.5$ square units.

For a triangle with vertices (x_1, y_1), (x_2, y_2), (x_3, y_3), the area is $\tfrac{1}{2}\,|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$, where the vertical bars denote the absolute value. If the result is zero the three points are collinear, which is a standard way to test whether three points lie on a straight line. Another test is to compute the gradients AB and BC; if they are equal, A, B and C lie on one line, since B is shared.

WORKED EXAMPLE 9.7

Find the area of the triangle with vertices A(1, 1), B(5, 1) and C(3, 6), and show that D(−1, −3) , A and the point E(3, 5) are collinear.

Solution: Area = ½ |1(1 − 6) + 5(6 − 1) + 3(1 − 1)| = ½ |−5 + 25 + 0| = 10 square units. Check: AB is horizontal with length 4 and C is 5 units above it, so ½ × 4 × 5 = 10. For collinearity, the gradient of DA is (1 − (−3))/(1 − (−1)) = 4/2 = 2, and the gradient of AE is (5 − 1)/(3 − 1) = 4/2 = 2. The gradients are equal and A is shared, so D, A and E lie on one line.

Distance from a point to a line

The shortest distance from the point (p, q) to the line ax + by + c = 0 is |ap + bq + c| / √(a² + b²). This is the length of the perpendicular from the point to the line. It appears in the examination less often than the earlier results, but is worth knowing for the strongest candidates. For instance, the distance from (1, 2) to 3x + 4y − 20 = 0 is |3 + 8 − 20|/√(9 + 16) = 9/5 = 1.8 units.

9.5 The circle

A circle is the set of points at a fixed distance, the radius r, from a fixed point, the centre. If the centre is (a, b) then a general point (x, y) on the circle satisfies the distance formula, and squaring gives the standard equation.

Key point. A circle with centre (a, b) and radius r has equation $(x - a)^2 + (y - b)^2 = r^2$. If the centre is at the origin the equation is $x^2 + y^2 = r^2$. Expanding gives the general form $x^2 + y^2 + 2gx + 2fy + c = 0$, which has centre $(-g, -f)$ and radius $\sqrt{(g^2 + f^2 - c)}$.

Going from the general form back to the standard form is done by completing the square in x and in y, as in Chapter 5. The general form is a real circle only when $g^2 + f^2 - c$ is positive; if it is zero the "circle" is a single point and if it is negative there are no real points at all.

WORKED EXAMPLE 9.8

Find the equation of the circle with centre (3, −2) and radius 5, in standard and general form. Then find the centre and radius of $x^2 + y^2 - 4x + 6y - 12 = 0$.

Solution: The standard form is $(x - 3)^2 + (y + 2)^2 = 25$. Expanding: $x^2 - 6x + 9 + y^2 + 4y + 4 = 25$, which gives $x^2 + y^2 - 6x + 4y - 12 = 0$. For the second part, complete the squares: $(x^2 - 4x + 4) + (y^2 + 6y + 9) = 12 + 4 + 9 = 25$, so $(x - 2)^2 + (y + 3)^2 = 25$. The centre is (2, −3) and the radius is 5. Using the formula instead: $2g = -4$ so $g = -2$, $2f = 6$ so $f = 3$, centre $(-g, -f) = (2, -3)$, and $r = \sqrt{(4 + 9 + 12)} = 5$. Both methods agree.

Lines and circles

To find where a line meets a circle, substitute the equation of the line into that of the circle to obtain a quadratic equation in one variable. If the discriminant is positive there are two intersection points (the line is a secant), if it is zero there is one (the line is a tangent), and if it is negative there are none. The tangent at a point on a circle is perpendicular to the radius drawn to that point, so its gradient is the negative reciprocal of the gradient of the radius.

WORKED EXAMPLE 9.9

(a) Find the points where the line $y = x + 1$ meets the circle $x^2 + y^2 = 25$, and the length of the chord. (b) Find the equation of the tangent to the circle $(x - 3)^2 + (y + 2)^2 = 25$ at the point $(6, 2)$.

Solution: (a) Substituting: $x^2 + (x + 1)^2 = 25$, so $2x^2 + 2x + 1 = 25$, that is, $2x^2 + 2x - 24 = 0$, or $x^2 + x - 12 = 0$. Factorising, $(x + 4)(x - 3) = 0$, so $x = -4$ or $x = 3$. The points are $(-4, -3)$ and $(3, 4)$; check: $16 + 9 = 25$ and $9 + 16 = 25$. The chord length is $\sqrt{[(3 + 4)^2 + (4 + 3)^2]} = \sqrt{98} = 7\sqrt{2} = 9.90$ units. (b) First confirm that $(6, 2)$ is on the circle: $(6 - 3)^2 + (2 + 2)^2 = 9 + 16 = 25$. The radius from the centre $(3, -2)$ to $(6, 2)$ has gradient $(2 + 2)/(6 - 3) = 4/3$, so the tangent has gradient $-3/4$. Then $y - 2 = -(3/4)(x - 6)$, so $4y - 8 = -3x + 18$, giving $3x + 4y - 26 = 0$. Check: $18 + 8 - 26 = 0$.

9.6 Strategies and common errors

Many coordinate geometry questions can be solved in two ways, one algebraic and one by drawing, and using the second to check the first is excellent practice. Always draw a rough sketch, even a poor one; a sketch will reveal, for instance, that a calculated gradient should be negative or that a point cannot be inside a circle. When working with equations, keep the form the question asks for: if the answer is wanted as $ax + by + c = 0$ with whole-number coefficients, clear the fractions before you finish.

The commonest errors are these. First, confusing the signs in the circle equation: the centre of $(x + 2)^2 + (y - 5)^2 = 9$ is $(-2, 5)$, not $(2, -5)$. Second, forgetting to take the square root of the right-hand side when reading the radius: $x^2 + y^2 = 49$ has radius 7, not 49. Third, using the wrong ratio order in the section formula; checking against the midpoint case or computing distances afterwards protects against this. Fourth, writing a perpendicular gradient with the wrong sign: the negative reciprocal of 2 is $-1/2$, and the negative reciprocal of $-3/5$ is $5/3$.

Exercise 9.1

1. Find the distance between $(-1, 2)$ and $(5, 10)$.
2. Find the midpoint of the line joining $(-3, 4)$ and $(7, -8)$.
3. Find the gradient of the line joining $(2, -1)$ and $(6, 7)$.
4. Find the equation of the line with gradient -2 passing through $(3, 1)$.

5. For what value of k are the lines $2x + ky = 5$ and $3x - 6y = 1$ parallel?
6. Find the equation of the line through (0, 5) perpendicular to $y = 3x + 1$.
7. Find the point of intersection of $3x + 2y = 13$ and $x - y = 1$.
8. Find the centre and radius of the circle $x^2 + y^2 + 8x - 6y = 0$.
9. Find the equation of the circle with diameter joining (1, 2) and (7, 10).
10. Find the area of the triangle with vertices (0, 0), (6, 0) and (2, 5).
11. Find the point dividing the line joining (1, −2) and (7, 4) in the ratio 1 : 2.
12. Find the equation of the perpendicular bisector of the line joining (0, 0) and (4, 6).

Answers to Exercise 9.1

1. 10 units: $\sqrt{(6^2 + 8^2)} = \sqrt{100}$.
2. (2, −2): $(-3 + 7)/2 = 2$ and $(4 - 8)/2 = -2$.
3. 2: $(7 + 1)/(6 - 2) = 8/4$.
4. $y = -2x + 7$: $y - 1 = -2(x - 3)$.
5. $k = -4$: gradients are $-2/k$ and $1/2$, so $-2/k = 1/2$.
6. $x + 3y = 15$: the gradient is $-1/3$ and the y-intercept is 5, so $y = -x/3 + 5$.

7. (3, 2): put $x = y + 1$ to get $3y + 3 + 2y = 13$, so $y = 2$ and $x = 3$. Check $9 + 4 = 13$.
8. Centre $(-4, 3)$, radius 5: $r^2 = 16 + 9 + 0 = 25$.
9. $(x - 4)^2 + (y - 6)^2 = 25$: the centre is $(4, 6)$ and the diameter is $\sqrt{(36 + 64)} = 10$, so $r = 5$.
10. 15 square units: base 6, height 5, so $\frac{1}{2} \times 6 \times 5$.
11. $(3, 0)$: $x = (2 \times 1 + 1 \times 7)/3 = 3$ and $y = (2 \times (-2) + 1 \times 4)/3 = 0$.
12. $2x + 3y = 13$: the midpoint is $(2, 3)$, the gradient of the segment is $3/2$, so the bisector has gradient $-2/3$, and $3y - 9 = -2x + 4$. Check: $4 + 9 = 13$.

Chapter summary

- Distance $= \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$; midpoint = average of coordinates; gradient $= (y_2 - y_1)/(x_2 - x_1)$.
- A point dividing AB in ratio $m : n$ has coordinates $((n\,x_1 + m\,x_2)/(m + n), (n\,y_1 + m\,y_2)/(m + n))$.
- Line forms: $y = mx + c$, $y - y_1 = m(x - x_1)$, $x/a + y/b = 1$ and $ax + by + c = 0$ (gradient $-a/b$).
- Parallel lines have equal gradients; perpendicular lines have gradients whose product is -1.
- Intersections are found by solving simultaneously; zero area of a triangle means collinear points.
- Circle: $(x - a)^2 + (y - b)^2 = r^2$; general form $x^2 + y^2 + 2gx + 2fy + c = 0$ has centre $(-g, -f)$ and radius $\sqrt{(g^2 + f^2 - c)}$.

- A tangent is perpendicular to the radius at the point of contact; the discriminant of the substituted quadratic tells whether a line is a secant, tangent or misses the circle.

Chapter 10: Calculus: Differentiation and Integration

Calculus is the mathematics of change. Algebra and geometry describe quantities that are fixed; calculus describes how quantities vary, how fast they vary, and how small changes accumulate into a total. When a driver on the Lagos–Ibadan expressway glances at the speedometer, the figure shown is a rate of change of distance with respect to time. When a farmer wonders what dimensions give the largest enclosure for a fixed length of fencing, the question is one of maximisation. When an engineer needs the area bounded by a curved dam wall, the tool is integration. Calculus handles all three with two linked operations, **differentiation**, which finds rates of change and gradients, and **integration**, which reverses differentiation and measures areas. The UTME confines itself to algebraic and simple trigonometric functions, so the rules in this chapter, applied with care, cover the whole syllabus.

10.1 The derivative and its meaning

The gradient of a straight line is constant, but the gradient of a curve changes from point to point. To define the gradient of a curve at a point P, join P to a nearby point Q on the curve. The line PQ is a chord whose gradient is easy to calculate. As Q slides along the curve towards P, the chord turns and approaches a limiting position, the tangent at P. The

gradient of that tangent is the **derivative** of the function at P. If the function is y = f(x), the derivative is written dy/dx or f′(x), and is read "dy by dx". It tells us the rate at which y changes with respect to x.

For a curve such as $y = x^2$, take a point with x-coordinate x and a nearby point with x-coordinate x + h. The chord gradient is $[(x + h)^2 - x^2]/h = (2xh + h^2)/h = 2x + h$. As h shrinks to zero, this tends to 2x, so the gradient of $y = x^2$ at any point is 2x. The same limiting process proves the general rule below, which spares us from working from first principles each time.

> **Key point.** If $y = ax^n$, where a and n are constants, then $dy/dx = anx^{n-1}$. A constant differentiates to 0. The derivative of a sum or difference is the sum or difference of the derivatives. Also: d/dx (sin x) = cos x and d/dx (cos x) = −sin x, with x in radians.

In words, multiply by the power, then reduce the power by one. This works for negative and fractional powers, provided you first write roots and reciprocals as powers: √x = x^(1/2), 1/x = x^{-1}, $1/x^2 = x^{-2}$, and so on. For example, $y = 3x^4 - 5x^2 + 2x - 7$ has derivative $12x^3 - 10x + 2$, where the term 2x differentiated as $2x^1$ gives $2 \times 1 \times x^0 = 2$ and the constant vanishes.

10.2 Further rules of differentiation

Most functions in the examination are not simple powers, but combinations of simpler functions. Three rules deal with them.

The **chain rule** handles a function of a function. If y is a function of u and u is a function of x, then dy/dx = (dy/du) × (du/dx). In practice, differentiate the outer function leaving the inner one intact, then multiply by the derivative of the inner function. Thus for y = (3x − 2)⁵, the outer power gives 5(3x − 2)⁴ and the inner function 3x − 2 has derivative 3, so dy/dx = 15(3x − 2)⁴. Similarly, d/dx [sin (ax)] = a cos (ax) and d/dx [cos (ax)] = −a sin (ax).

The **product rule** handles a product of two functions of x. If y = uv, where u and v are functions of x, then dy/dx = u (dv/dx) + v (du/dx). The **quotient rule** handles a ratio. If y = u/v, then dy/dx = [v (du/dx) − u (dv/dx)]/v². Observe that in the quotient rule the order matters, because of the subtraction: it is "bottom times derivative of top, minus top times derivative of bottom, all over bottom squared".

WORKED EXAMPLE 10.1

Differentiate (a) $y = (2x + 1)(x^2 - 3)$, (b) $y = x^2/(x + 1)$, (c) $y = (3x - 2)^5 + \sin 3x$.

Solution: (a) Let $u = 2x + 1$ and $v = x^2 - 3$, so $du/dx = 2$ and $dv/dx = 2x$. Then $dy/dx = (2x + 1)(2x) + (x^2 - 3)(2) = 4x^2 + 2x + 2x^2 - 6 = 6x^2 + 2x - 6$. Check by expanding first: $y = 2x^3 + x^2 - 6x - 3$, so $dy/dx = 6x^2 + 2x - 6$. Both methods agree. (b) Let $u = x^2$ and $v = x + 1$, so $du/dx = 2x$ and $dv/dx = 1$. Then $dy/dx = [(x + 1)(2x) - x^2(1)]/(x + 1)^2 = (2x^2 + 2x - x^2)/(x + 1)^2 = (x^2 + 2x)/(x + 1)^2$. (c) By the chain rule, the first term gives $15(3x - 2)^4$ and the second gives $3 \cos 3x$, so $dy/dx = 15(3x - 2)^4 + 3 \cos 3x$.

10.3 Tangents, normals and rates of change

The gradient of a curve at a point is the gradient of the tangent there. Once you have it, the equation of the tangent follows from the point-gradient form $y - y_1 = m(x - x_1)$, as in Chapter 9. The **normal** to a curve at a point is the line through that point perpendicular to the tangent, so its gradient is the negative reciprocal of the tangent gradient.

WORKED EXAMPLE 10.2

Find the gradient of $y = x^2 - 4x + 1$ at $x = 3$, and the equations of the tangent and normal at that point.

Solution: $dy/dx = 2x - 4$, which equals 2 at $x = 3$. When $x = 3$, $y = 9 - 12 + 1 = -2$, so the point is $(3, -2)$. The tangent is $y + 2 = 2(x - 3)$, so $y = 2x - 8$. Check: at $x = 3$, $2(3) - 8 = -2$. The normal has gradient $-1/2$: $y + 2 = -(1/2)(x - 3)$, so $2y + 4 = -x + 3$, giving $x + 2y + 1 = 0$. Check: $3 - 4 + 1 = 0$.

The derivative dy/dx is a **rate of change**: the amount by which y changes for each unit increase in x. When both variables change with time, the chain rule connects their rates: $dA/dt = (dA/dr) \times (dr/dt)$. The method is to write down a formula linking the quantities, differentiate it with respect to the variable that is not time, and multiply by the known rate.

WORKED EXAMPLE 10.3

The radius of a circular oil slick on a lagoon is increasing at 2 cm/s. How fast is its area increasing when the radius is 5 cm?

Solution: $A = \pi r^2$, so $dA/dr = 2\pi r$. Then $dA/dt = (dA/dr)(dr/dt) = 2\pi r \times 2 = 4\pi r$. When $r = 5$, $dA/dt = 20\pi$ cm²/s, which is 62.8 cm²/s using $\pi = 3.14$.

In kinematics, if s is the displacement of a particle from a fixed point at time t, then the velocity is $v = ds/dt$ and the acceleration is $a = dv/dt = d^2s/dt^2$. Two derivatives in succession give the **second derivative**, written d^2y/dx^2 or $f''(x)$.

WORKED EXAMPLE 10.4

A particle moves so that $s = t^3 - 6t^2 + 9t$ metres after t seconds. Find when it is at rest, its position at those times, and its acceleration at $t = 4$.

Solution: $v = ds/dt = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$. The particle is at rest when $v = 0$, that is at $t = 1$ s and $t = 3$ s. At $t = 1$, $s = 1 - 6 + 9 = 4$ m; at $t = 3$, $s = 27 - 54 + 27 = 0$ m. The acceleration is $a = dv/dt = 6t - 12$, so at $t = 4$, $a = 12$ m/s².

10.4 Maxima and minima

At the top of a hill or the bottom of a valley the tangent to a curve is horizontal, so the gradient is zero. Points where $dy/dx = 0$ are called **stationary points**. To decide what kind of stationary point you have, use the second derivative. If $d^2y/dx^2 > 0$ at the point, the curve is bending upward, and the point is a **minimum**. If $d^2y/dx^2 < 0$, the curve bends downward and the point is a **maximum**. If the second derivative is zero the test fails and one must examine the sign of dy/dx on either side, as the point may be a point of inflexion.

WORKED EXAMPLE 10.5

Find the stationary points of $y = x^3 - 6x^2 + 9x + 1$ and determine their nature.

Solution: $dy/dx = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3)$. Stationary points occur at $x = 1$ and $x = 3$. At $x = 1$, $y = 1 - 6 + 9 + 1 = 5$; at $x = 3$, $y = 27 - 54 + 27 + 1 = 1$. The second derivative is $d^2y/dx^2 = 6x - 12$. At $x = 1$ it is -6, which is negative, so $(1, 5)$ is a maximum. At $x = 3$ it is 6, which is positive, so $(3, 1)$ is a minimum.

Many practical problems ask for the largest or smallest value of a quantity. The approach is always the same: express the quantity in terms of a single variable, differentiate, set the derivative to zero, solve, check with the second derivative that you have the right kind of stationary point, and finally state the answer in the units of the question.

WORKED EXAMPLE 10.6

A farmer in Ibadan has 40 m of fencing and uses an existing wall for one side of a rectangular vegetable plot, so only three sides need fencing. Find the dimensions giving the greatest area.

Solution: Let the two sides perpendicular to the wall have length x metres. The side parallel to the wall is 40 − 2x metres. The area is $A = x(40 - 2x) = 40x - 2x^2$. Then $dA/dx = 40 - 4x$, which is zero when $x = 10$. The second derivative is −4, which is negative, so this is a maximum. The dimensions are 10 m by 20 m and the greatest area is $10 \times 20 = 200$ m².

10.5 Integration as the reverse of differentiation

If differentiating F(x) gives f(x), then integrating f(x) gives back F(x). Because the derivative of a constant is zero, many functions have the same derivative, differing only by a constant; so an **indefinite integral** always carries an arbitrary constant C, called the constant of integration.

Key point. $\int ax^n\,dx = a\,x^{n+1}/(n+1) + C$, for any $n \neq -1$: raise the power by one, then divide by the new power. Also $\int \sin x\,dx = -\cos x + C$ and $\int \cos x\,dx = \sin x + C$. More generally, $\int \cos(ax)\,dx = (1/a)\sin(ax) + C$ and $\int \sin(ax)\,dx = -(1/a)\cos(ax) + C$. The integral of a sum is the sum of the integrals.

Always check an integral by differentiating your answer: you should return to the original function. For instance, $\int (6x^2 - 4x + 3)\,dx = 2x^3 - 2x^2 + 3x + C$, and differentiating $2x^3 - 2x^2 + 3x$ gives $6x^2 - 4x + 3$, as it should. Roots and reciprocals must again be written as powers before integrating: $\int (1/x^2)\,dx = \int x^{-2}\,dx = -x^{-1} + C = -1/x + C$.

If extra information is supplied, such as a point through which the curve passes, the constant can be found. This is how the original function is recovered from its gradient.

WORKED EXAMPLE 10.7

The gradient of a curve is $dy/dx = 2x + 3$ and the curve passes through the point (1, 6). Find its equation.

Solution: Integrating, $y = x^2 + 3x + C$. Since the curve passes through (1, 6), $6 = 1 + 3 + C$, so $C = 2$. The equation is $y = x^2 + 3x + 2$. Check: at $x = 1$, $y = 6$.

10.6 Definite integrals and areas

A **definite integral** has limits and produces a number rather than a function. If F(x) is any integral of f(x), then the integral of f from x = a to x = b is $\int_a^b f(x)\,dx = F(b) - F(a)$. The constant C cancels in the subtraction, so it is not needed. The working is conventionally set out with square brackets: [F(x)] evaluated from a to b.

The significance of the definite integral is geometric. If f(x) is positive between a and b, the integral $\int_a^b f(x)\,dx$ equals the area bounded by the curve y = f(x), the x-axis and the vertical lines x = a and x = b. If the curve lies below the x-axis, the integral is negative, and the area is its absolute value. When a curve crosses the axis between the limits, split the integral at the crossing points, take each piece separately as a positive area, and add them; otherwise positive and negative regions cancel and understate the true area.

WORKED EXAMPLE 10.8

Evaluate $\int_1^3 (x^2 + 2)\,dx$, and find the area enclosed by the curve $y = 4 - x^2$ and the x-axis.

Solution: For the first part, the integral is $[x^3/3 + 2x]$ from 1 to 3. At x = 3: 9 + 6 = 15. At x = 1: 1/3 + 2 = 7/3. The difference is 15 − 7/3 = 38/3 = 12.67 (to 2 d.p.). For the second part, the curve meets the x-axis where $4 - x^2 = 0$, so at x = −2 and x = 2, and it lies above the axis between them. The area is $\int_{-2}^2 (4 - x^2)\,dx = [4x - x^3/3]$ from −2 to 2. At x = 2: 8 − 8/3 = 16/3. At x = −2: −8 + 8/3 = −16/3. The difference is 16/3 − (−16/3) = 32/3 = 10.67 square units.

WORKED EXAMPLE 10.9

Evaluate $\int_0^{(\pi/2)} \cos x \, dx$ and $\int \cos 2x \, dx$.

Solution: $\int_0^{(\pi/2)} \cos x \, dx = [\sin x]$ from 0 to $\pi/2 = \sin(\pi/2) - \sin 0 = 1 - 0 = 1$. For the second, $\int \cos 2x \, dx = (1/2) \sin 2x + C$. Differentiating $(1/2) \sin 2x$ gives $(1/2) \times 2 \cos 2x = \cos 2x$, confirming the result.

10.7 Strategies and common errors

Several errors recur in scripts. In differentiation, candidates forget to rewrite roots and fractions as powers, or forget to apply the chain rule to a bracket raised to a power. A reliable cure is to ask, before writing the derivative, "is the thing inside the bracket just x?" If not, multiply by its derivative. In integration, the most frequent slip is dropping the constant C from an indefinite integral, followed by raising the power but dividing by the old power instead of the new. In definite integrals, sign errors arise when substituting negative limits; use brackets around every substituted value. Finally, remember that trigonometric calculus requires radians: the rule $d/dx (\sin x) = \cos x$ is false if x is measured in degrees.

When a question asks you to "find the maximum value" rather than "the value of x at the maximum", substitute your x-value back into the original function to obtain it. And in tangent problems, make sure you substitute the x-value into the original curve, never into the derivative, to find the y-coordinate of the point of contact.

Exercise 10.1

1. Differentiate $5x^3 - 4x^2 + x - 9$.
2. Differentiate $y = \sqrt{x} + 1/x$.
3. Find the gradient of $y = 2x^3 - x$ at $x = 2$.
4. Differentiate $(x^2 + 1)(x - 3)$.
5. Differentiate $\sin 2x + \cos 3x$.
6. Find the coordinates and nature of the stationary point of $y = x^2 - 6x + 5$.
7. Find the equation of the tangent to $y = x^3$ at the point where $x = 2$.
8. Integrate $4x^3 - 3x^2 + 2$ with respect to x.
9. Evaluate $\int_0^2 (3x^2 + 1)\,dx$.
10. Find the area bounded by $y = x^2$, the x-axis and the line $x = 3$.
11. The edge of a cube is increasing at 0.5 cm/s. How fast is the volume increasing when the edge is 4 cm?
12. Two positive numbers add up to 12. Find the greatest possible value of their product.

Answers to Exercise 10.1

1. $15x^2 - 8x + 1$.
2. $1/(2\sqrt{x}) - 1/x^2$: write $y = x^{(1/2)} + x^{-1}$.
3. 23: $dy/dx = 6x^2 - 1$, so $24 - 1 = 23$.

4. $3x^2 - 6x + 1$: product rule gives $2x(x - 3) + (x^2 + 1)$; expanding first gives $x^3 - 3x^2 + x - 3$, which agrees.
5. $2\cos 2x - 3\sin 3x$.
6. $(3, -4)$, a minimum: $dy/dx = 2x - 6 = 0$ gives $x = 3$, $y = 9 - 18 + 5 = -4$, and the second derivative is 2, which is positive.
7. $y = 12x - 16$: the gradient is $3x^2 = 12$ and the point is $(2, 8)$, so $y - 8 = 12(x - 2)$.
8. $x^4 - x^3 + 2x + C$.
9. 10: $[x^3 + x]$ from 0 to 2 $= 8 + 2$.
10. 9 square units: $[x^3/3]$ from 0 to 3 $= 27/3$.
11. 24 cm^3/s: $V = x^3$, $dV/dt = 3x^2 \times dx/dt = 3 \times 16 \times 0.5$.
12. 36: with numbers x and $12 - x$, $P = 12x - x^2$, $dP/dx = 12 - 2x = 0$ gives $x = 6$, so $P = 36$ and the second derivative -2 confirms a maximum.

Chapter summary

- The derivative dy/dx is the gradient of the tangent and the rate of change of y with respect to x; for $y = ax^n$, $dy/dx = anx^{n-1}$.
- Chain rule: $dy/dx = (dy/du)(du/dx)$. Product rule: $(uv)' = u\,v' + v\,u'$. Quotient rule: $(u/v)' = (v\,u' - u\,v')/v^2$.
- $d/dx(\sin ax) = a\cos ax$ and $d/dx(\cos ax) = -a\sin ax$, with angles in radians.
- The tangent at a point has the gradient dy/dx there; the normal has the negative reciprocal gradient.

- Stationary points satisfy $dy/dx = 0$; $d^2y/dx^2 > 0$ gives a minimum and $d^2y/dx^2 < 0$ gives a maximum.
- In motion, $v = ds/dt$ and $a = dv/dt$.
- $\int ax^n\,dx = a\,x^{n+1}/(n+1) + C$; the definite integral $\int_a^b f(x)\,dx = F(b) - F(a)$ gives the area under the curve when f is positive on $[a, b]$.

Chapter 11: Statistics, Probability and Vectors

This final chapter gathers three topics that share a common purpose: making sense of information and of uncertainty. Statistics teaches us to summarise a mass of data with a few well-chosen numbers, to measure how widely the data are spread, and to read grouped data from tables and graphs. Probability measures how likely an event is, and it comes with counting techniques, permutations and combinations, for finding how many ways something can happen. Vectors, the third topic, describe quantities that have both size and direction and give geometry and mechanics a compact algebraic language. The three topics need little new machinery beyond what you already know, but they demand careful reading of the question and tidy organisation of the working. Read each problem twice, decide which tool it calls for, and write down the formula before substituting.

11.1 Measures of central tendency

A single number that represents a whole set of data is called an **average** or measure of central tendency. Three are in common use. The **mean** is the sum of all the values divided by the number of values, written $\bar{x} = \Sigma x / n$. The **median** is the middle value when the data are arranged in

order; if there is an even number of values it is the mean of the two middle ones. The **mode** is the value that occurs most often; a set may have more than one mode or none.

Each has a role. The mean uses every value and is the usual choice, but a few extreme values can distort it: one very rich person in a village raises the mean income without changing what a typical household earns. The median resists such outliers. The mode is the only average that makes sense for non-numerical categories, such as the most popular brand of garri.

WORKED EXAMPLE 11.1

The scores of ten students in a test are 12, 15, 11, 18, 14, 15, 9, 16, 15 and 10. Find the mean, median and mode.

Solution: The sum is 12 + 15 + 11 + 18 + 14 + 15 + 9 + 16 + 15 + 10 = 135, so the mean is 135/10 = 13.5. In order the scores are 9, 10, 11, 12, 14, 15, 15, 15, 16, 18. The two middle values are the fifth and sixth, 14 and 15, so the median is 14.5. The value 15 occurs three times, more than any other, so the mode is 15.

11.2 Grouped data

When there are many observations they are sorted into **classes** and the number of observations in each class, the **frequency** f, is recorded. Individual values are no longer known, so we estimate using the **class**

mid-value x, the average of the lower and upper class boundaries. The estimated mean is $\bar{x} = \Sigma fx/\Sigma f$. For the median and mode we work within the class where they must lie.

> **Key point.** For grouped data with class width c: Median = L + $[(n/2 - F)/f] \times c$, where L is the lower boundary of the median class, F the cumulative frequency before that class, f the frequency of the class and n the total frequency. Mode = L + $[\Delta_1/(\Delta_1 + \Delta_2)] \times c$, where L is the lower boundary of the modal class (the class of highest frequency), Δ_1 is the modal frequency minus the frequency of the class before it, and Δ_2 is the modal frequency minus the frequency of the class after it.

The cumulative frequencies are plotted against upper class boundaries to draw the **ogive** or cumulative frequency curve. The median is read at n/2 on the vertical axis, the lower quartile at n/4 and the upper quartile at 3n/4. The difference between the upper and lower quartiles is the **interquartile range**. A **histogram** shows frequency by the area of bars, so with unequal class widths the heights must be frequency densities (frequency divided by class width); with equal widths the heights can simply be the frequencies. A **pie chart** divides a circle into sectors in proportion to the frequencies, the angle of each sector being (frequency/total) × 360°.

WORKED EXAMPLE 11.2

The marks of 25 candidates are grouped as follows.

Mark	0–10	10–20	20–30	30–40	40–50
Frequency	2	5	9	6	3

Estimate the mean, median and mode.

Solution: The mid-values are 5, 15, 25, 35 and 45. Then $\Sigma fx = 2(5) + 5(15) + 9(25) + 6(35) + 3(45) = 10 + 75 + 225 + 210 + 135 = 655$, and $\Sigma f = 25$, so the mean is $655/25 = 26.2$. The cumulative frequencies are 2, 7, 16, 22, 25. Since $n/2 = 12.5$, the median class is 20–30, with $L = 20$, $F = 7$, $f = 9$ and $c = 10$. Median $= 20 + [(12.5 - 7)/9] \times 10 = 20 + 55/9 = 26.1$ (to 1 d.p.). The modal class is 20–30 (highest frequency 9). $\Delta_1 = 9 - 5 = 4$ and $\Delta_2 = 9 - 6 = 3$, so Mode $= 20 + [4/7] \times 10 = 20 + 5.71 = 25.7$.

11.3 Measures of dispersion

Two classes may have the same mean mark yet very different spreads: in one every pupil is near the average, in the other the marks range from near zero to near full marks. The simplest measure of spread is the **range**, the largest value minus the smallest, but it uses only two values. A better measure uses every value. The **variance** is the mean of the squared deviations from the mean, $\sigma^2 = \Sigma(x - \bar{x})^2/n$, and the **standard deviation** is its positive square root, σ. For grouped data, $\sigma^2 = \Sigma f(x -$

$\bar{x})^2/\Sigma f$. An equivalent working formula is $\sigma^2 = \Sigma x^2/n - \bar{x}^2$ (and $\sigma^2 = \Sigma fx^2/\Sigma f - \bar{x}^2$ for grouped data), which is often quicker. The standard deviation is in the same units as the data, which is why it is preferred to the variance for interpretation.

WORKED EXAMPLE 11.3

Find the variance and standard deviation of 2, 4, 4, 4, 5, 5, 7, 9.

Solution: The sum is 40 and $n = 8$, so the mean is 5. The deviations are $-3, -1, -1, -1, 0, 0, 2, 4$ and their squares are 9, 1, 1, 1, 0, 0, 4, 16, which total 32. Variance $= 32/8 = 4$ and the standard deviation is $\sqrt{4} = 2$. Check with the working formula: $\Sigma x^2 = 4 + 16 + 16 + 16 + 25 + 25 + 49 + 81 = 232$, so $\sigma^2 = 232/8 - 25 = 29 - 25 = 4$. The two methods agree.

11.4 Probability

An experiment is any process with an uncertain outcome, such as tossing a coin or selecting a student. The set of all possible outcomes is the **sample space**, and an **event** is a collection of outcomes. When all outcomes are equally likely, the probability of an event E is $P(E) = (\text{number of outcomes in E})/(\text{total number of outcomes})$. A probability is always between 0 and 1, where 0 means impossible and 1 means certain. The probability that an event does not happen is $1 - P(E)$, and this complement rule often shortens a calculation dramatically.

Events are **mutually exclusive** if they cannot occur together, and then P(A or B) = P(A) + P(B). For events that may overlap, the general addition rule is P(A or B) = P(A) + P(B) − P(A and B). Events are **independent** if the occurrence of one does not affect the probability of the other, and then P(A and B) = P(A) × P(B). If the events are dependent, as when items are drawn without replacement, the probability of the second event must be recalculated using the changed contents: P(A and B) = P(A) × P(B given A).

WORKED EXAMPLE 11.4

A bag contains 5 red, 3 blue and 2 green beads. Two beads are drawn one after the other without replacement. Find the probability that (a) both are red, (b) they are of different colours.

Solution: There are 10 beads at first. (a) P(red then red) = 5/10 × 4/9 = 20/90 = 2/9. (b) Use the complement. P(both blue) = 3/10 × 2/9 = 6/90 and P(both green) = 2/10 × 1/9 = 2/90. The probability that both are the same colour is 20/90 + 6/90 + 2/90 = 28/90, so the probability of different colours is 1 − 28/90 = 62/90 = 31/45.

WORKED EXAMPLE 11.5

A card is drawn from a standard pack of 52. Find the probability that it is a heart or a king. A fair die and a fair coin are also thrown together; find the probability of an even number and a head.

Solution: There are 13 hearts and 4 kings, and one card, the king of hearts, is in both groups. So P(heart or king) = 13/52 + 4/52 − 1/52 = 16/52 = 4/13. For the die and coin, the events are independent, so P(even and head) = 3/6 × 1/2 = 1/4.

11.5 Permutations and combinations

To count outcomes without listing them, we use the multiplication principle: if one choice can be made in m ways and a second in n ways, then the two together can be made in m × n ways. The **factorial** of a positive integer n is n! = n × (n − 1) × ... × 2 × 1, with 0! = 1. It counts the ways of arranging n different objects in a row.

Key point. Permutations (order matters): ${}^{n}P_{r} = n!/(n - r)!$. Combinations (order does not matter): ${}^{n}C_{r} = n!/[r!(n - r)!]$. Arrangements of n objects of which p are alike, q are alike and so on: n!/(p! q! ...).

The difference between the two is the question: does the order of selection create a different outcome? Electing a president and a secretary from a group of students is a permutation, since swapping roles gives a different result; choosing a committee of two is a combination, since the committee {Ada, Musa} is the same as {Musa, Ada}.

WORKED EXAMPLE 11.6

(a) In how many ways can the letters of IBADAN be arranged? (b) A team of 3 is to be chosen from 8 players. In how many ways? (c) A committee of 2 boys and 2 girls is chosen from 5 boys and 4 girls. How many committees are possible?

Solution: (a) There are 6 letters of which A occurs twice, so the number of arrangements is $6!/2! = 720/2 = 360$. (b) ${}^{8}C_{3} = (8 \times 7 \times 6)/(3 \times 2 \times 1) = 336/6 = 56$. (c) The boys can be chosen in ${}^{5}C_{2} = 10$ ways and the girls in ${}^{4}C_{2} = 6$ ways, so the committee can be formed in $10 \times 6 = 60$ ways.

11.6 Vectors

A vector has magnitude and direction. In the plane we write vectors in component form using the unit vectors **i** and **j**, of length 1 along the x-axis and y-axis. The vector $\mathbf{a} = 3\mathbf{i} + 4\mathbf{j}$ can also be written as a column $(3, 4)$. Adding and subtracting vectors means adding and subtracting components, and multiplying by a scalar multiplies each component. The **magnitude** (length) of $\mathbf{a} = x\mathbf{i} + y\mathbf{j}$ is $|\mathbf{a}| = \sqrt{(x^2 + y^2)}$. A **unit vector**

in the direction of a is a/|a|. If A and B have position vectors a and b relative to the origin, then the vector from A to B is AB = b − a, and its length is the distance between the points.

Two vectors are **parallel** if one is a scalar multiple of the other. The **scalar product** (dot product) of $a = (a_1, a_2)$ and $b = (b_1, b_2)$ is $a\cdot b = a_1b_1 + a_2b_2 = |a||b| \cos\theta$, where θ is the angle between the vectors. Two non-zero vectors are perpendicular exactly when a·b = 0. Rearranging gives the angle: $\cos\theta = a\cdot b/(|a||b|)$.

WORKED EXAMPLE 11.7

Given a = 3**i** + 4**j** and b = **i** − 2**j**, find |a|, a unit vector along a, a + b and 2a − 3b.

Solution: |a| = √(9 + 16) = 5, so a unit vector along a is (3**i** + 4**j**)/5 = 0.6**i** + 0.8**j**. Next a + b = 4**i** + 2**j**. For 2a − 3b: 2a = 6**i** + 8**j** and 3b = 3**i** − 6**j**, so 2a − 3b = 3**i** + 14**j**.

WORKED EXAMPLE 11.8

Find the angle between p = (2, 3) and q = (4, −1), and the value of k for which (k, 3) is perpendicular to (2, −4).

Solution: p·q = 2(4) + 3(−1) = 8 − 3 = 5. |p| = √13 and |q| = √17, so $\cos\theta = 5/\sqrt{221} = 5/14.866 = 0.3363$, giving θ = 70.3°. For the second part, perpendicularity means the scalar product is zero: 2k + 3(−4) = 0, so k = 6.

WORKED EXAMPLE 11.9

A is the point (1, 2) and B is (4, 6). Find the vector AB and its length, and the bearing of B from A if **i** points east and **j** points north.

Solution: AB = (4 − 1)**i** + (6 − 2)**j** = 3**i** + 4**j** and |AB| = 5 units. The direction makes an angle of $\tan^{-1}(3/4) = 36.9°$ east of north, so the bearing of B from A is 036.9°.

11.7 Strategies and common errors

In statistics, check that frequencies total the stated number and that you divide by Σf, not by the number of classes. Take care to identify the correct median class by comparing n/2 with the cumulative column. In probability, decide first whether the draws are with or without replacement, because that determines whether the second denominator is reduced. Use the complement rule whenever "at least one" appears: P(at least one) = 1 − P(none). In counting problems, ask whether order matters and whether any objects are identical. In vectors, never add magnitudes when you mean to add vectors; add the components and then find the magnitude of the result. A magnitude is always positive, and the scalar product of two vectors is a number, not a vector.

Exercise 11.1

1. Find the mean, median and mode of 3, 5, 5, 6, 11.

2. Find the mean of the data: value 10, 20, 30, 40 with frequencies 2, 3, 4, 1.
3. The mean of five numbers is 8. A sixth number is added and the mean becomes 9. Find the sixth number.
4. Find the variance and standard deviation of 1, 3, 5, 7.
5. A bag holds 4 white and 6 black balls. Two are drawn without replacement. Find the probability that both are black.
6. Two fair dice are thrown. Find the probability that the sum is 7.
7. In how many ways can the letters of KADUNA be arranged?
8. In how many ways can a team of 3 be chosen from 7 players?
9. If P(A) = 0.3 and P(B) = 0.6, and A and B are mutually exclusive, find P(A or B). If they were independent, what would P(A and B) be?
10. If $\mathbf{a} = 2\mathbf{i} + \mathbf{j}$ and $\mathbf{b} = -\mathbf{i} + 3\mathbf{j}$, find $|\mathbf{a} + \mathbf{b}|$.
11. Find k such that $(3, k)$ is perpendicular to $(4, 6)$.
12. Find the unit vector in the direction of $5\mathbf{i} - 12\mathbf{j}$.

Answers to Exercise 11.1

1. Mean 6, median 5, mode 5: the sum is 30, and 5 is the middle and the repeated value.
2. 24: $\Sigma fx = 20 + 60 + 120 + 40 = 240$ and $\Sigma f = 10$.
3. 14: the new total is $6 \times 9 = 54$ and the old total was $5 \times 8 = 40$.

4. Variance 5 and SD $\sqrt{5} = 2.24$: the mean is 4, the squared deviations are 9, 1, 1, 9 = 20, and 20/4 = 5.
5. 1/3: $6/10 \times 5/9 = 30/90$.
6. 1/6: the pairs (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) give 6 of the 36 outcomes.
7. 360: 6!/2! because A repeats.
8. 35: ${}^{7}C_{3} = (7 \times 6 \times 5)/6$.
9. 0.9 and 0.18: 0.3 + 0.6 for exclusive events, and 0.3×0.6 for independent events.
10. $\sqrt{17}$: $a + b = \mathbf{i} + 4\mathbf{j}$, so the magnitude is $\sqrt{(1 + 16)}$.
11. $k = -2$: $12 + 6k = 0$.
12. $(5\mathbf{i} - 12\mathbf{j})/13$: the magnitude is $\sqrt{(25 + 144)} = 13$.

Chapter summary

- Mean = $\Sigma x/n$ (or $\Sigma fx/\Sigma f$ for grouped data); median is the middle value; mode is the most frequent value.
- Grouped median = $L + [(n/2 - F)/f]c$; grouped mode = $L + [\Delta_1/(\Delta_1 + \Delta_2)]c$; the ogive gives the median and quartiles.
- Variance $\sigma^2 = \Sigma(x - \bar{x})^2/n = \Sigma x^2/n - \bar{x}^2$, and standard deviation is its square root.
- P(E) = favourable/total outcomes, P(not E) = 1 − P(E); for exclusive events P(A or B) = P(A) + P(B); for independent events P(A and B) = P(A)P(B).

- ${}^{n}P_{r} = n!/(n - r)!$ and ${}^{n}C_{r} = n!/[r!(n - r)!]$; repeated objects reduce the arrangements by dividing by the factorials of the repeats.
- For $a = x\mathbf{i} + y\mathbf{j}$, $|a| = \sqrt{(x^2 + y^2)}$; $AB = b - a$; $a \cdot b = a_1b_1 + a_2b_2 = |a||b| \cos \theta$, and $a \cdot b = 0$ means perpendicular.